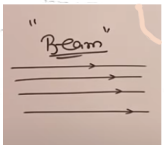
Geometrical optics

Optics → Study of light

→ In geometrical optics, light behaves like a ray.

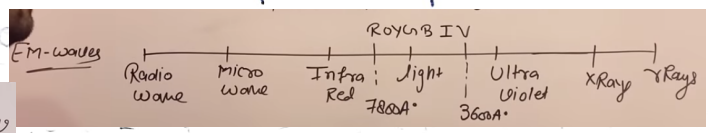
Rectilinear propagation of light: Light will keep on travelling along straight line in a homogenous medium, until it encounters a hindrance.

→ A collection of ray's is called a beam.

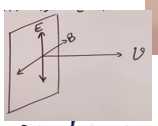


Light: It is part of electromagnetic wave whose wavelength lies b/w 3600Å to 7800Å in vacuum/air.

→ It is part of em wave which is responsible for our vision.



Properties of light:



→ Light is EM wave

→ Speed of light in air/vacuum ≈ $c = 3\times10^8$ m/s

→ In any other medium, speed of light = $v = \frac{c}{n} = \frac{3\times10^8 \text{ m/s}}{n}$ (n = Refractive index of medium)

$n = \frac{c}{v}$ = Refractive index

→ ratio

→ unitless, dimensionless

→ $n_{water} = 4/3$ $n_{glass} = 3/2$

→ $n \geq 1$, in general

→ $n_{meta\text{-}materials} = -ve$

$v_{water} = \frac{3\times10^8}{4/3} = \frac{9}{4}\times10^8$ m/s

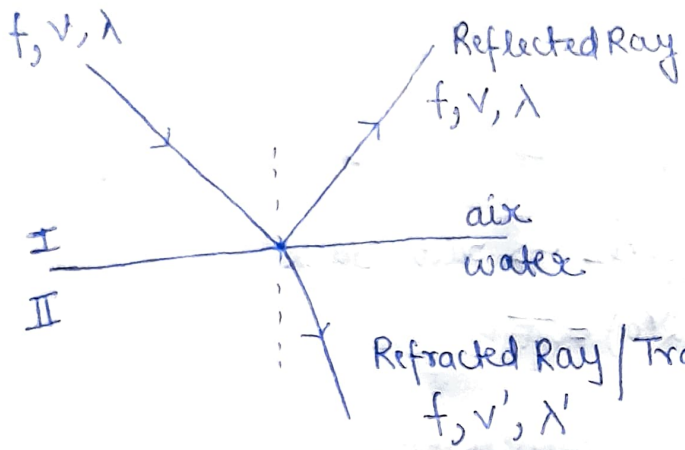
$v_{glass} = \frac{3\times10^8}{3/2} = 2\times10^8$ m/s

→ $\boxed{v = f\lambda}$ v = speed of light, λ = wavelength of light, f = frequency of light

→ Particle of light is known as photon/quanta.

→ Momentum of photon = $P = \frac{h}{\lambda}$ $h = 6.626\times10^{-34}$ J-sec (Planck's const.)

→ Energy = $E = hf = \frac{hc}{\lambda}$



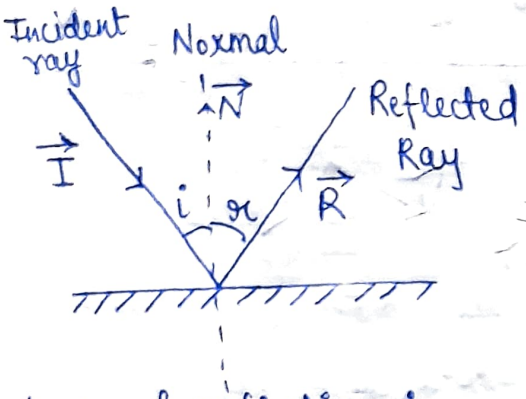
→ f is source dependent, not medium dependent

→ Speed of light is medium dependent

→ λ of light is also medium dependent.

$$\lambda = \frac{v}{f}$$

Reflection of light:



→ If ray strikes a surface and returns back to the medium, this phenomenon is known as reflection.

i = angle of incidence

r = angle of reflection

Laws of reflection:

① Incident ray, normal and reflected ray, all must lie in same plane (coplanar)

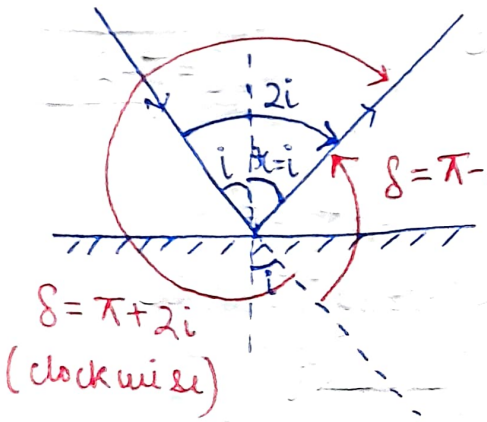
$$(\vec{I} \times \vec{R}) \cdot \vec{N} = 0$$

$(\vec{I} \times \vec{R}) \cdot \vec{N} = 0$

$(\vec{R} \times \vec{N}) \cdot \vec{I} = 0$

② Angle of reflection (r) = Angle of incidence (i)

Angle of deviation (δ):

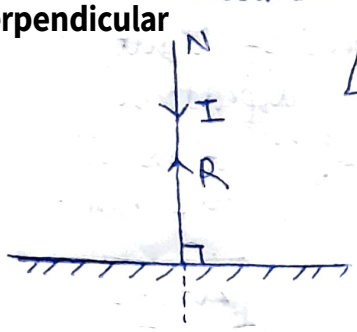


★ we prefer $\delta = \pi - 2i$ more.

Special Cases :

① Normal Incidence :

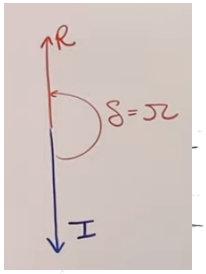
perpendicular

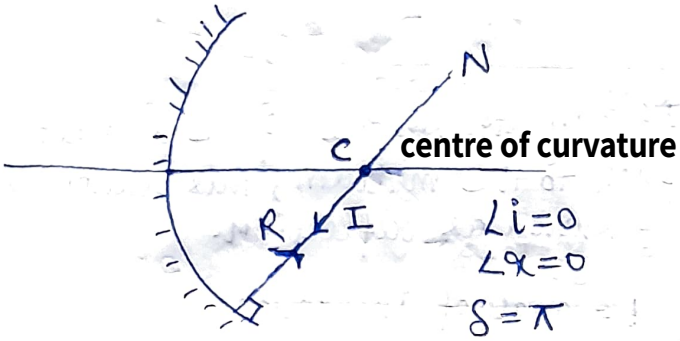


$\angle i = 0 = r$

"Ray retraces its path back"

$\delta = \pi - 2(0) = \pi$





$\angle i = 0$

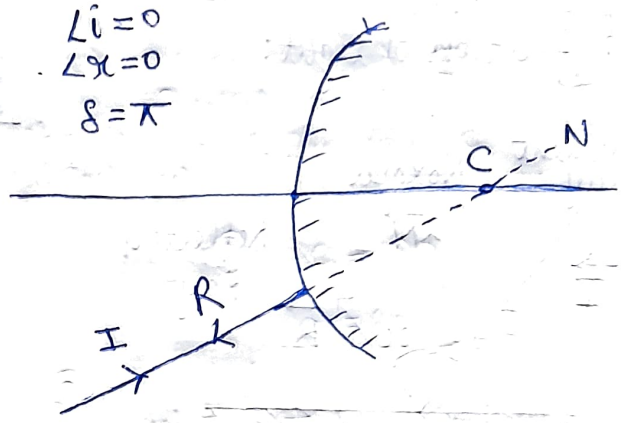
$\angle r = 0$

$\delta = \pi$

$\angle i = 0$

$\angle r = 0$

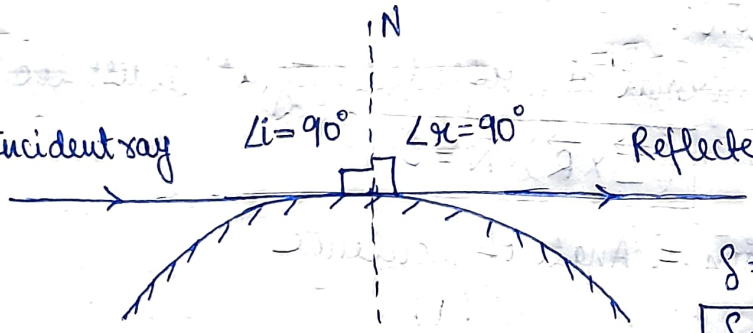
$\delta = \pi$



② Grazing Incidence :

↓

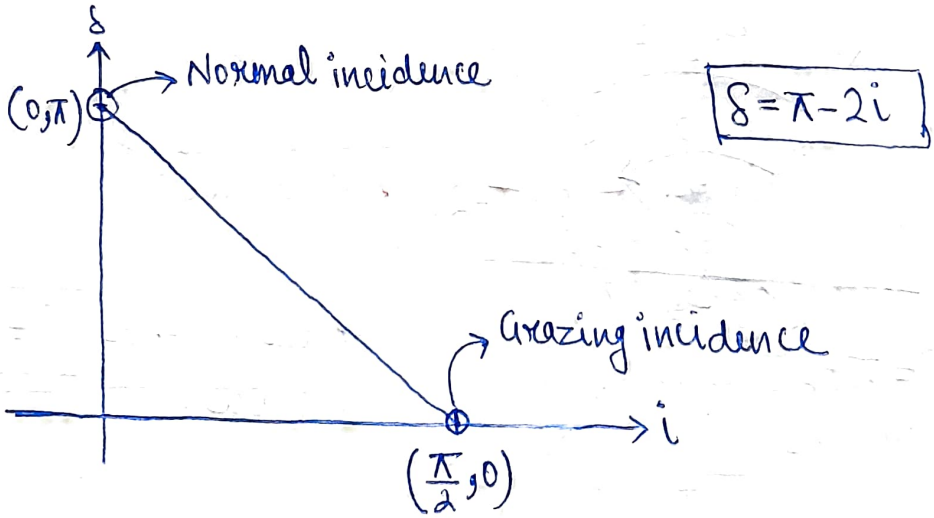
Just touch



$\delta = \pi - 2(\pi/2)$

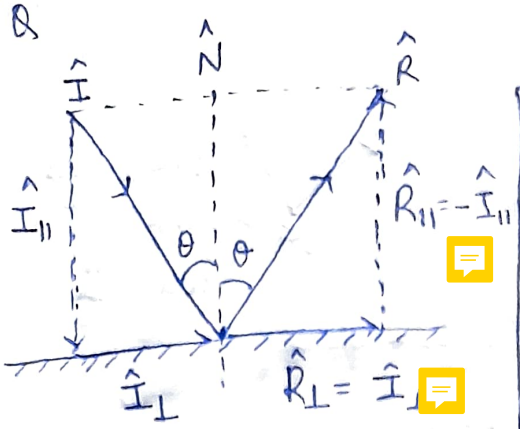
$\boxed{\delta = 0}$

Q Draw graph between δ and i



$\boxed{\delta = \pi - 2i}$

Find unit vector ($\hat{R}$) along reflected ray, in terms of $\hat{I}$ and $\hat{N}$.

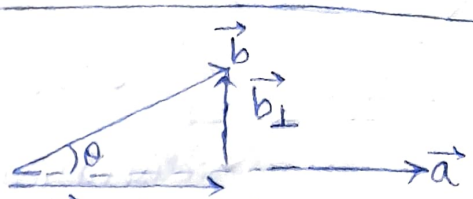


$$\hat{I}_{\parallel} = \frac{(\hat{N}\cdot\hat{I})}{(1)^2}\hat{N}$$

$$\boxed{\hat{I}_{\parallel} = (\hat{N}\cdot\hat{I})\hat{N}}$$

$$\boxed{\hat{I}_{\perp} = \hat{I} - (\hat{N}\cdot\hat{I})\hat{N}}$$

#Note:



$$\vec{b}_{\parallel} = (b\cos\theta)\hat{a}$$

$$= \frac{ab\cos\theta}{a}\cdot\frac{\vec{a}}{a}$$

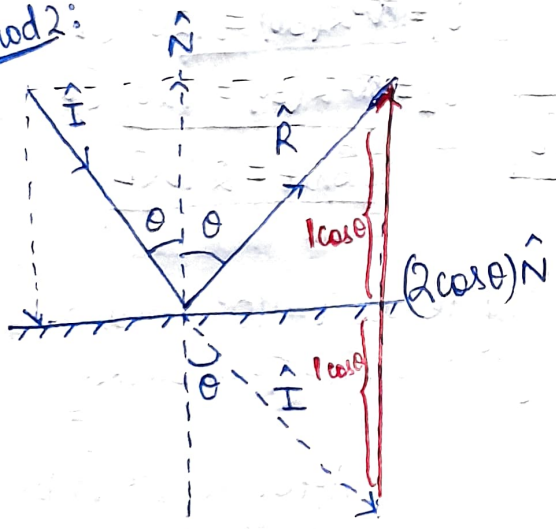
$$\boxed{\vec{b}_{\parallel} = \frac{(\vec{a}\cdot\vec{b})\vec{a}}{a^2}}$$

$$\vec{b}_{\parallel} + \vec{b}_{\perp} = \vec{b}$$

$$\boxed{\vec{b}_{\perp} = \vec{b} - \frac{(\vec{a}\cdot\vec{b})\vec{a}}{a^2}}$$

$$\hat{R} = \hat{R}_{\perp} + \hat{R}_{\parallel} = \{\hat{I} - (\hat{N}\cdot\hat{I})\hat{N}\} - (\hat{N}\cdot\hat{I})\hat{N} \Rightarrow \boxed{\hat{R} = \hat{I} - 2(\hat{N}\cdot\hat{I})\hat{N}}$$

Method 2:

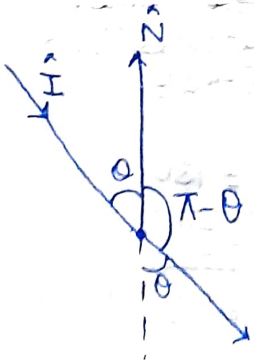


Triangle law of vector:

$$\hat{R} = \hat{I} + (2\cos\theta)\hat{N}$$

From (1)

$$\boxed{\hat{R} = \hat{I} - 2(\hat{I}\cdot\hat{N})\hat{N}}$$



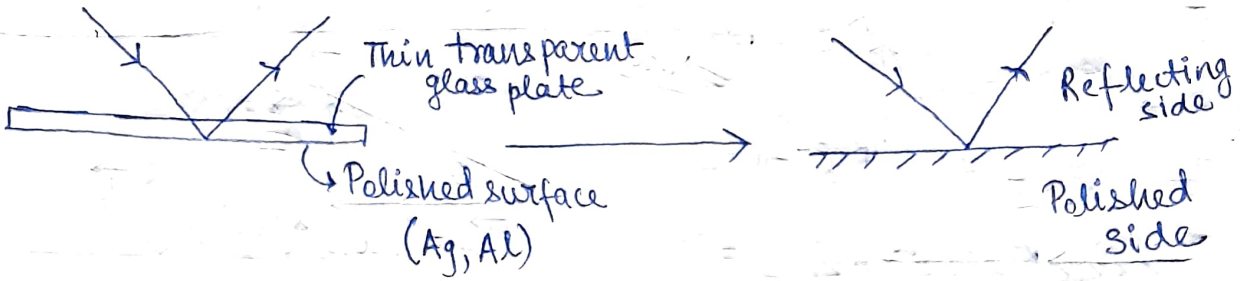
Angle b/w $\hat{I}$ and $\hat{N}$:

$$\hat{I}\cdot\hat{N} = (1)(1)\cos(\pi-\theta)$$

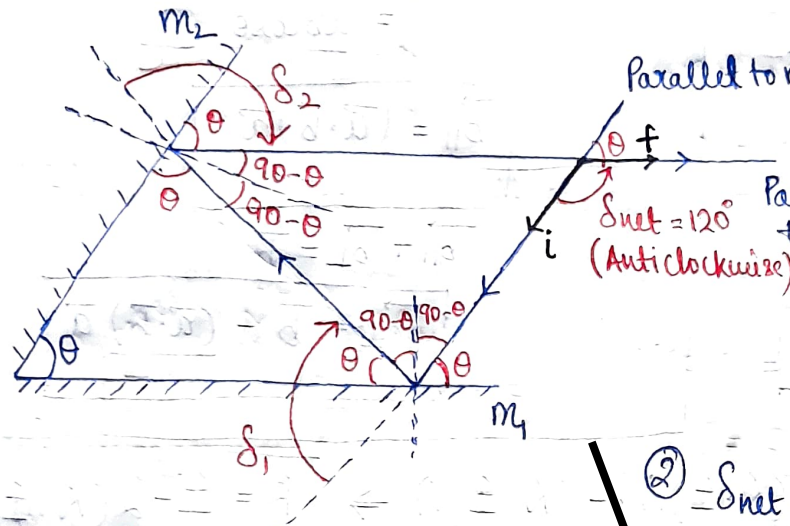
$$\hat{I}\cdot\hat{N} = -\cos\theta$$

$$\cos\theta = -(\hat{I}\cdot\hat{N}) \quad \text{—(1)}$$

Plane mirror:



Q



Find ① θ ② δ_{net}

① $3\theta = 180$

$\boxed{\theta = 60°}$

② $\delta_{net} = \delta_1 + \delta_2$ (clockwise)

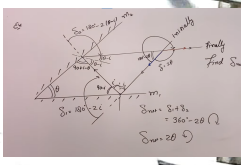
$\delta_1 = 180 - 2(30) = 120°$ ↻

$\delta_2 = 180 - 2(30) = 120$ ↻

$\boxed{\delta_{net} = 240 ↻}$

We can also say δ_{net} is angle b/w i and f.

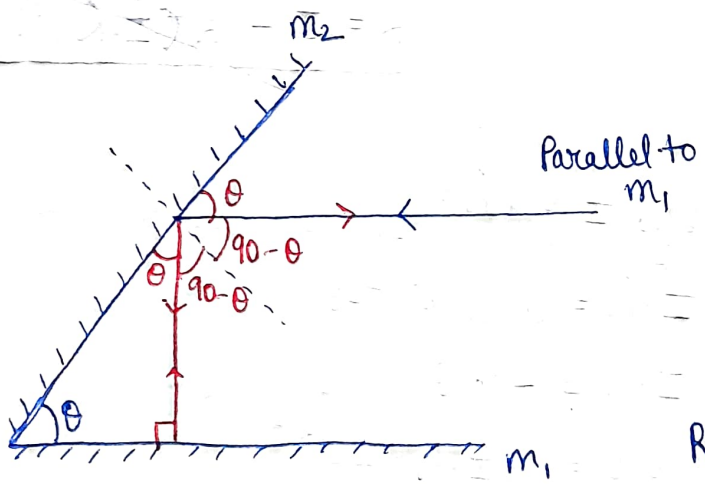
$\boxed{\delta_{net} = 120° ↺}$ or $\boxed{\delta_{net} = 240° ↻}$



Principle of Reversibility:

When light strikes a mirror normally ie at $i = 10°$, then ray suffers a deviation of 180° and ray retraces its path back.

Q



After reflection by m_1 for 1st time, the ray retraces its path back. Find θ.

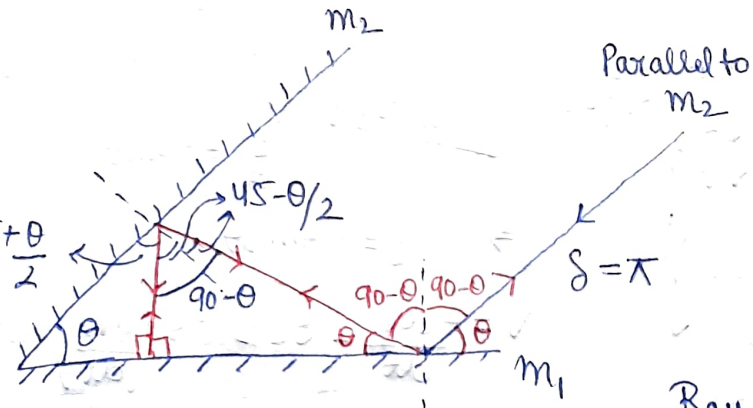
$2\theta + 90 = 180$

$\boxed{\theta = 45°}$

Ray undergoes 3 reflections

$\delta_{net} = \pi$

Q



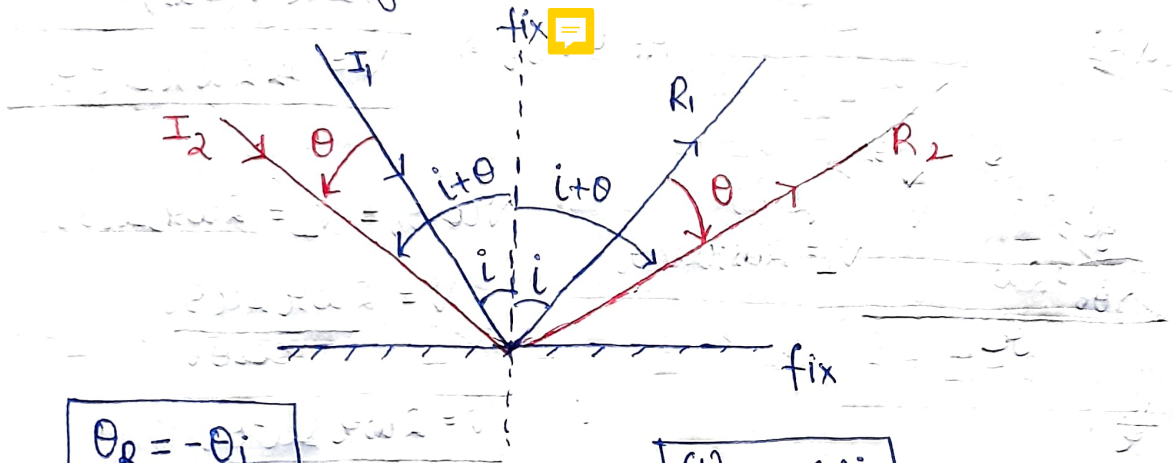
After 2nd reflection by m_1, the ray starts to retrace its path back. Find θ.

$$\theta + 90° + 45 + \frac{\theta}{2} = 180°$$

$$\boxed{\theta = 30°}$$

Ray suffers 5 reflections

Rotation of incident ray:

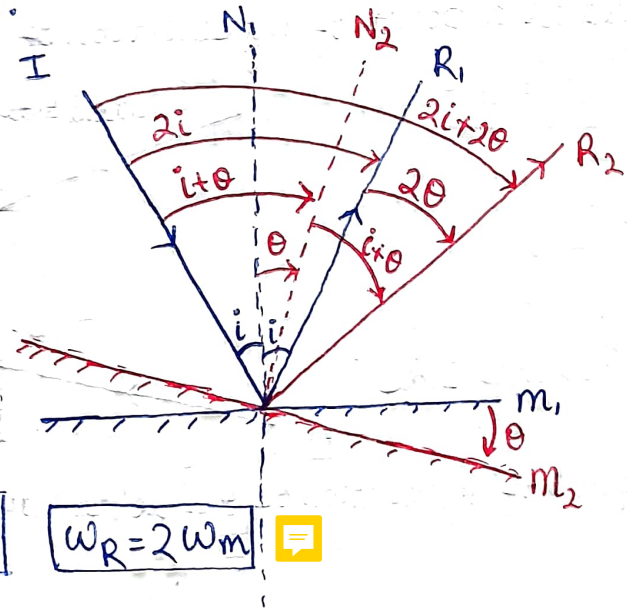


$$\boxed{\theta_R = -\theta_i}$$

$$\boxed{\omega_R = -\omega_i}$$

(-ve indicates opposite sense)

Rotation of mirror:

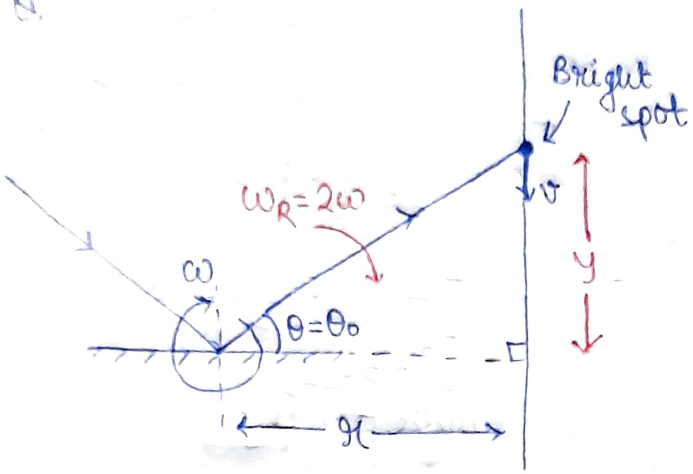


$$\therefore \boxed{\theta_R = 2\theta_m} \quad \boxed{\omega_R = 2\omega_m}$$

* If incident ray and mirror both are rotating

$$\boxed{\theta_R = 2\theta_m - \theta_i} \quad \boxed{\omega_R = 2\omega_m - \omega_i}$$

Q.



Find speed of bright spot at this instant.

Method I:

$$\frac{y}{x} = \tan\theta$$

$$\Rightarrow y = x\tan\theta$$

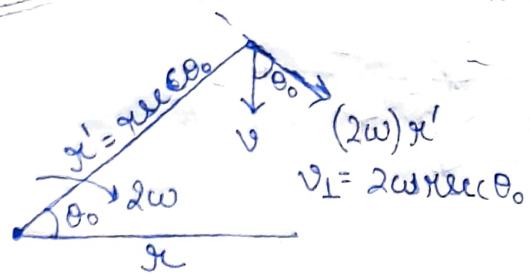
$$v = -\frac{dy}{dt} = x\sec^2\theta\left(\frac{d\theta}{dt}\right)$$

$$\Rightarrow v = x\sec^2\theta(+2\omega) \quad \left\{2\omega = \frac{d\theta}{dt}\right\}$$

$$v = x\sec^2\theta(2\omega)$$

At $\theta = \theta_0 \Rightarrow$ $\boxed{v = 2\omega x\sec^2\theta_0}$

Method 2:



$$v\cos\theta_0 = v_\perp = 2\omega x\sec\theta_0$$

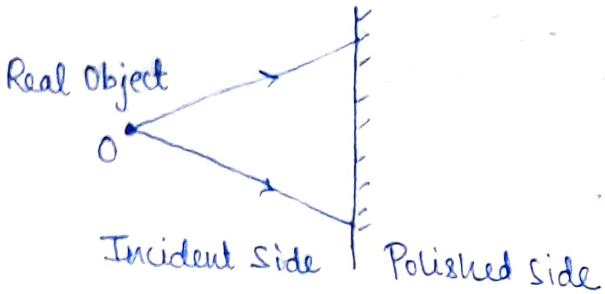
$$\therefore v = \frac{2\omega x\sec\theta_0}{\cos\theta_0}$$

$$\therefore \boxed{v = 2\omega x\sec^2\theta_0}$$

$$\boxed{\omega = \frac{v_\perp}{x}}$$

Object and image:

Object: It is defined as a point of intersection of incident rays.



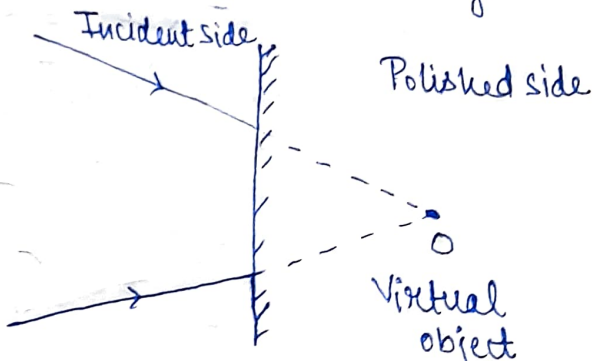
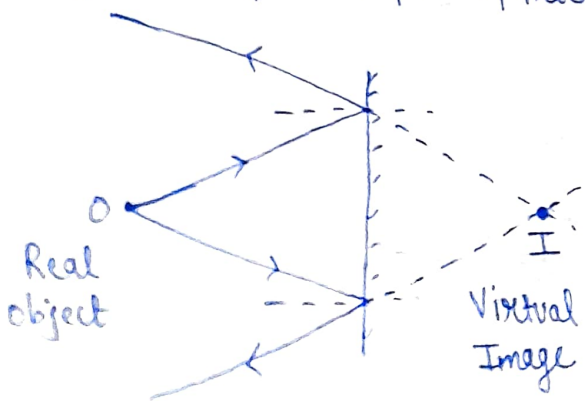
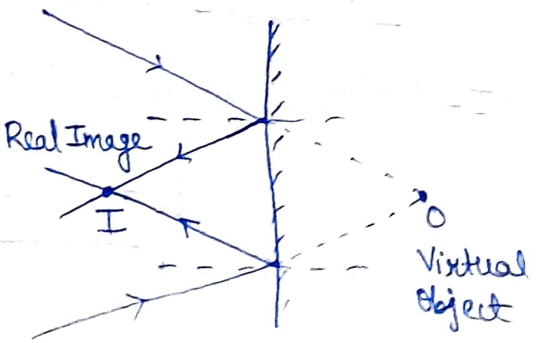
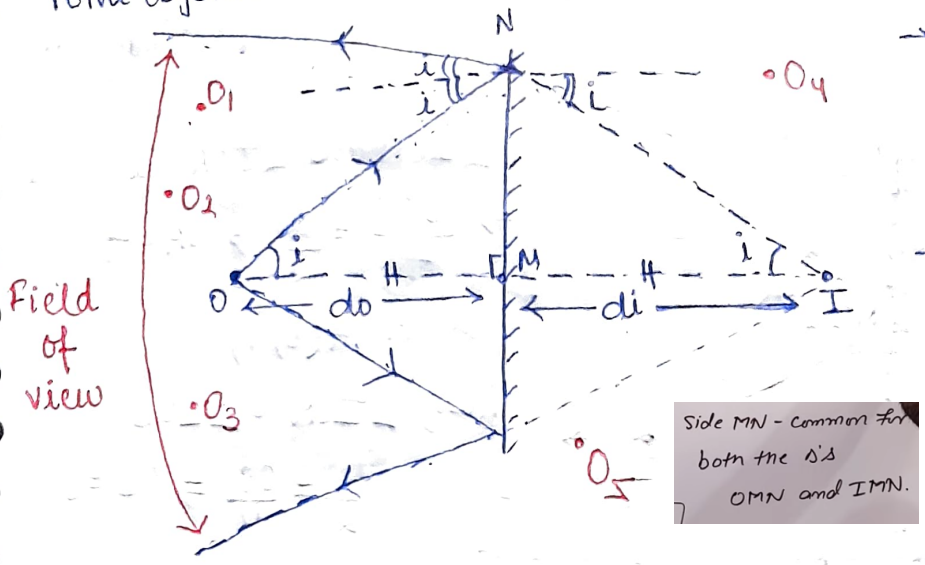


Image: It is defined as point of intersection of reflected rays or refracted rays.





Point object :



→ O_1, O_2 and O_3 receive reflected ray and hence can observe image I.

→ O_4 and O_5 didnt receive any reflected ray and hence cant observe the image I.

→ O and I are equidistant from plane mirror. ($d_o = d_i$)

→ line OI is ⊥ to the mirror.

∴ Plane mirror is perpendicular bisector to the line segment OI.

→ Any ray coming from O will appear to be passing through I after reflection.

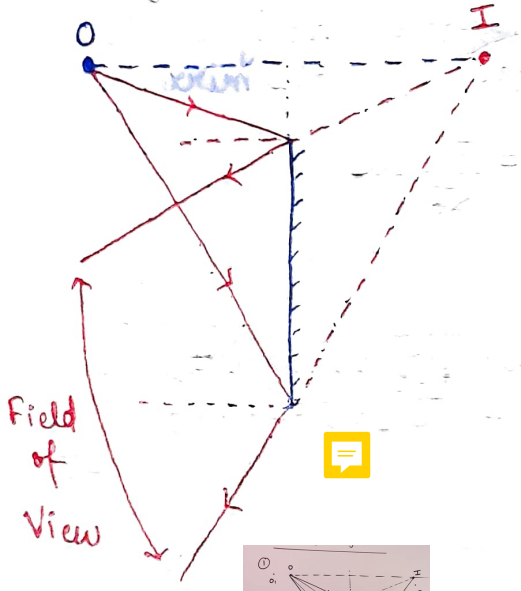
Field of view : It is the region in which if an observer is present, he can see the image.

OR

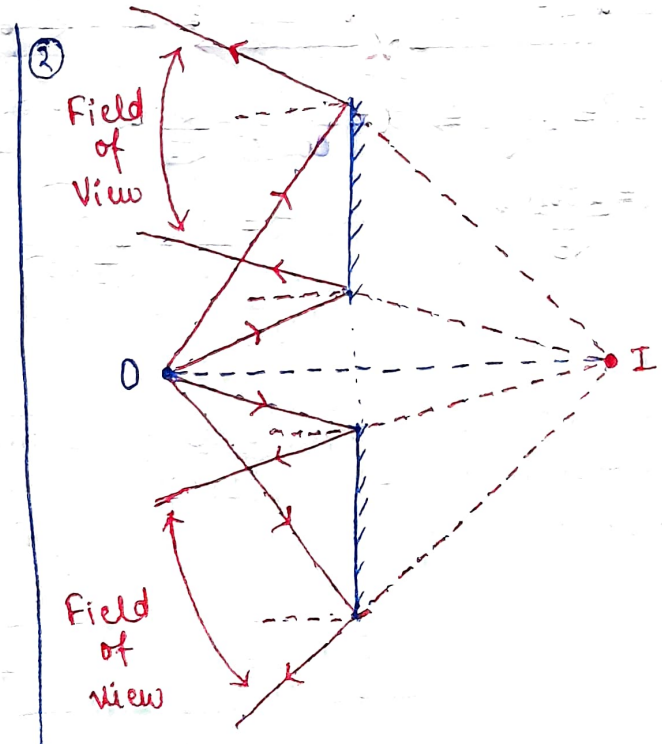
It is the region consisting all reflected rays.

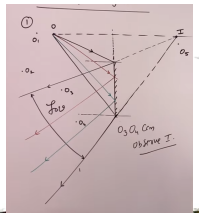
Q Draw field of view -

①

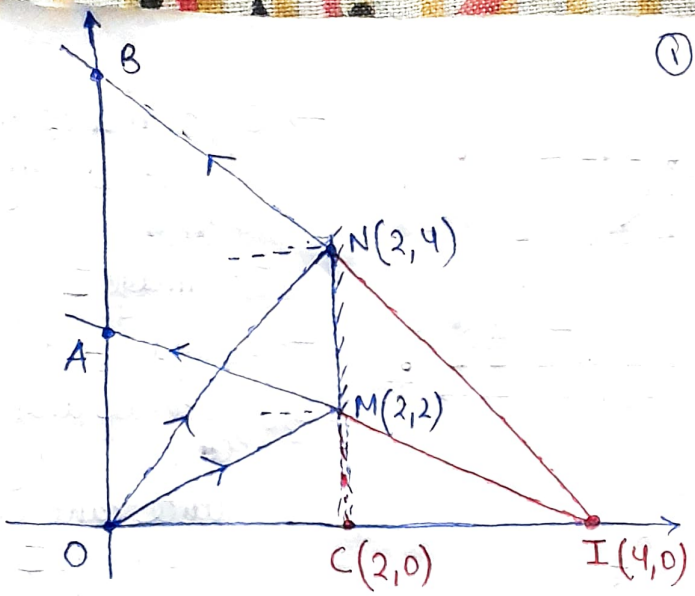


②





Q



① Find the region on y-axis in which reflected rays are present.

ΔICM and ΔIOA are similar

$\Rightarrow \frac{OA}{4} = \frac{2}{2} \Rightarrow OA = 4$

$\therefore A(0,4)$

Also, $\Delta ICN \sim \Delta IOB$

$\frac{4}{2} = \frac{OB}{4} \Rightarrow OB = 8$

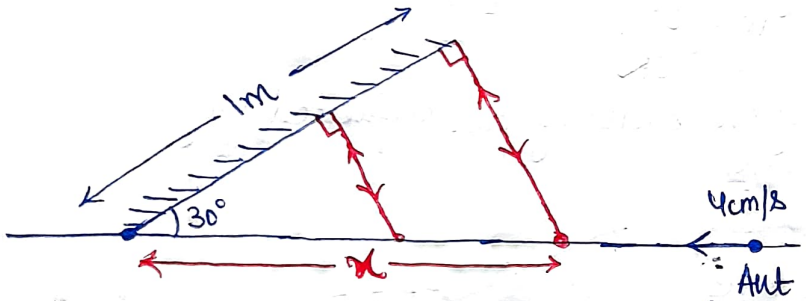
$\therefore B(0,8)$

$\therefore$ Field of view is from $(0,4)$ to $(0,8)$

② If an ant moves on the y-axis with speed 4cm/s, for how much time will it observe the image.

$$t = \frac{d}{v} = \frac{4m}{4cm/s} = \frac{4m}{0.04m/s} = 100\,sec$$

Q



For how long ant can see its own image in the mirror?

Concept: For ant to see its own image, rays from ant should come back to ant after reflection.

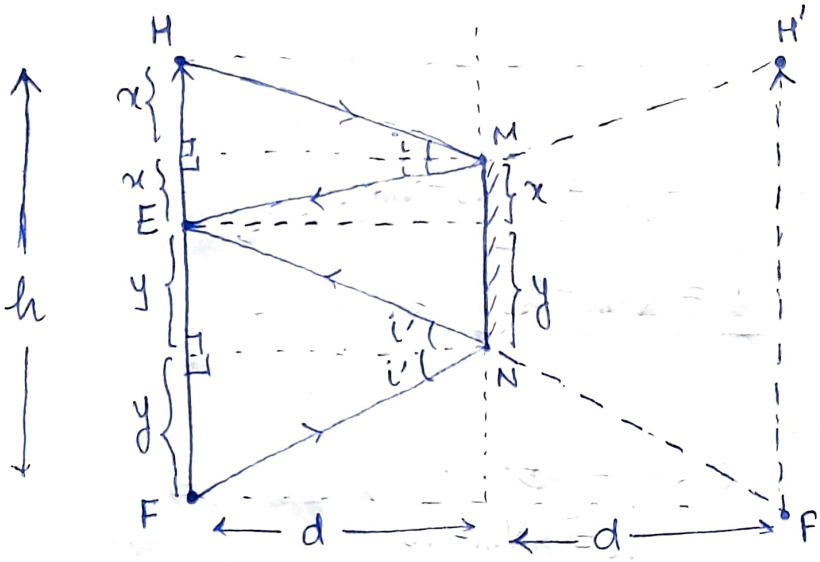
$\therefore$ Rays should strike the mirror normally.

$\cos 30° = \frac{1m}{x} \Rightarrow x = \frac{1}{\cos 30°} = \frac{2}{\sqrt{3}} m$

$x = \frac{200cm}{\sqrt{3}}$

Now, $t = \frac{d}{v} = \frac{200}{\sqrt{3} \times 4} = \frac{50}{\sqrt{3}}\,sec$

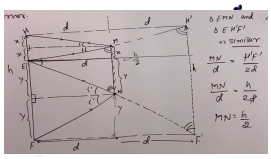
* Q Show that the minimum size of a plane mirror, required to see the full image of an observer is half the size of that observer.

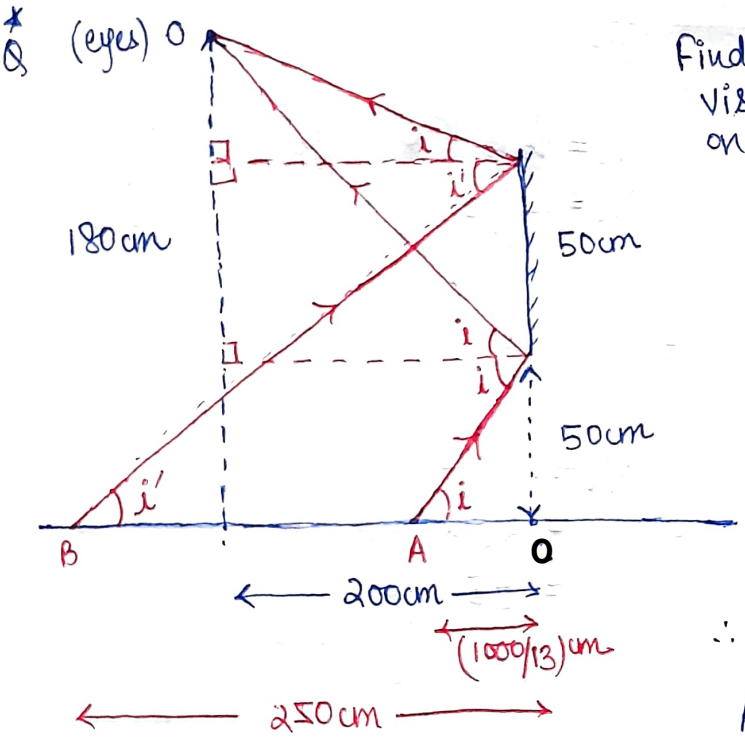


H = Head
F = Feet
E = Eyes
Height of man = h
Height of mirror = x+y
We know,
$2x+2y = h$
$2(x+y) = h$

$$\boxed{x+y = \frac{h}{2}}$$

Also, height of eyes from foot = 2y
Then, height above the foot at which mirror should be kept = y





Find the length of the floor visible to observer due to reflection from mirror.

$$\tan i = \frac{50\,cm}{OA} = \frac{130\,cm}{200\,cm}$$

$$\therefore OA = \frac{200\times 5}{13} = \frac{1000}{13}\,cm$$

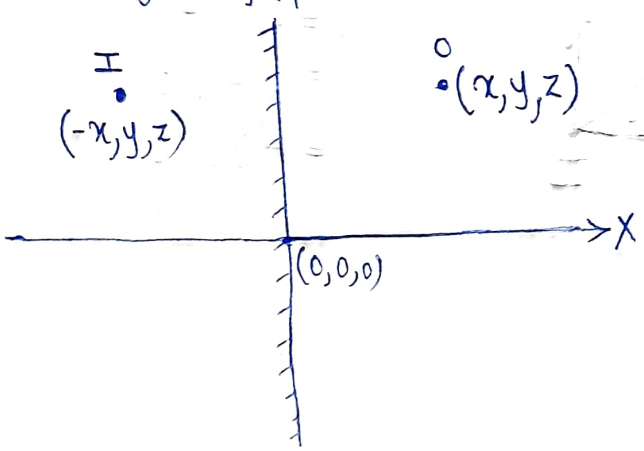
$$\tan i' = \frac{80\,cm}{200\,cm} = \frac{100\,cm}{OB}$$

$$\therefore OB = \frac{100\times 200}{80} = 250\,cm$$

$\therefore$ Length of floor visible = AB

$$AB = \left(250 - \frac{1000}{13}\right)\,cm$$

Velocity of Image:



① In the plane parallel to the mirror, object and image have same motion.

$\therefore$

$$\boxed{(V_i)_y = (V_o)_y}$$

$$\boxed{(V_i)_z = (V_o)_z}$$

② ⊥ to mirror:

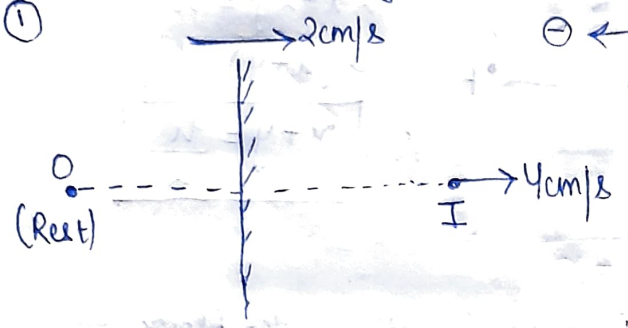
$$x_{im} = -x_{om}$$

$$\Rightarrow [V_{im} = -V_{om}]_x \quad = \quad V_i - V_m = -(V_o - V_m)$$

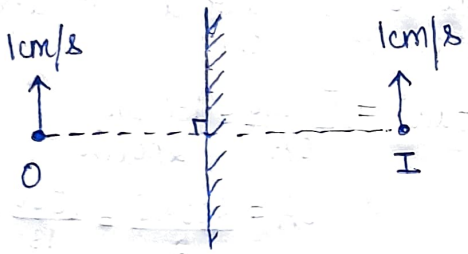
$$\therefore \star \boxed{V_i = 2V_m - V_o} \qquad \boxed{a_i = 2a_m - a_o}$$

only for plane ⊥ to the mirror

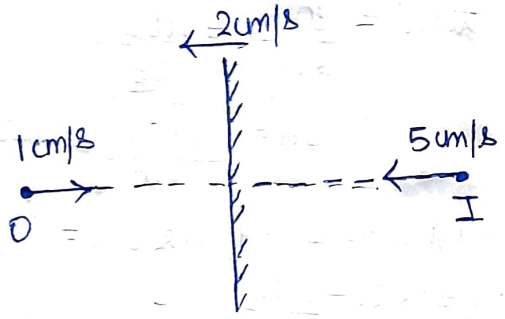
Q Find velocity of image.

① 

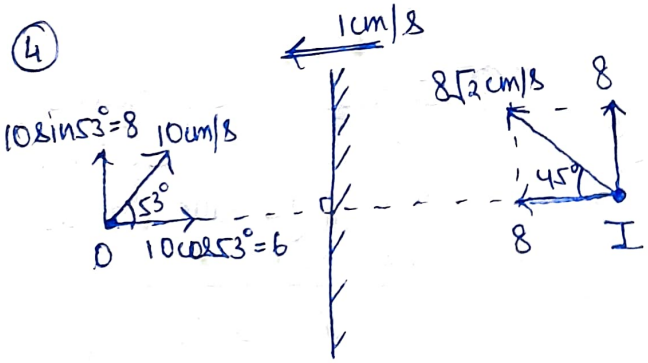
$V_i = 2V_m - V_o$
$= 2(2) - (0)$
$= 4\,cm/s$

② 

$V_i = V_o$
$V_i = 1\,cm/s$

③ 

$V_i = 2V_m - V_o$
$= 2(-2) - (1)$
$= -5\,cm/s$

④ 

$V_i = 2V_m - V_o$
$= 2(-1) - (6)$
$= -8\,cm/s$

Q A mirror parallel to XY plane is moving with velocity $\vec{V}_m = (3\hat{i} - 4\hat{j} + 5\hat{k})$ m/s and an object in front of mirror is moving with velocity $\vec{V}_o = (2\hat{i} + 4\hat{j} - \hat{k})$ m/s. Find velocity of image.

Parallel to plane of mirror,

$V_i = V_o$

$\Rightarrow (\vec{V}_i)_x = 2\hat{i}$

$(\vec{V}_i)_y = 4\hat{j}$

⊥ to mirror,

$V_i = 2V_m - V_o$

$= 2(5) - (-1)$

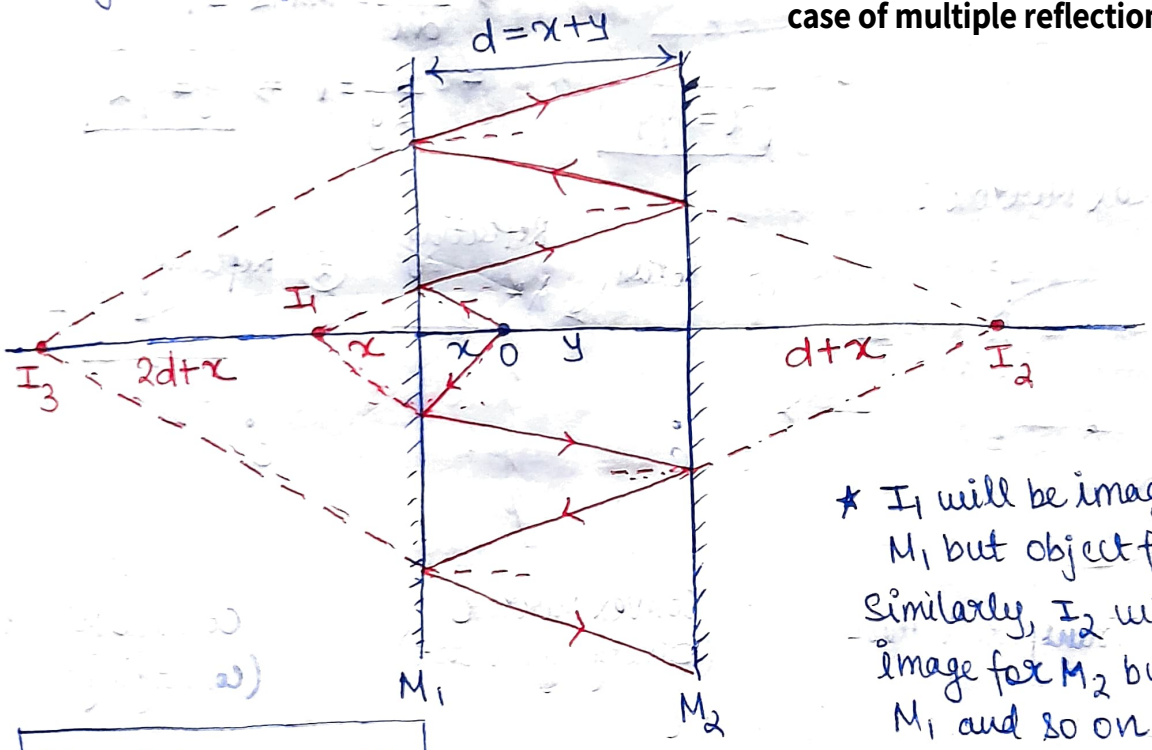
$= 11$

$\therefore (\vec{V}_i)_z = 11\hat{k}$

$\therefore \vec{V}_i = (2\hat{i} + 4\hat{j} + 11\hat{k})$ m/s

Image formed by two parallel plane mirrors:

case of multiple reflection



★ I_1 will be image for M_1 but object for M_2 Similarly, I_2 will be image for M_2 but object M_1 and so on.

No. of images = ∞

If rays strike M_1 first (from O)

M_1 M_2

$x \rightarrow d+x$

$2d+x \leftarrow$

$\rightarrow 3d+x$

$4d+x \leftarrow$

$\rightarrow 5d+x$

position of images

If rays from O strike M_2 first

M_1 M_2

$d+y \leftarrow y$

$\rightarrow 2d+y$

$3d+y \leftarrow$

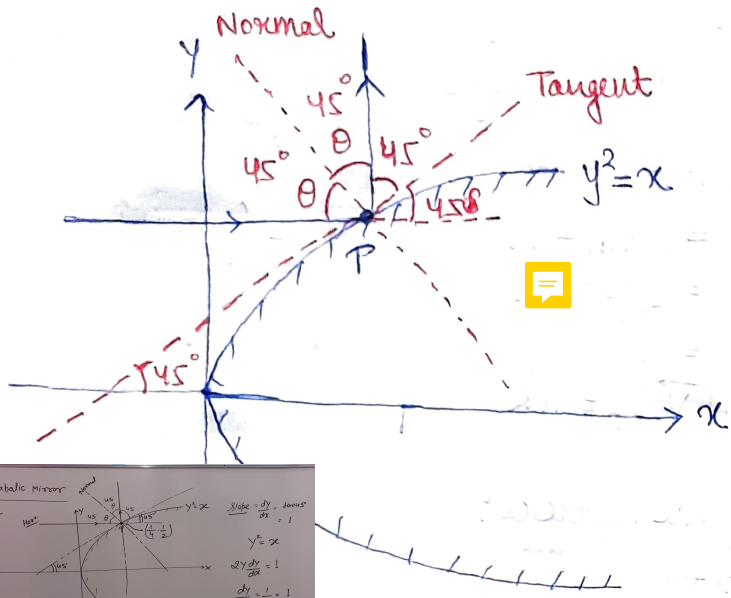
$\rightarrow 4d+y$

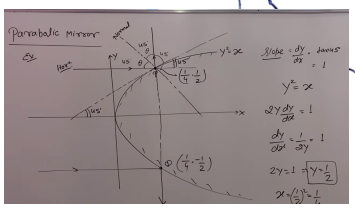
$5d+y \leftarrow$

⋮

Parabolic mirror:

Q





Find coordinates of point P, if ray becomes vertical up after reflection.

$2\theta = 90 \Rightarrow \theta = 45^\circ$

Now, slope $= \frac{dy}{dx} = \tan\theta$

$y^2 = x$

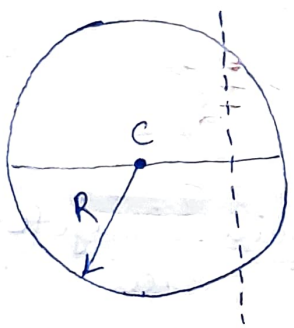
$2y\frac{dy}{dx} = 1 \Rightarrow \frac{dy}{dx} = \frac{1}{2y}$

$\Rightarrow \frac{dy}{dx} = \tan 45^\circ = 1$

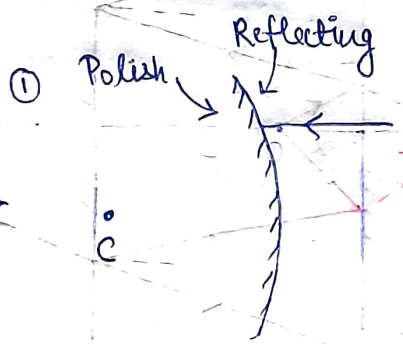
$= \frac{1}{2y} = 1 \Rightarrow \boxed{y = 1/2}$

$\boxed{x = 1/4}$

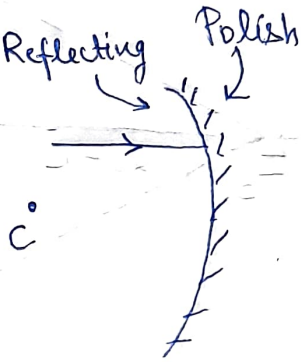
Spherical Mirror:



Hollow Transparent glass

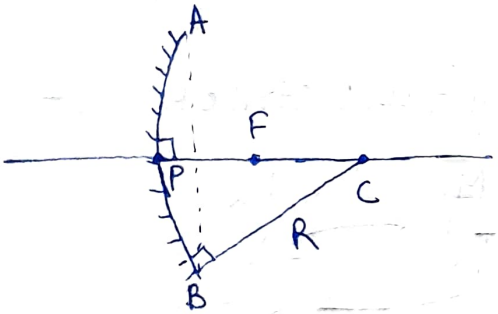


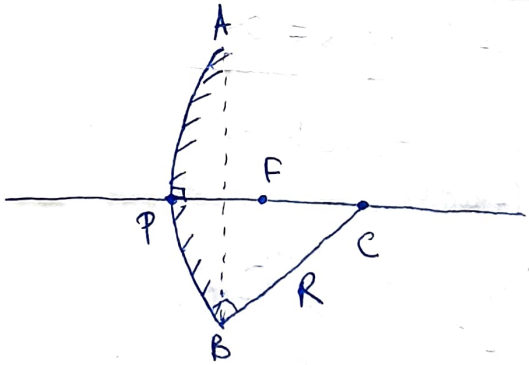
Convex Mirror
(Diverging mirror)



Concave Mirror
(Converging mirror)

Some definition of spherical mirror:





① Center of Curvature (C): It is centre of hollow sphere from which mirror is cut.

② Radius of curvature (R): It is radius of hollow sphere from which mirror is cut.

It is separation b/w C and surface of mirror

* Any ray passing through C is always normal to the mirror.

③ Pole (P): It is point on surface of mirror about which coordinates of object and image can be written easily. Generally, it is mid point of surface of mirror.

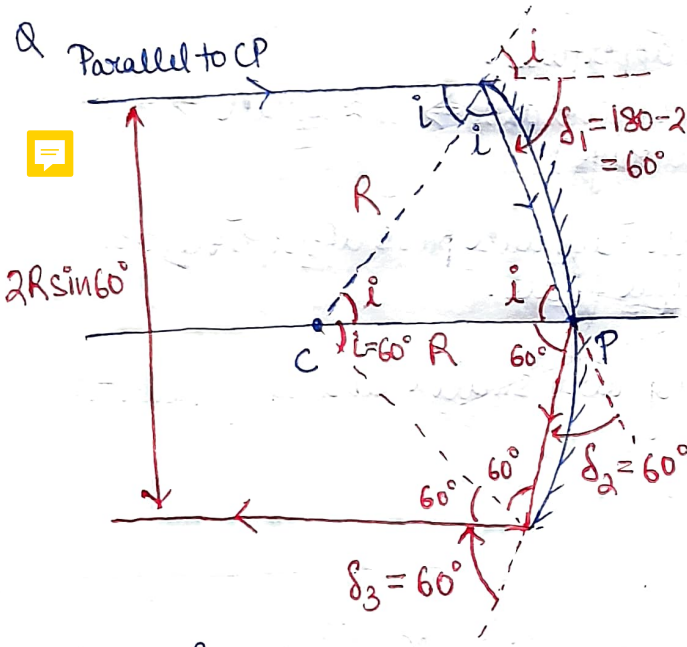
④ Principal axis / Optical axis: Line joining C and pole P is known as principal axis. It is always normal to the mirror.

⑤ Aperture Diameter: AB, diameter of portion of mirror surface which is available for reflection is known as aperture diameter.

⑥ Principal focus (F) = Midpoint of PC

Focal length = CF = FP = R/2

Q



① Find angle of incidence i, if after reflection, ray strikes pole P.

② Also draw ray diagram after reflection from P and find total number of reflections.

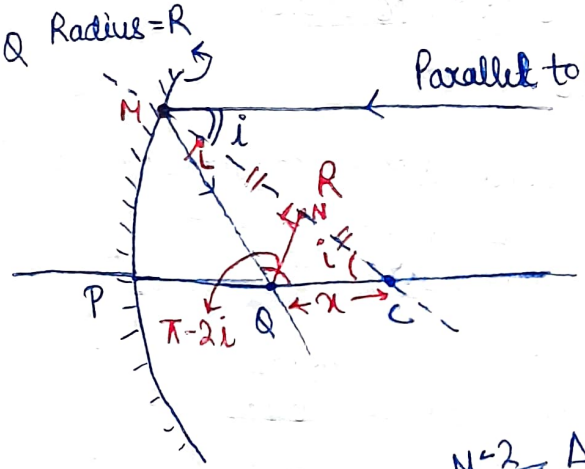
③ Also find deviation of each ray and net deviation.

① $3i = 180° \Rightarrow i = 60°$

② Total number of reflection = 3

③ $\delta_{net} = \delta_1 + \delta_2 + \delta_3 = 180°$

Q



Find CQ.

M-1 Sine rule (ΔMQC)

$$\frac{x}{\sin i} = \frac{R}{\sin(\pi - 2i)} = \frac{R}{\sin 2i}$$

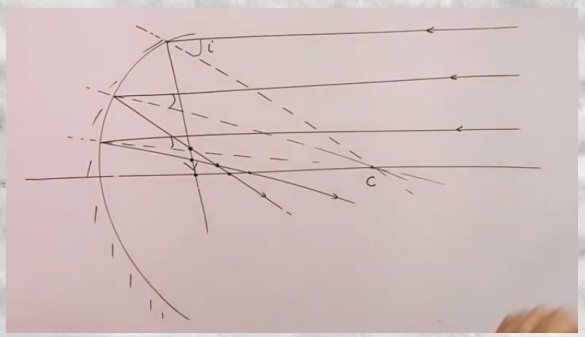
$$x = \frac{R}{2\sin i \cos i} \cdot \sin i = \frac{R}{2\cos i}$$

M-2 $\Delta CQN \cong \Delta NQM$

$$\Rightarrow \cos i = \frac{R/2}{x} \Rightarrow \boxed{x = \frac{R}{2\cos i}}$$

Spherical Aberration : aberration
defect

→ It is incapability of spherical mirrors to focus reflected rays at a single point.

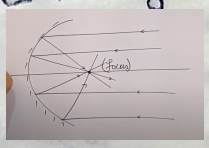


$$CQ = \frac{R}{2\cos i}$$

As $i \uparrow$, $\cos i \downarrow \Rightarrow CQ \uparrow$

How to remove spherical aberration?

① Use parabolic mirror → All the rays after reflection passes through focus



② Use spherical mirror of small apperture.

Small aperture ⇒ Rays are near to principal axis ⇒ $i \approx$ small → $\sin i \to 0$

If $i \to 0 \Rightarrow \cos i \to 1$ (paraxial rays)

$\therefore \boxed{CQ = \frac{R}{2}}$ Point from which rays will pass after reflection will be principal focus.

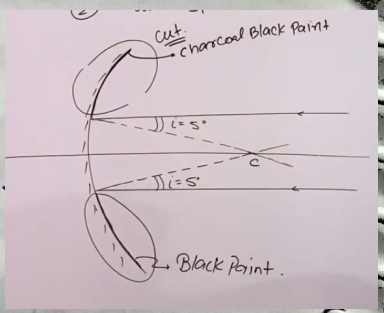
Paraxial Rays : They are rays having very small i and very near to principal axis.

We can make aperture small by :

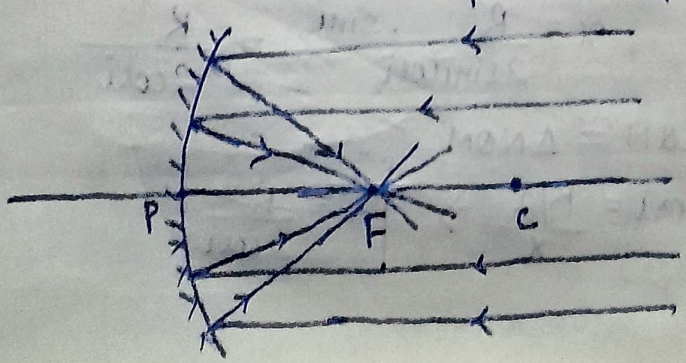
① Cutting the unnecessary part.

② Applying charcoal black paint on the extra portion.

③ Removing polish of the extra portion.

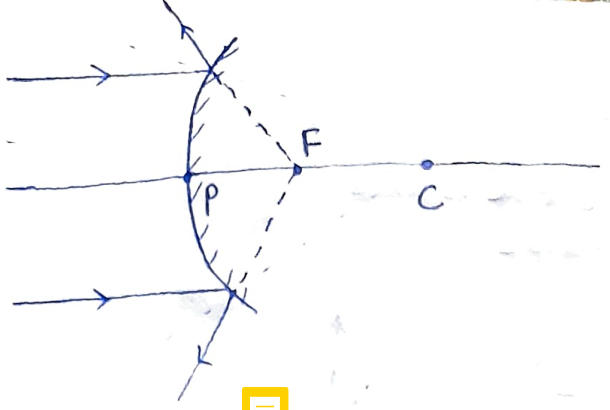


Principal focus : It is a point at which reflected rays converge or a point from which reflected rays seem to diverge if incident beam is parallel to principal axis.



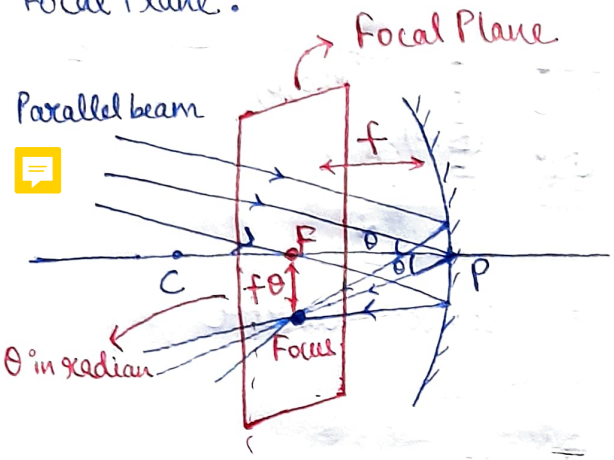
Concave mirrors
(converging mirror)
Real principal focus

Focal length = $f = CF = FP = \frac{R}{2}$



Convex Mirror
Diverging Mirror
Virtual Principal focus
Focal length = $f = \frac{R}{2}$

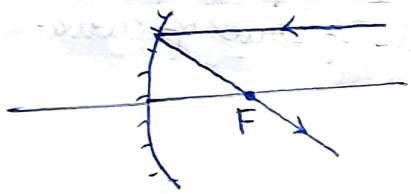
Focal Plane:

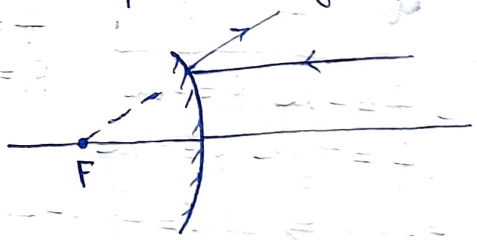


→ Focal plane is ⊥ to principal axis
→ Away from mirror by $f = R/2$
→ It has all the foci
→ Intersection of principal axis and focal plane is principal focus (F).

Ray Tracing:

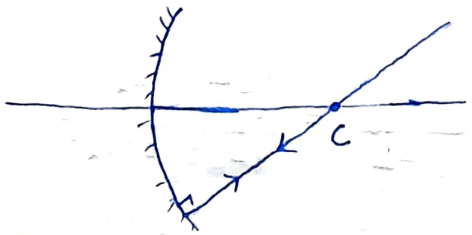
① Any ray parallel to principal axis will pass through principal focus after reflection

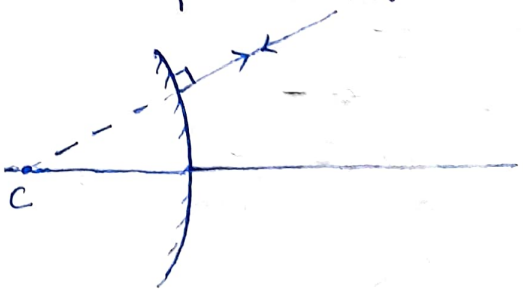




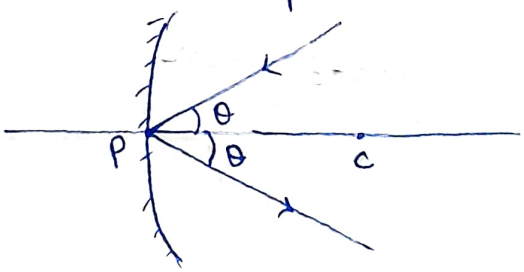
② If an incident ray passes through principal focus, then after reflection it will get parallel to principal axis.

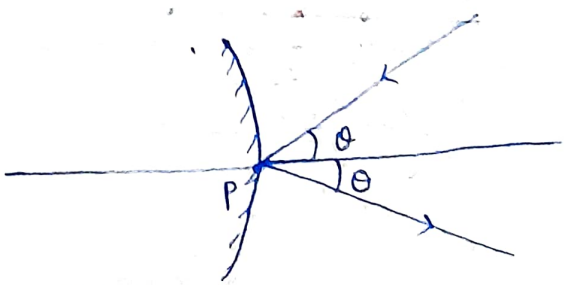
③ If incident ray passes through C, then it will strike the mirror normally and it will retrace its path back.





④ Reflection at pole:





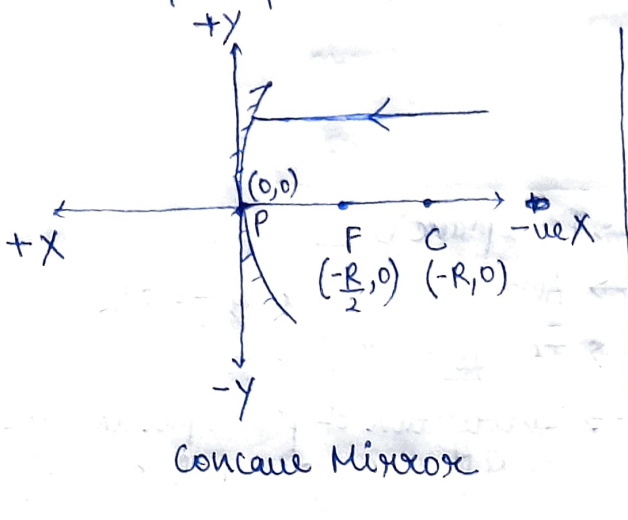
Sign Convention:

① Pole: It is origin

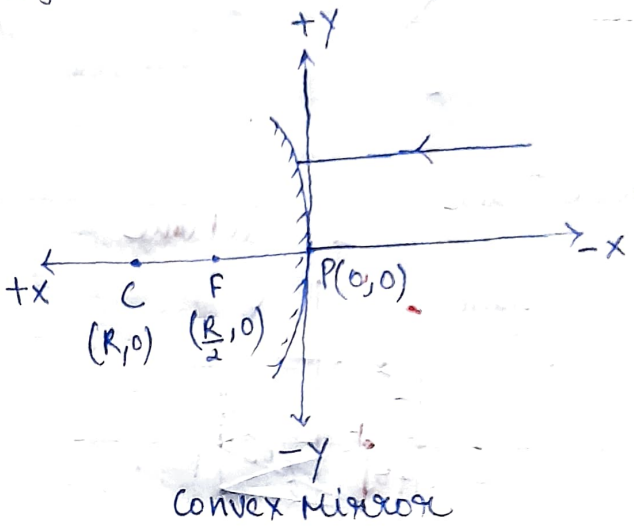
② Principal axis: X-axis

③ Direction of incident ray will provide +ve X-axis.

④ Above principal axis, we have +ve y-axis



Concave Mirror



Convex Mirror

Mirror Formula:

→ It relates X-coordinates of Object and Image.

$$\frac{1}{v}+\frac{1}{u}=\frac{1}{f}=\frac{2}{R}$$

u = X-coordinate of object
v = X-coordinate of image
f = X-coordinate of principal focus
R = X-coordinate of center of curvature

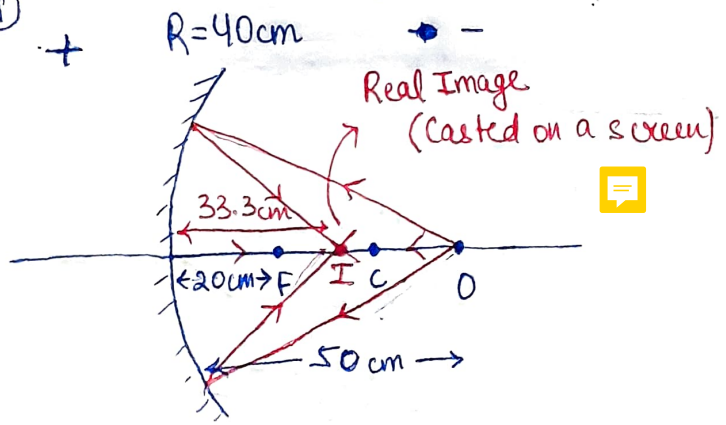
→ For concave, f = -ve, R = -ve
For convex, f = +ve, R = +ve

Now, $\frac{1}{v}+\frac{1}{u}=\frac{1}{f} \Rightarrow \frac{1}{v}=\frac{1}{f}-\frac{1}{u}$

$$\therefore * \quad v=\frac{fu}{u-f} \qquad f=\frac{uv}{u+v} \qquad u=\frac{fv}{v-f}$$

Q Find position of image.

①

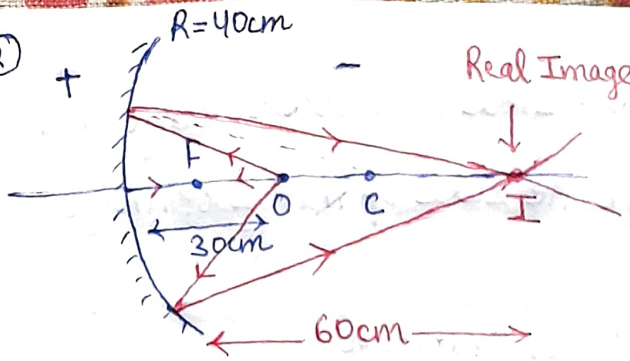


$$v=\frac{uf}{u-f}=\frac{(-50)(-20)}{(-50)-(-20)}$$

$$v=\frac{50\times 20}{-30}=\frac{-100}{3}\text{ cm}$$

$$v=-33.33\text{cm}$$

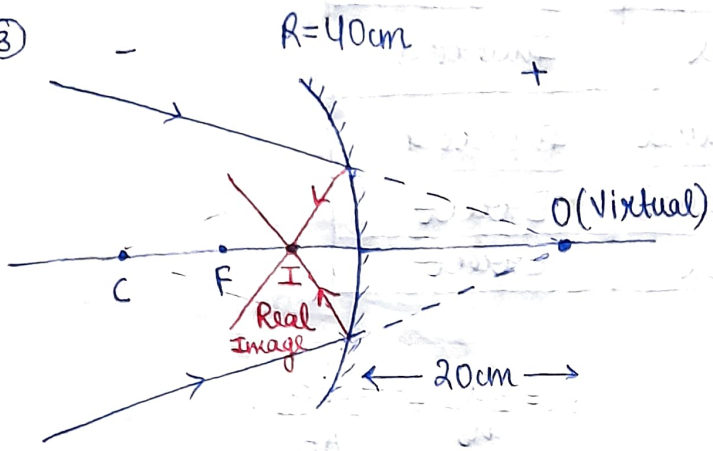
(2)



$$v = \frac{fu}{u-f} = \frac{(-20)\times(-30)}{(-30)-(-20)}$$

$$v = \frac{+20\times 30}{-10} = -60\text{cm}$$

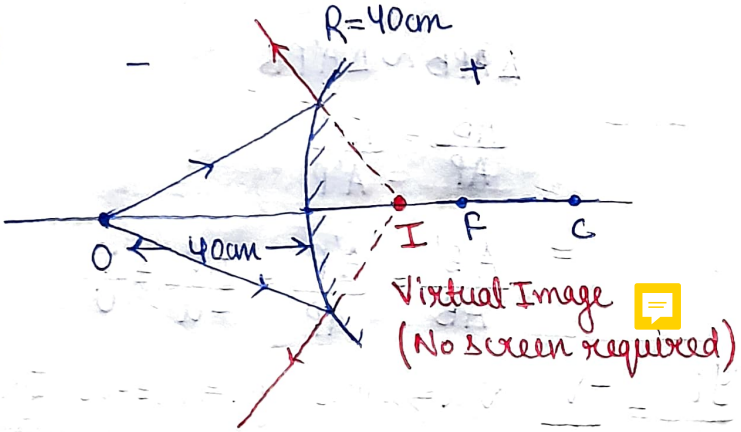
(3)



$$v = \frac{fu}{u-f} = \frac{(-20)(20)}{(20)-(-20)}$$

$$v = -\frac{20\times 20}{40_2} = -10\text{cm}$$

(4)

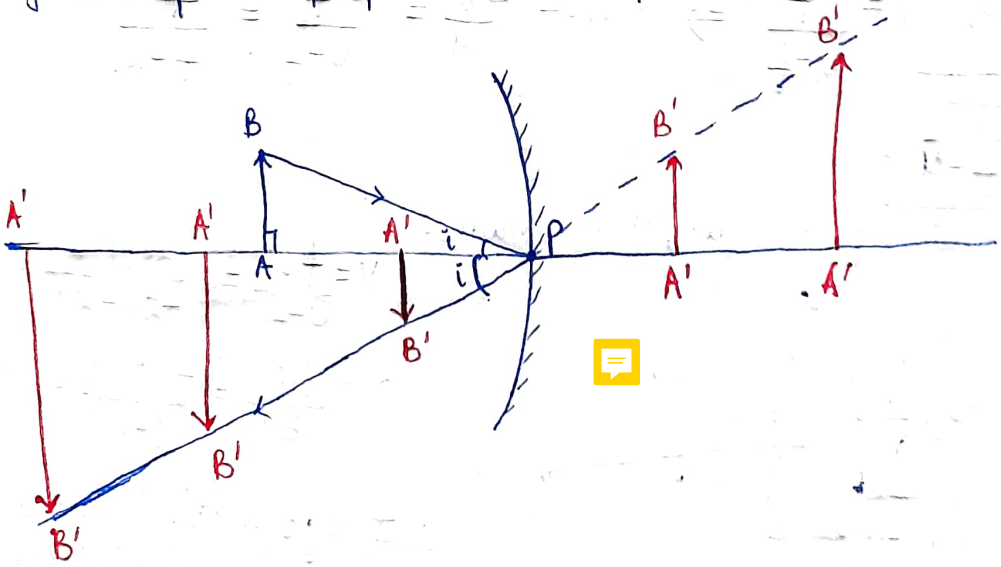


$$v = \frac{fu}{u-f} = \frac{(20)(-40)}{-40-(20)}$$

$$v = \frac{+20\times 40}{+60} = +\frac{40}{3}\text{cm}$$

Lateral / Transverse Magnification:

→ Object is placed perpendicular to principal axis.

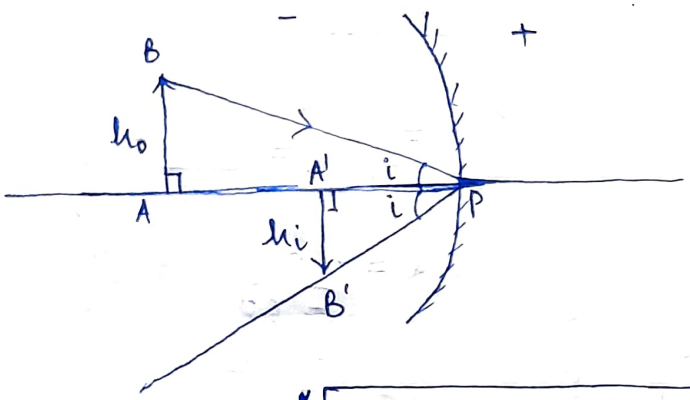


→ Farther the image, larger it is.

→ If object and image are on same side of mirror, image will be ~~erect~~ inverted

→ If object and image are on opposite side of mirror, image is erect

Object	Image	Configuration	
Real	Real	Inverted	both on the same side of mirror
Virtual	Virtual	Inverted	
Real	Virtual	Erect	both on opposite sides of the mirror
Virtual	Real	Erect	



$$m = \frac{h_i}{h_o} = \frac{A'B'}{AB}$$

$\Delta APB \sim \Delta A'PB'$

$$\frac{AB}{AP} = \frac{A'B'}{A'P}$$

$$= \frac{A'B'}{AB} = \frac{A'P}{AP} = \frac{-v}{-u} = \frac{v}{u}$$

$$\therefore * \quad \boxed{m = \frac{h_i}{h_o} = \frac{y_i}{y_o} = \frac{-v}{u}}$$

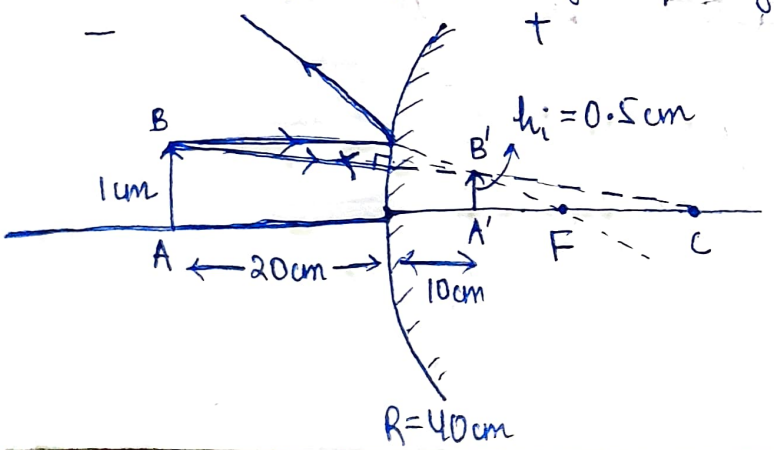
v, u ⇒ same sign ⇒ m = −ve ⇒ Inverted

v, u ⇒ opp. sign ⇒ m = +ve ⇒ Erect

Also, we know, $v = \frac{uf}{u-f} \Rightarrow m = \frac{-v}{u} = \frac{-f}{u-f} = \frac{f}{f-u}$

$$\therefore * \quad \boxed{m = \frac{f}{f-u}}$$

Q Find position of image and height of image.

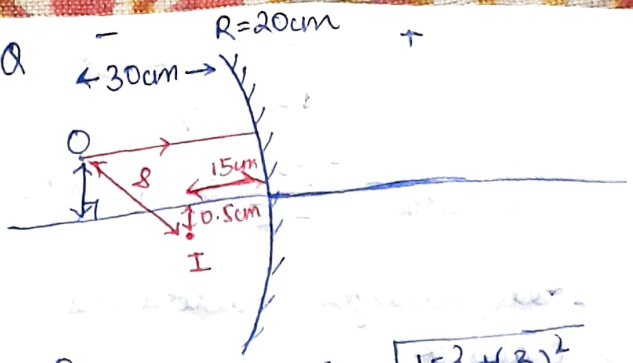


$$v = \frac{fu}{u-f} = \frac{(20)(-20)}{-20-20} = 10\text{cm}$$

$$m = \frac{h_i}{1\text{cm}} = \frac{-(10\text{cm})}{(-20)} \Rightarrow h_i = \frac{1}{2}\text{cm}$$

∴ Image is diminished, erect and virtual

Q



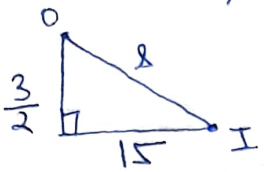
Find separation b/w O and I

$$v = \frac{fu}{u-f} = \frac{(-10)(-30)}{-30-(-10)}$$

$$v = -15\text{cm}$$

$$m = \frac{h_i}{h_o} = \frac{-v}{u} \Rightarrow \frac{h_i}{1\text{cm}} = \frac{-(-15)}{-(30)}$$

$$h_i = \frac{-1}{2}\text{ cm}$$



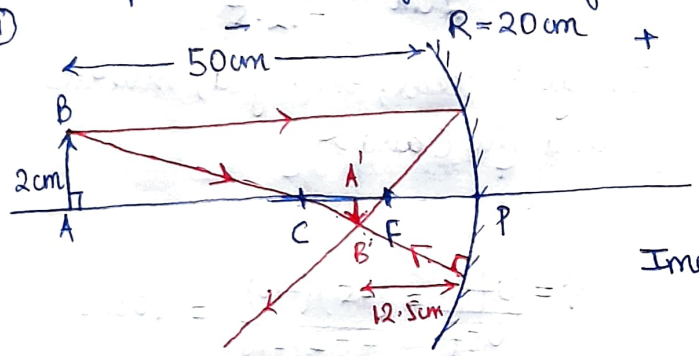
$$s = \sqrt{15^2 + \left(\frac{3}{2}\right)^2}$$

$$= \sqrt{225 + \frac{9}{4}}$$

$$= \sqrt{\frac{909}{4}}\text{ cm}$$

Q Find position and height of image

①

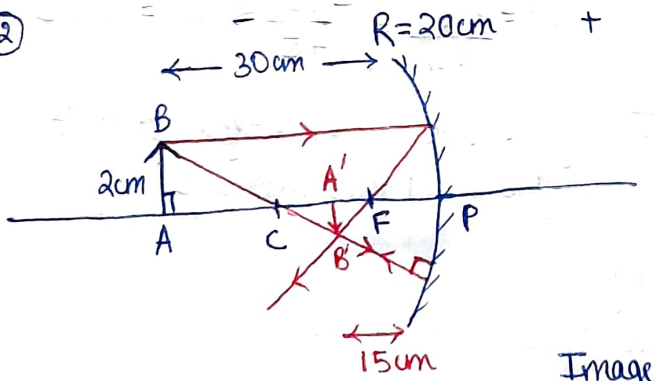


$$v = \frac{uf}{u-f} = \frac{(-50)(-10)}{-50-(-10)} = -12.5\text{cm}$$

$$m = \frac{h_i}{2\text{cm}} = \frac{-(-12.5)}{(-50)} \Rightarrow h_i = \frac{-1}{2}\text{cm}$$

Image is, (a) b/w C and F
(b) diminished (c) inverted
(d) Real

②



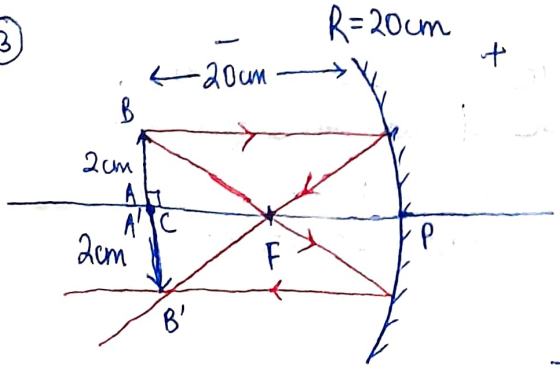
$$v = \frac{uf}{u-f} = \frac{(-30)(-10)}{-30-(-10)} = -15\text{cm}$$

$$m = \frac{h_i}{2\text{cm}} = \frac{-(-15)}{-30}$$

$$h_i = -1\text{cm}$$

Image is Real, diminished, inverted and b/w C and F

③

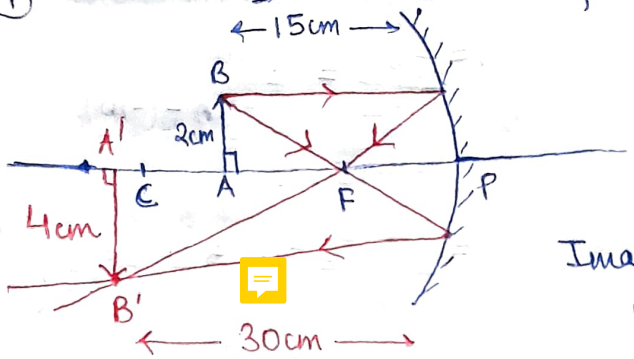


$$v = \frac{uf}{u-f} = \frac{(-20)(-10)}{-20-(-10)} = -20\text{cm}$$

$$m = \frac{h_i}{h_o} = -\frac{(-20)}{-20} \Rightarrow h_i = -2\text{cm}$$

Image is real, same size, inverted and at C.

(4)

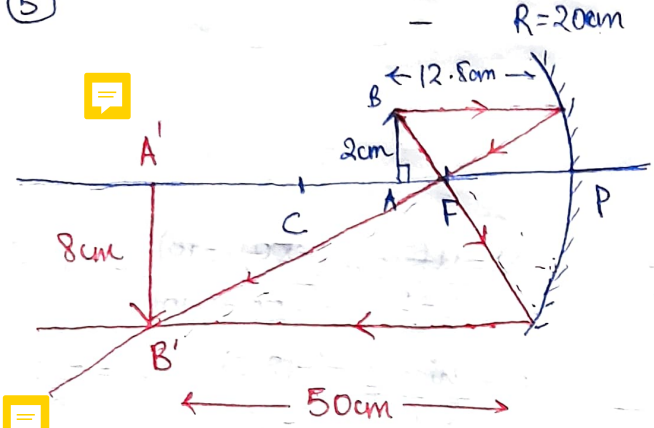


$$v = \frac{uf}{u-f} = \frac{(-15)(-10)}{-15-(-10)} = -30\,cm$$

$$m = \frac{h_i}{2cm} = \frac{-(-30)}{-15} \Rightarrow h_i = -4cm$$

Image is real, enlarged, inverted and b/w C and ∞

(5)

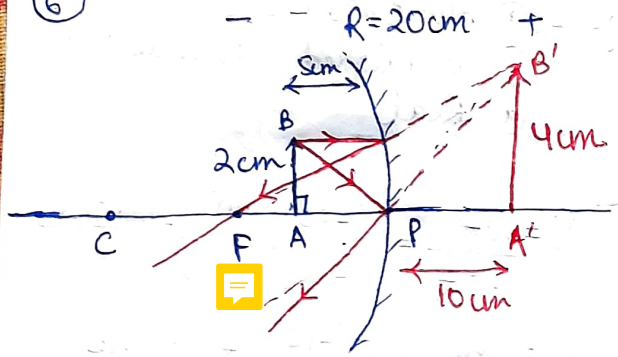


$$v = \frac{uf}{u-f} = \frac{(-12.5)(-10)}{-12.5-(-10)} = -50\,cm$$

$$m = \frac{h_i}{2cm} = \frac{-(-50)}{-12.5} \Rightarrow h_i = -8cm$$

Image is real, enlarged, inverted and b/w C and ∞

(6)



$$v = \frac{uf}{u-f} = \frac{(-5)(-10)}{-5-(-10)} = 10\,cm$$

$$m = \frac{h_i}{2cm} = \frac{-10}{-5} \Rightarrow h_i = 4cm$$

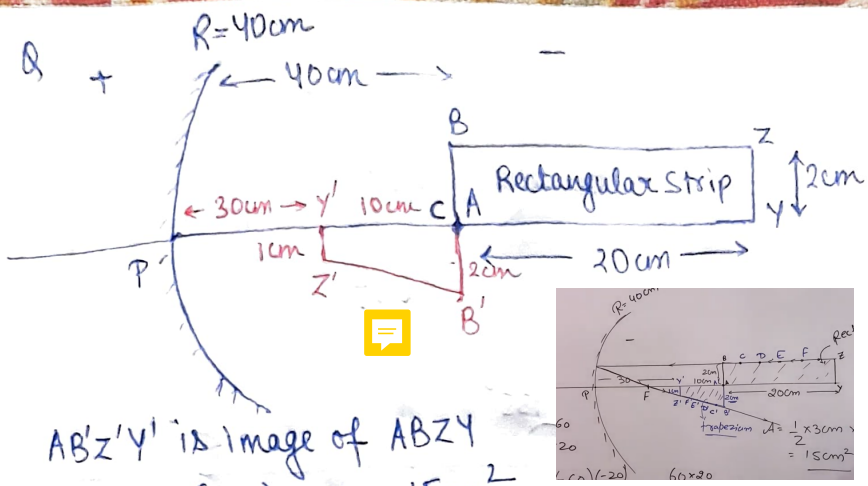
Image is virtual, enlarged, erect and behind the mirror

From the above examples, we can say

→ Object and image have velocity in opposite direction

→ Object or image : Farther ⇒ larger ⇒ Faster

Note: If we want to use concave mirror as shaving mirror, we would need a virtual image of our face for which our face must be b/w focus and pole. The image formed will be erect and magnified.



$AB'Z'Y'$ is image of ABZY

$\text{Area} = \frac{1}{2}(2+1)\times 10 = 15\text{cm}^2$

Find area of its image.

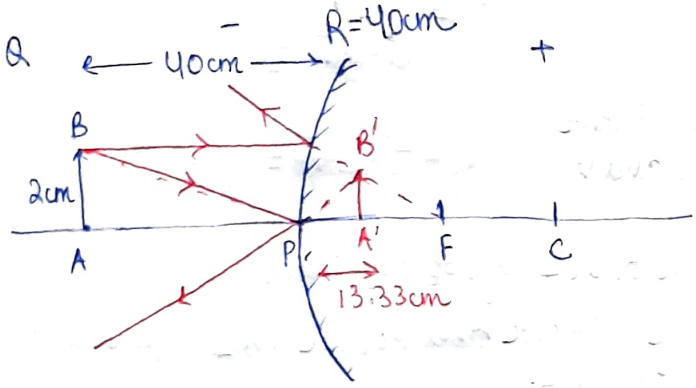
For YZ, $u = -60\text{cm}$

$f = -20\text{cm}$

$$v = \frac{uf}{u-f} = \frac{(-60)(-20)}{-60-(-20)}$$

$$v = -30\text{cm}$$

$$m = \frac{hi}{ho} = \frac{-(-30)}{-60} \Rightarrow hi = -1$$



Find position and height of image.

$$v = \frac{uf}{u-f} = \frac{-40\times 20}{-40-20}$$

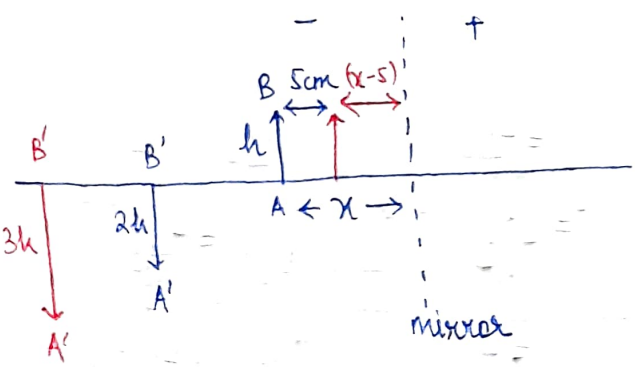
$$v = \frac{40}{3} = 13.33\text{cm}$$

$$m = \frac{hi}{ho} = \frac{-(40/3)}{-40} \Rightarrow hi = \frac{2}{3}\text{cm}$$

Image is virtual, diminished, erect and behind the mirror.

Note: We use convex mirrors as rear view mirrors in vehicles as it provides virtual and erect image, Also it gives a diminished image which provides larger field of view.

★ Q A candle in front of a mirror has image of twice the size on screen. If candle is shifted by 5cm, the image become thrice the size on screen. Find focal length of mirror.



$$m = \frac{f}{f-u}$$

When O is at x, $m = -2$

$$-2 = \frac{f}{f-(-x)} \Rightarrow -2f-2x = f$$

$$\boxed{3f = -2x} \quad -(1)$$

When O is at x-5, $m = -3$

$$-3 = \frac{f}{f-(-(x-5))} \Rightarrow -3f-3x+15 = f$$

From (1)

$$-3f+3\left(\frac{-3}{2}f\right)+15 = f$$

$$\Rightarrow 15 = 4f+\frac{9}{2}f$$

$$\Rightarrow 15 = -\frac{1}{2}f$$

$$\boxed{f = -30\text{cm}} \quad (f = -\text{ve} \Rightarrow \text{Concave mirror})$$

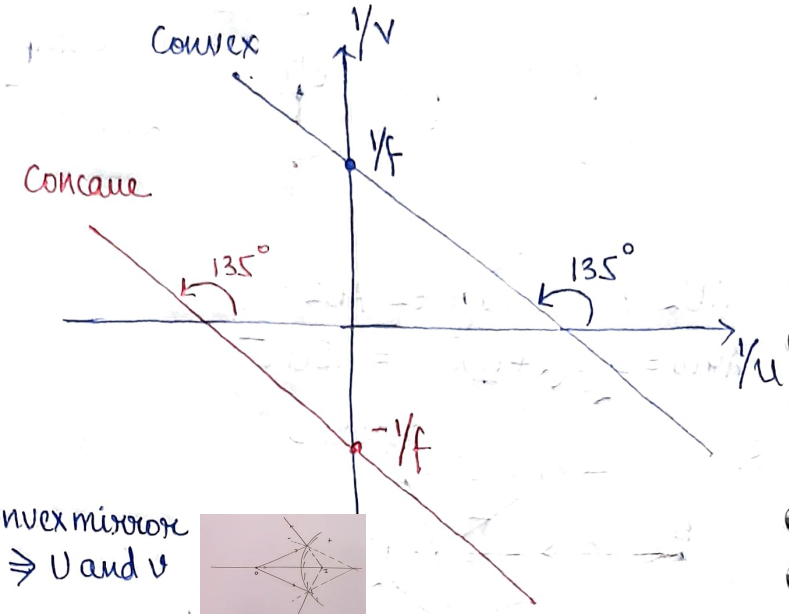
Q Draw graph b/w $\frac{1}{u}$ and $\frac{1}{v}$ for ① Convex mirror
② Concave mirror

$$\frac{1}{v}+\frac{1}{u}=\frac{1}{f}$$

$$\frac{1}{v}=-\frac{1}{u}+\frac{1}{f}$$

$\equiv y = mx + c$

For concave $f = -ve$
For convex $f = +ve$

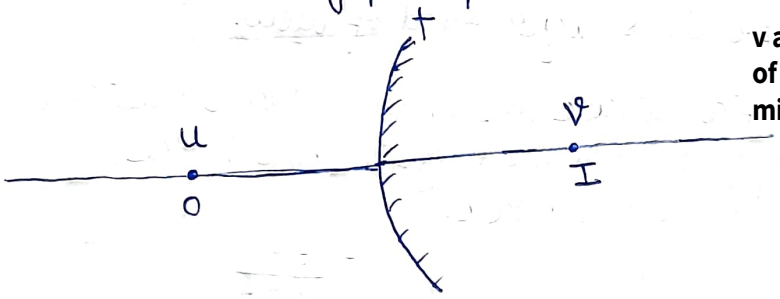


→ We observe that graph of convex mirror does not lie in 3rd quadrant ⇒ u and v cant be -ve together.
∴ Convex mirror will never form Real Image of Real Object.

→ We observe that graph of concave mirror does not lie in 1st quadrant ⇒ u and v cant be positive together.
∴ Concave mirror will never form Virtual image of Virtual object.

Velocity of image:

ⓐ Object is moving along principal axis:-



v and u are the coordinates of object and image w.r.t the mirror

We know, $v^{-1} + u^{-1} = f^{-1}$

Diff. both sides wrt time

$$= -\frac{1}{v^2}\left(\frac{dv}{dt}\right) - \frac{1}{u^2}\left(\frac{du}{dt}\right) = 0$$

$$= \frac{dv}{dt} = -\frac{v^2}{u^2}\left(\frac{du}{dt}\right)$$

$$\therefore \boxed{V_{im} = -\frac{v^2}{u^2} V_{om}}$$

$$\boxed{V_i - V_m = -\frac{v^2}{u^2}(V_o - V_m)}$$

Also, $m = -\frac{v}{u}$

$$\Rightarrow m^2 = \frac{v^2}{u^2}$$

$$\therefore \boxed{V_{im} = -m^2 V_{om}}$$

$$\boxed{V_i - V_m = -m^2(V_o - V_m)}$$

Also,

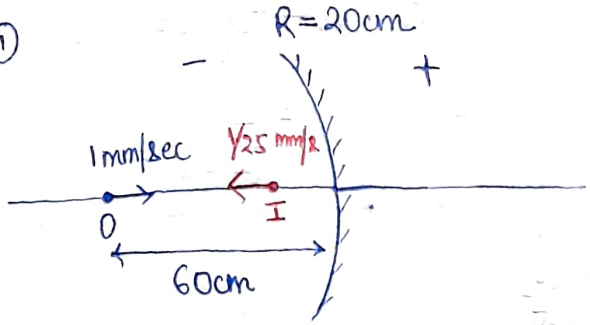
$$\boxed{V_{im} = -\left(\frac{f}{f-u}\right)^2 V_{om}}$$

If $V_m = 0$

$$\boxed{V_i = -\frac{v^2}{u^2} V_o}$$

$$\boxed{V_i = -m^2 V_o}$$

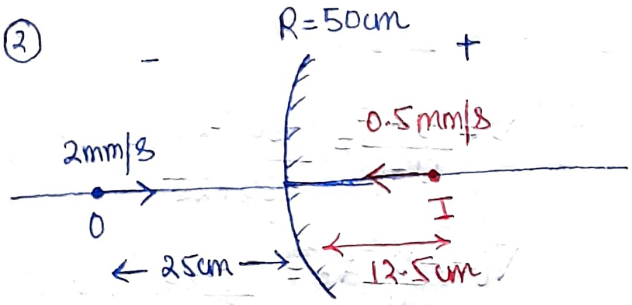
Q Find velocity of image.

① 

$$v = \frac{uf}{u-f} = \frac{(-60)(-10)}{-60-(-10)} = -12\,cm$$

$$V_i - \cancel{V_m}^{0} = -\frac{v^2}{u^2}\left(V_o - \cancel{V_m}^{0}\right)$$

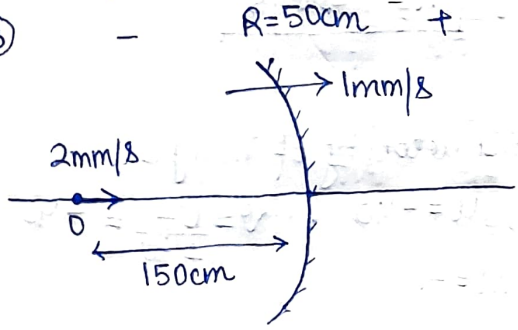
$$V_i = -\frac{(-12)^2}{(-60)^2}\left(+1\,mm/sec\right) = -\frac{1}{25}\,mm/s$$

② 

$$v = \frac{uf}{u-f} = \frac{(-25)(25)}{-25-(25)} = +12.5\,cm$$

$$V_i = -\frac{(12.5)^2}{(-25)^2}\left(+2\,mm/s\right) = -0.5\,mm/s$$

* Image and object will have velocity in opposite direction wrt mirror.

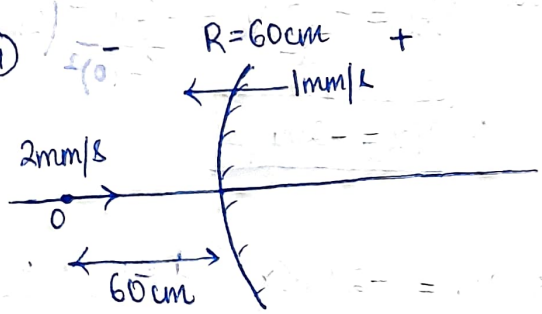
③ 

$$V_i - V_m = -\left(\frac{f}{f-u}\right)^2\left(V_o - V_m\right)$$

$$= V_i - 1mm/s = -\left(\frac{-25}{-25+150}\right)^2\left(2mm/s - 1mm/s\right)$$

$$= V_i - 1mm/s = -\frac{1}{25}$$

$$\therefore \boxed{V_i = \frac{24}{25}\,mm/sec}$$

④ 

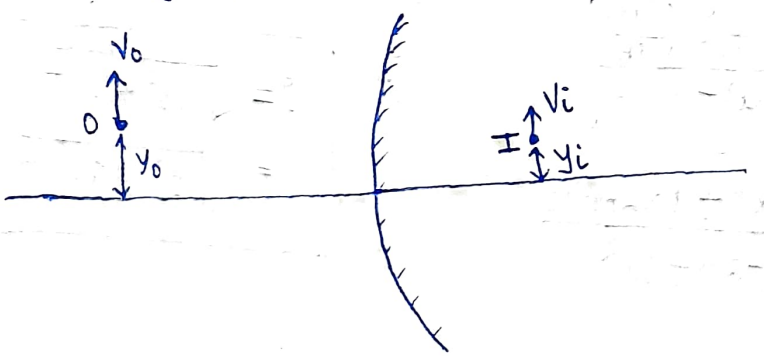
$$V_i - V_m = -\left(\frac{f}{f+u}\right)^2\left(V_o - V_m\right)$$

$$= V_i - (-1mm/s) = -\left(\frac{30}{30+60}\right)^2\left(2mm/s - (-1mm/s)\right)$$

$$= V_i + 1mm/s = -\frac{1}{3}\,mm/s$$

$$= \boxed{V_i = -\frac{4}{3}\,mm/s}$$

⑤ Object is moving perpendicular to principal axis :-



$$\Rightarrow \frac{y_i}{y_o} = m = \frac{-v}{u} = \frac{f}{f-u}$$

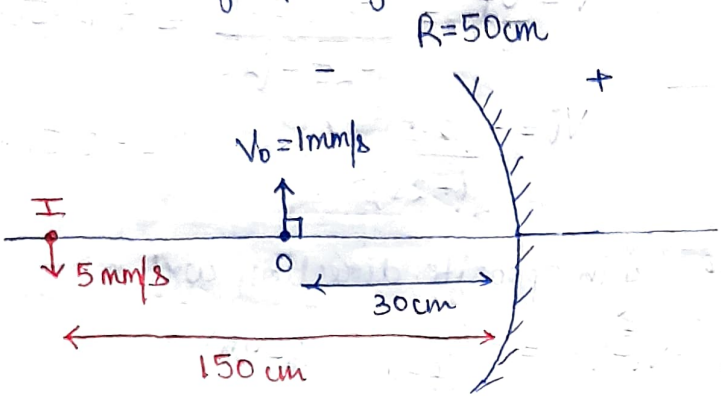
$$= y_i = m y_o = -\frac{v}{u} y_o = \frac{f}{f-u} y_o$$

Diff wrt time

$$\frac{dy_i}{dt} = -\frac{v}{u}\frac{dy_o}{dt} = m\frac{dy_o}{dt} = \left(\frac{f}{f-u}\right)\frac{dy_o}{dt}$$

$$\therefore \boxed{V_i = -\frac{v}{u} V_o = m V_o = \left(\frac{f}{f-u}\right) V_o}$$

Q Find velocity of image.

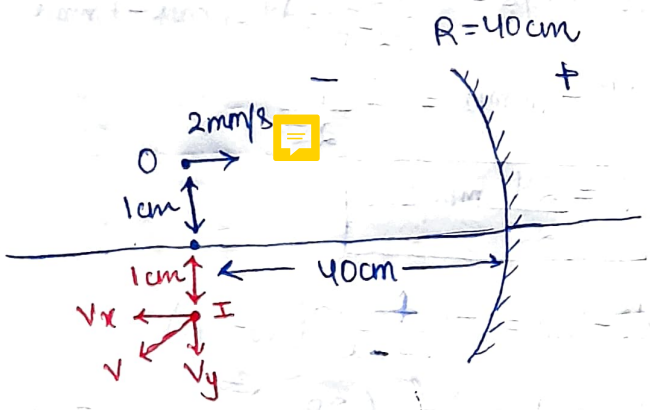


$$v = \frac{uf}{u-f} = \frac{(-30)(-25)}{-30+25} = -150\text{cm}$$

$$V_i = -\frac{v}{u} V_o = -\frac{(-150)}{(-30)} (1\text{mm/s})$$

$$\boxed{V_i = -5\text{mm/s}}$$

** Q



Find velocity of image.

$u = -40$; $v = \frac{uf}{f-u} = -40$

$m = -1$

Now, $V_x = -\frac{v^2}{u^2} V_o = -\frac{(-40)^2}{(-40)^2} (2\text{mm/s})$

$$\therefore \boxed{V_x = -2\text{mm/s}}$$

Now, $y_i = m y_o = \frac{f}{f-u} y_o$

Diff wrt time.

$$\frac{dy_i}{dt} = f\frac{d}{dt}\left(\frac{y_o}{f-u}\right)$$

$$= \frac{dy_i}{dt} = f\left(\frac{(f-u)\frac{dy_o}{dt} + y_o\left(\frac{dv}{dt}\right)}{(f-u)^2}\right)$$

$$= V_y = -20 \times \left(\frac{20\times 0 + 1\times 2\text{mm/s}}{(-20+40)^2}\right)$$

$$\Rightarrow V_y = -20\text{cm} \times \left(\frac{1\text{cm} \times 2\text{mm/s}}{400\text{cm}^2}\right)$$

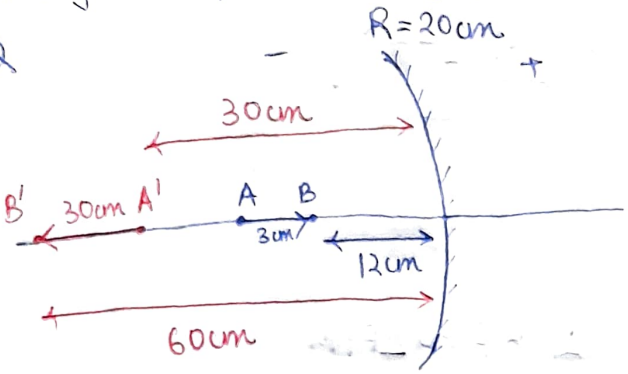
$$\boxed{V_y = -0.1\text{mm/s}}$$

$$\therefore \boxed{v = \sqrt{V_x^2 + V_y^2} = \sqrt{2^2 + (0.1)^2}\ \text{mm/s}}$$

Longitudinal Magnification:

→ Object is placed along principal axis.

Q



Find length of image and magnification.

B: $u = -12\text{cm}$, $f = -10\text{cm}$

$$v = \frac{(-12)(-10)}{-12+10} = -60\text{cm}$$

A: $u = -15\text{cm}$, $f = -10\text{cm}$

$$v = \frac{(-15)(-10)}{-15+10} = -30\text{cm}$$

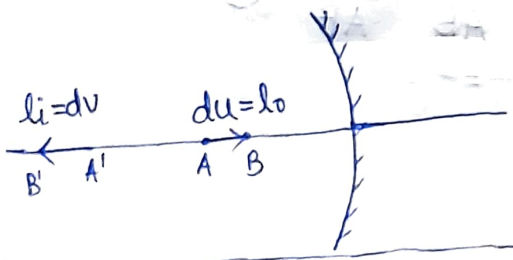
$\therefore$ Length of image $= A'B' = 30\text{cm}$

$$m = -\frac{A'B'}{AB} = \frac{-30}{3} = -10$$

Note:

$$\boxed{m_L = -\frac{l_i}{l_o} = \frac{-|v_2 - v_1|}{|u_2 - u_1|}}$$

* If object is very small:



$$\boxed{m = \frac{dv}{du} = \frac{l_i}{l_o}}$$

Now, we know, $v^{-1} + u^{-1} = f^{-1}$

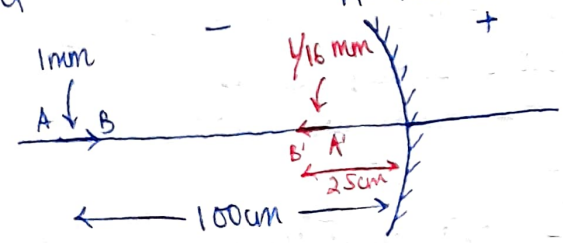
Diff both sides

$$= \frac{-1}{v^2}dv - \frac{1}{u^2}du = 0$$

$$\boxed{\frac{dv}{du} = -\frac{v^2}{u^2}}$$

$$\therefore \boxed{m_L = \frac{dv}{du} = -\frac{v^2}{u^2} = -m_T^2 = -\left(\frac{f}{f-u}\right)^2}$$

Q



Find length of image.

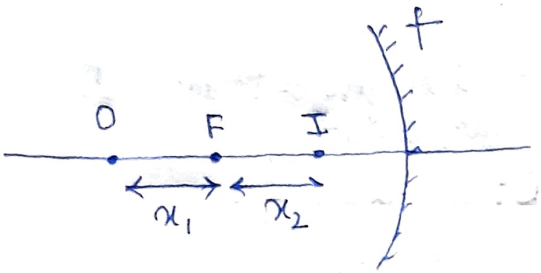
$u = -100\text{cm}$, $f = -20\text{cm}$

$$v = \frac{uf}{u-f} = \frac{(-100)(-20)}{-100+20} = -25\text{cm}$$

Now, $m_L = \frac{l_i}{l_o} = \frac{-v^2}{u^2} \Rightarrow \frac{l_i}{1\text{mm}} = -\frac{(-25)^2}{(-100)^2} = \frac{-1}{16}$

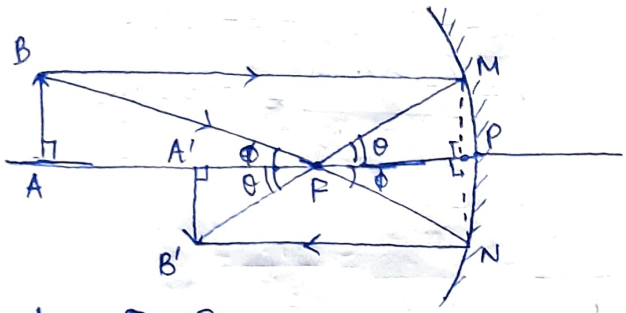
$$\therefore \boxed{l_i = -\frac{1}{16}\text{mm}}$$

Newtons formula for mirror:



* $\boxed{x_1 x_2 = f^2}$

Derivation:



$\Delta A'B'F \sim \Delta FPM$

$\Rightarrow \frac{A'B'}{A'F} = \frac{PM}{FP} = \frac{AB}{FP}$

$\Rightarrow \frac{A'B'}{AB} = \frac{A'F}{FP}$ — (1)

$\Delta ABF \sim \Delta NPF$

$\Rightarrow \frac{AB}{AF} = \frac{NP}{FP} = \frac{A'B'}{FP}$

$\Rightarrow \frac{A'B'}{AB} = \frac{FP}{AF}$ — (2)

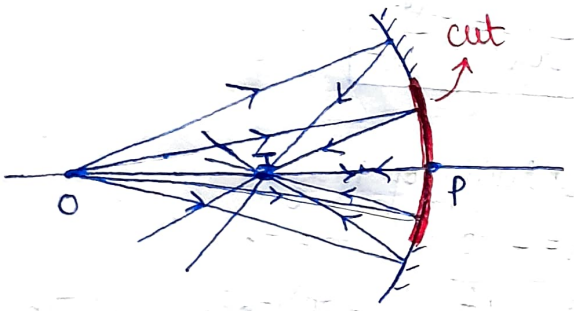
Now, (1) = (2)

$\frac{A'F}{FP} = \frac{FP}{AF}$

$= (A'F)(AF) = (FP)^2$

$= \boxed{(x_2)(x_1) = f^2}$

Mirror is cut:



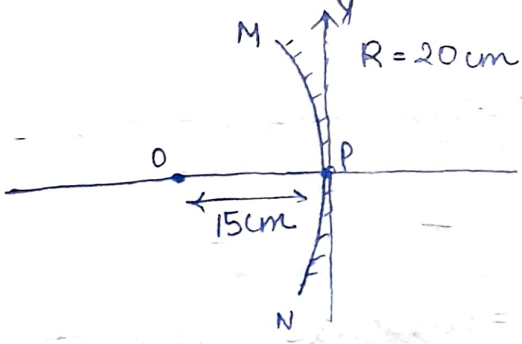
Initially rays from object O strikes the mirror and forms image.

Now, cut the red portion of the mirror, so ray falling on that portion will not reflect now.

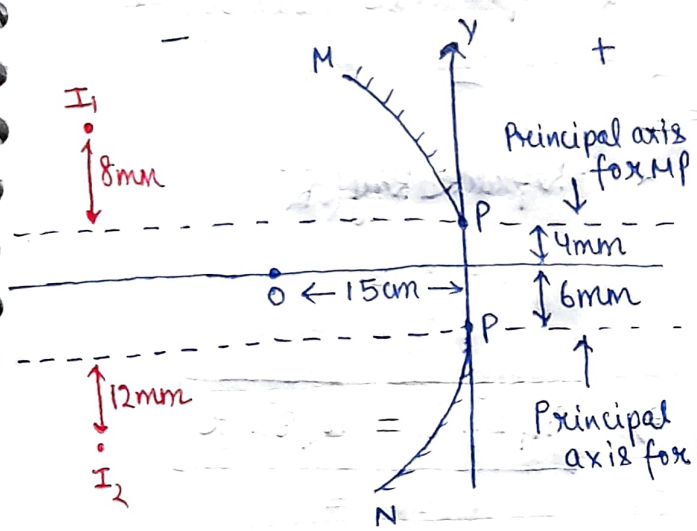
→ If mirror is cut, then position of image will remain unchanged but intensity of Image will decrease.

→ Position of pole remains unchanged.

Q



Mirror is cut along pole.
PM is shifted by 4mm along +y axis
PN is shifted by 6mm along -y axis
Find separation b/w the two images.



$u = -15$, $f = -10$

$$v = \frac{uf}{u-f} = \frac{(-15)(-10)}{-15-(-10)} = -30\text{cm}$$

$$m = \frac{-v}{u} = \frac{-(-30)}{-15} = -2$$

$\therefore \boxed{m = -2}$

For MP : $m = \frac{y_i}{y_o} \Rightarrow y_i = my_o = (-2)(-4)$

$\boxed{y_i = 8\text{mm}}$

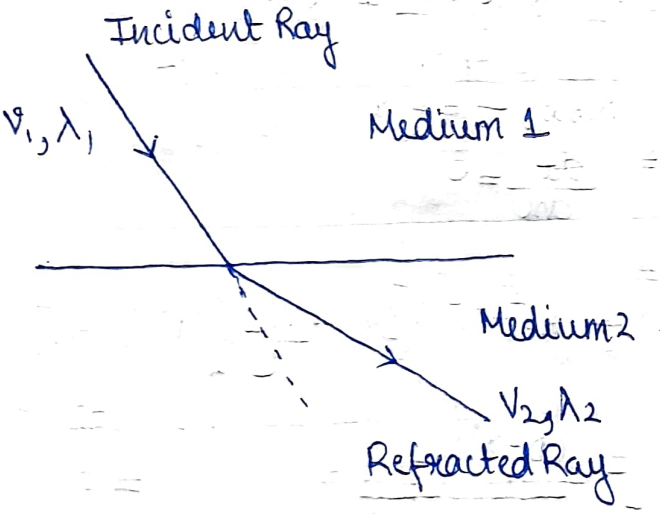
For PN : $y_i = my_o = (-2)(-6)$

$\boxed{y_i = 12\text{mm}}$

$I_1I_2 = 8+4+6+12 = 30\text{mm}$

$\therefore \boxed{I_1I_2 = 3\text{cm}}$

Refraction of light :



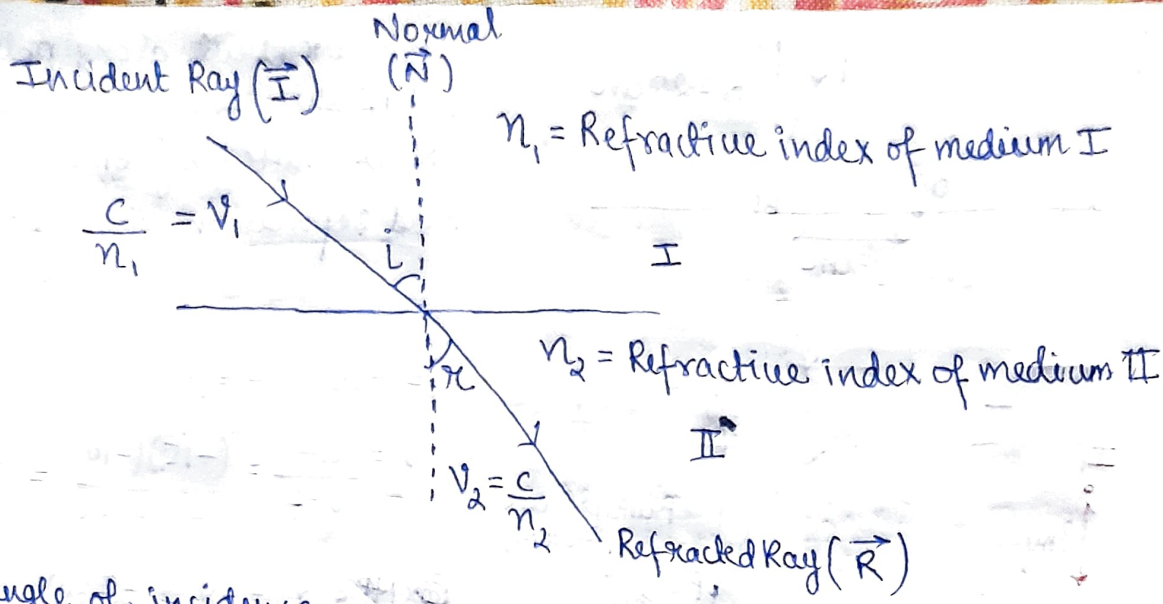
Light changes its velocity and wavelength when it travels from one medium to another. This phenomenon is known as refraction.

Generally, ray bends when it travells from one medium to another. except in case of normal incidence

Laws of Refraction :

① Incident ray, refracted ray and normal, all must lie in same plane.

$$\boxed{(\vec{I} \times \vec{R}) \cdot \vec{N} = 0}$$



n_1 = Refractive index of medium I

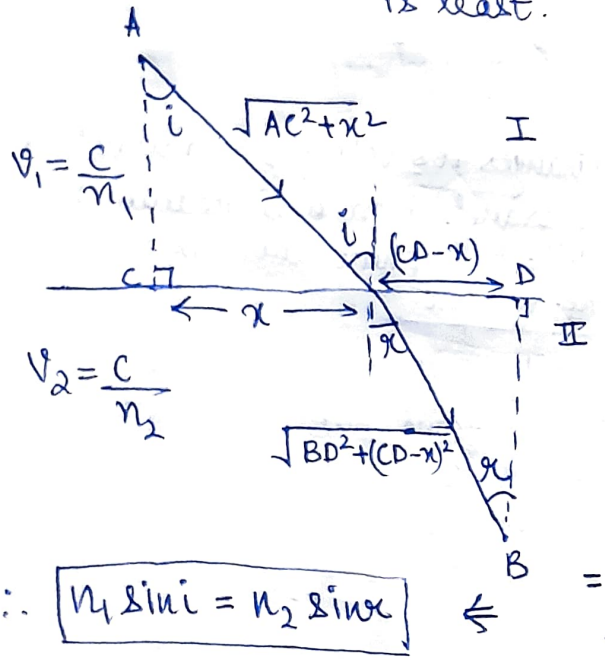
n_2 = Refractive index of medium II

(2) i = angle of incidence
r = angle of refraction

Snells law: $\boxed{\frac{\sin i}{\sin r} = \frac{v_1}{v_2} = \frac{c/n_1}{c/n_2} = \frac{n_2}{n_1}} \Rightarrow \boxed{n_1 \sin i = n_2 \sin r}$

Proof of snells law:

Fermat Principle: light will follow the path, along which time taken is least.



$t = t_1 + t_2$

$t = \frac{\sqrt{AC^2+x^2}}{c/n_1} + \frac{\sqrt{BD^2+(CD-x)^2}}{c/n_2}$

Now, $t = f(x)$

$\Rightarrow \frac{dt}{dx} = 0$

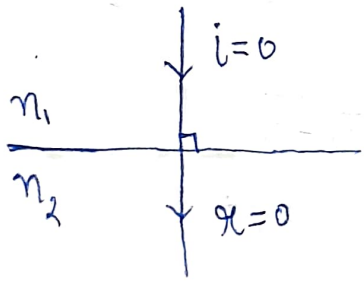
$= \frac{n_1}{c} \times \frac{1}{2}\left(AC^2+x^2\right)^{-1/2}(2x) + \frac{n_2}{c}\frac{1}{2}\left(BD^2+(CD-x)^2\right) \times 2(CD-x)(-1) = 0$

$= \frac{n_1 x}{\sqrt{AC^2+x^2}} = \frac{(CD-x)n_2}{\sqrt{BD^2+(CD-x)^2}}$

$\therefore \boxed{n_1 \sin i = n_2 \sin r} \Leftarrow$

Special Cases:

(1) Normal incidence:



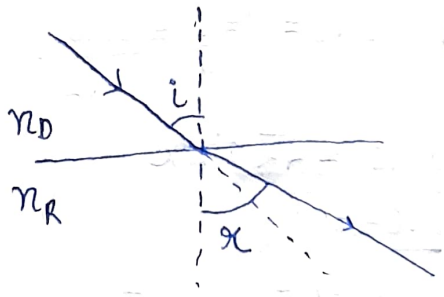
$i = 0$

$\Rightarrow n_1 \sin 0^\circ = n_2 \sin r$

$\Rightarrow r = 0$

$\therefore \delta = 0$

② Denser to Rarer



$n_D > n_R$

By Snells law, $n_D \sin i = n_R \sin r$

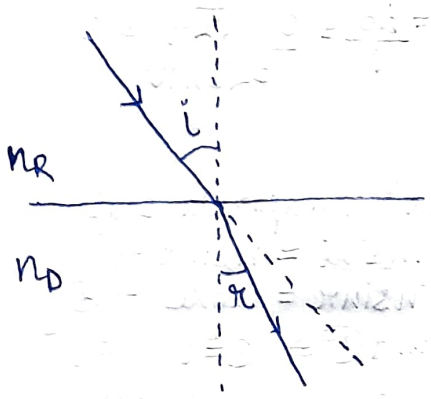
$\Rightarrow \sin r = \frac{n_D}{n_R} \sin i$

$\therefore \boxed{r > i} \quad \left(\frac{n_D}{n_R} > 1\right)$

$\therefore$ Ray deviates away from normal

$\boxed{\delta = r - i}$

③ Rarer to Denser:



$n_D > n_R$

By Snells law, $n_R \sin i = n_D \sin r$

$\Rightarrow \sin r = \frac{n_R}{n_D} \sin i$

$\therefore \boxed{r < i} \quad \left(\frac{n_R}{n_D} < 1\right)$

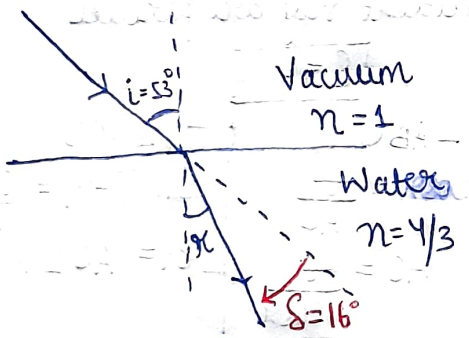
$\therefore$ Ray deviates towards normal

$\boxed{\delta = i - r}$

hence, for both cases:

$\boxed{\delta = |i - r|}$

Q



Find ① r ② δ

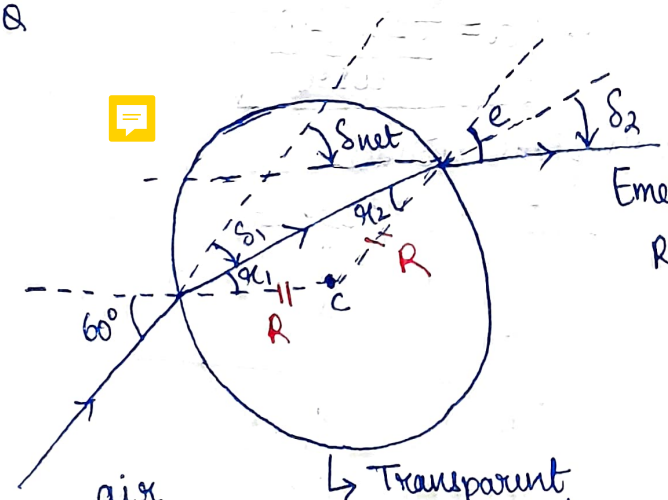
① $1 \sin 53° = \frac{4}{3} \sin r$

$= \frac{4}{5} = \frac{4}{3} \sin r \Rightarrow \sin r = \frac{3}{5}$

$\therefore \boxed{r = 37°}$

② $\delta = 53° - 37 = 16°$

Q



Find ① r_1, ② δ_1, ③ r_2, ④ e, ⑤ δ_2, ⑥ δ_{net}

① $1 \times \sin 60° = \sqrt{3} \sin r_1$

$= \frac{\sqrt{3}}{2} = \sqrt{3} \sin r_1$

$= \sin r_1 = \frac{1}{2} \Rightarrow \boxed{r_1 = 30°}$

② $\delta_1 = 60 - 30 = 30°$

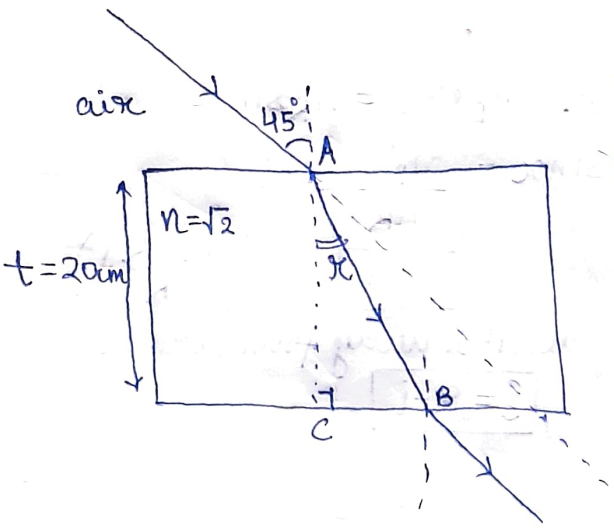
③ By geometry, $r_2 = 30°$

④ $\sqrt{3} \sin 30° = 1 \sin e \Rightarrow \boxed{e = 60°}$

⑤ $\delta_2 = 60 - 30 = 30°$

⑥ $\delta_{net} = \delta_1 + \delta_2 = 60°$

Q



Find time taken by light to cross the parallel slab.

Speed of light in slab = $V = \frac{c}{n} = \frac{3\times10^8 m/s}{\sqrt{2}}$

By snells law,

$1 \times \sin 45° = \sqrt{2} \sin r$

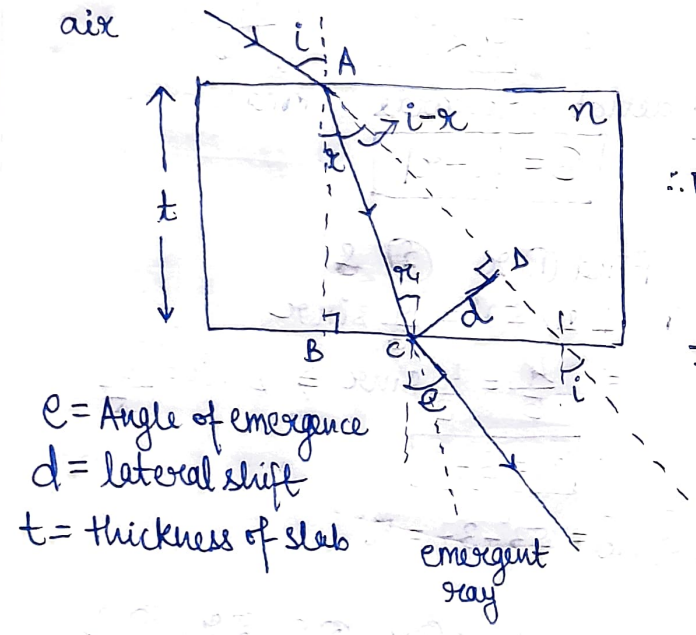
$$r = 30°$$

In ΔABC, $\cos 30° = \frac{AC}{AB} = \frac{t}{AB}$

$$AB = \frac{t}{\cos 30°} = \frac{0.2m \times 2}{\sqrt{3}} = \frac{0.4m}{\sqrt{3}}$$

So, Time = $T = \frac{AB}{V} = \frac{0.4\sqrt{2}}{3\sqrt{3}\times10^8}$ sec

Lateral Shift :



e = Angle of emergence

d = lateral shift

t = thickness of slab

Parallel transparent slab.

$1 \times \sin i = n \sin r$ — (1)

$n \sin r = 1 \sin e$ — (2)

From (1) and (2) $\Rightarrow e = i$

∴ Emergent ray and extended path of incident ray are parallel

$$\delta = 0$$

$\delta_1 = i - r$ clockwise

$\delta_2 = (i - r)$ Anticlockwise.

$\delta_{net} = \delta_1 + \delta_2 = 0$

In ΔABC

$\cos r = \frac{t}{AC}$

$AC = \frac{t}{\cos r}$

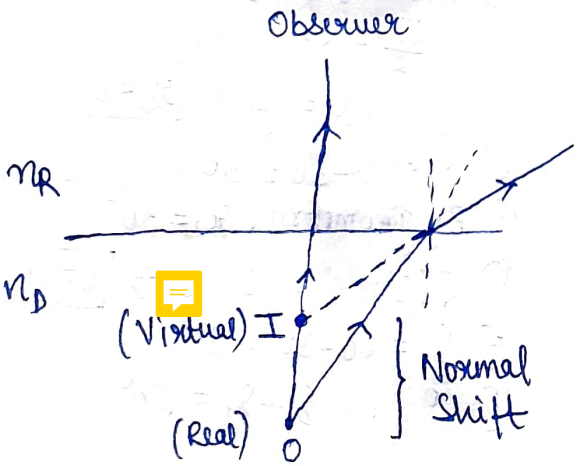
In ΔACD

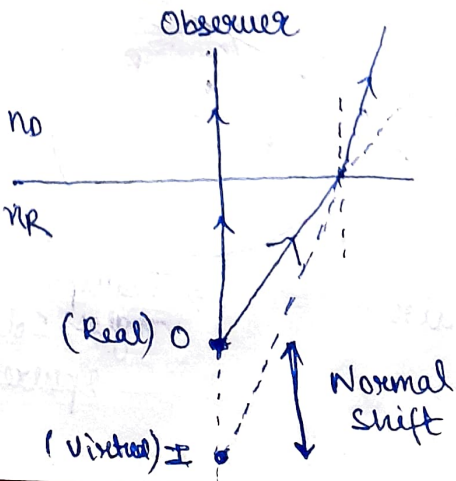
$\sin(i-r) = \frac{CD}{AC}$

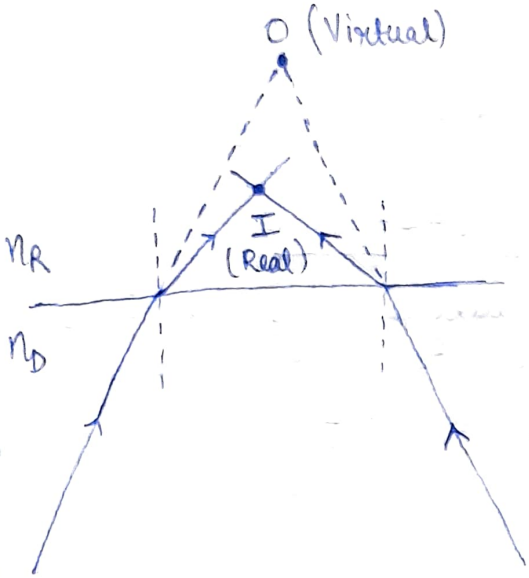
$d = AC \sin(i-r)$

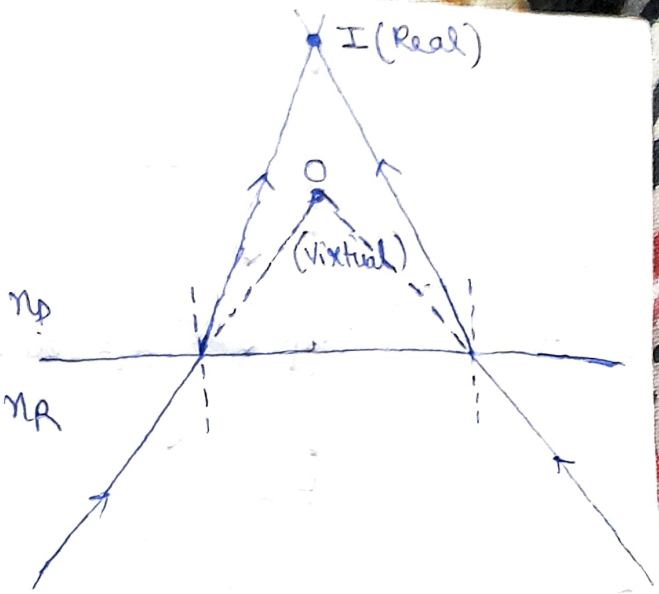
$$\therefore d = \frac{t \sin(i-r)}{\cos r}$$

Image Formation by refraction through plane surface :







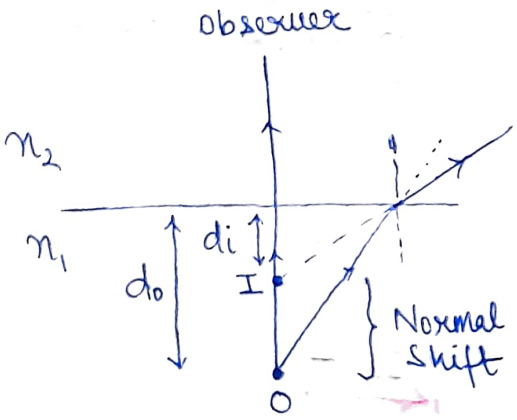


For refraction by plane surface:

① Object and image will be on same side of interface.

② For real object, we have virtual image and for virtual object, we have real image.

Position of image:



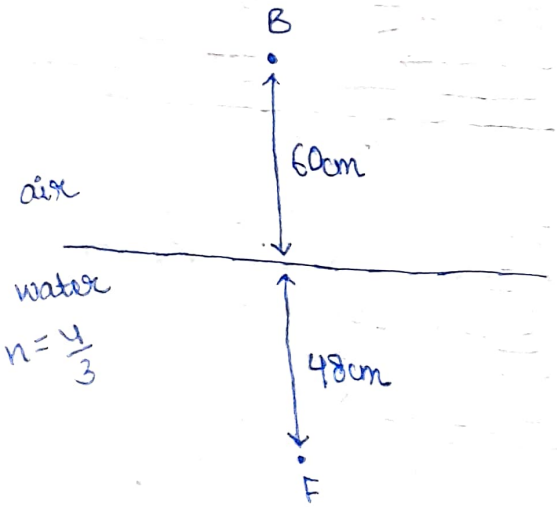
$$d_i = \frac{d_o}{n_{rel}} = \frac{d_o}{(n_1/n_2)} = \frac{d_o}{\left(\frac{n_{from}}{n_{to}}\right)}$$

$$\text{Normal shift} = d_o - d_i = d_o - \frac{d_o}{n_{rel}}$$

$$= d_o\left(1 - \frac{1}{n_{rel}}\right) = d_o\left(1 - \frac{1}{(n_1/n_2)}\right)$$

→ If Normal shift is -ve, shift is opposite to direction of incident ray

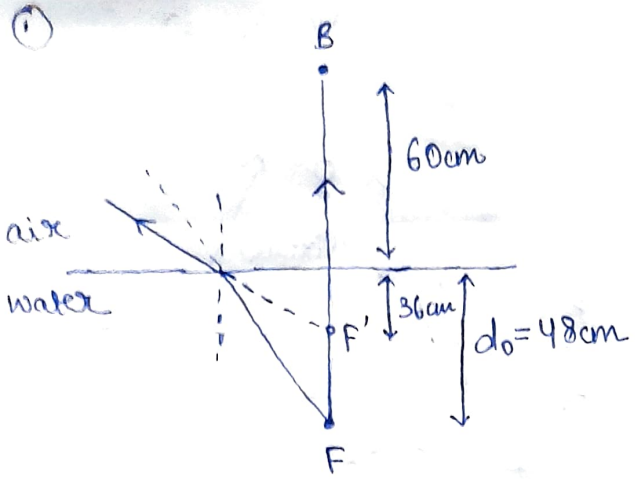
Q



① Find position of image of F as ~~seen~~ seen by B.

② Find position of image of B as seen by F.

①

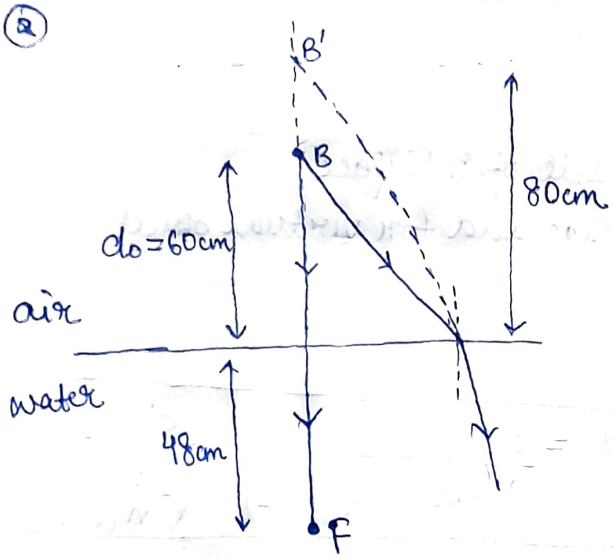


$$d_i = \frac{d_o}{n_{rel}} = \frac{48}{(4/3)/1} = 48 \times \frac{3}{4}$$

$$\boxed{d_i = 36\text{cm}}$$

∴ B will observe image of F at 36cm from interface and 96cm from itself.

②

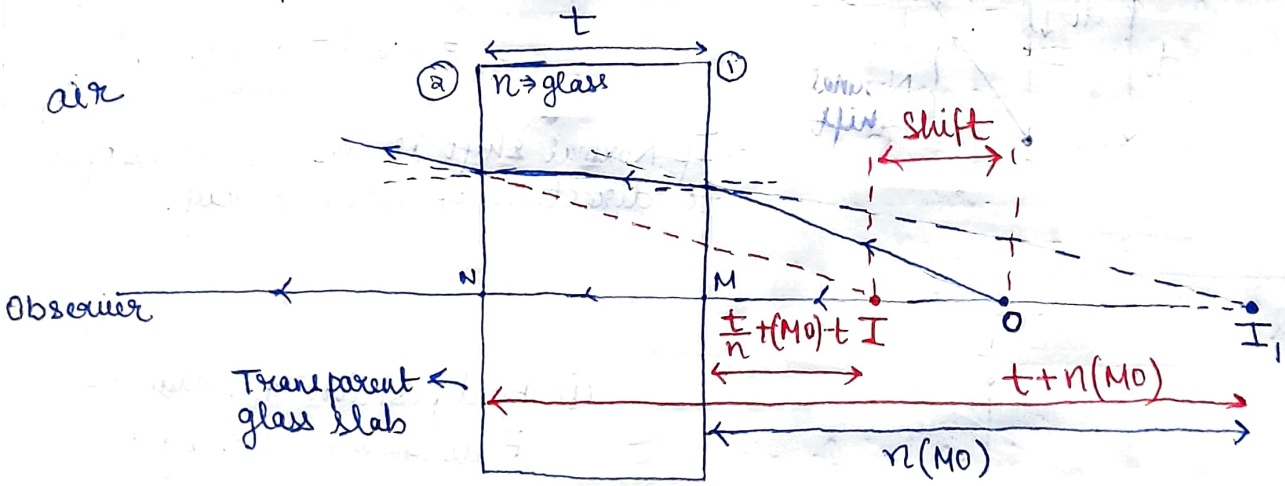


$$d_i = \frac{d_o}{n_{rel}} = \frac{60}{(1/4/3)} = 60 \times \frac{4}{3} = 80\text{cm}$$

$$\boxed{d_i = 80\text{cm}}$$

∴ F will observe image of B at 80cm from interface and 128cm from itself.

Normal Shift:



Surface ①

$MI_1 = \frac{MO}{(1/n)}$

$MI_1 = n(MO)$

Surface ②

I_1 = object

I = Image

$NI = \frac{t+n(MO)}{n}$

$NI = \frac{t}{n} + MO$

Shift:

$MO - MI$

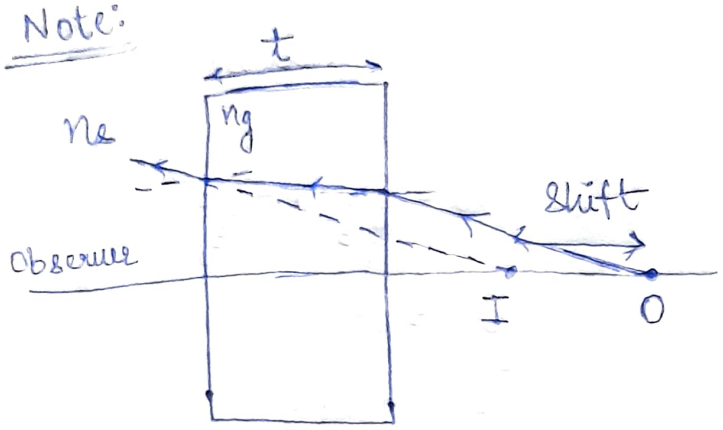
$= MO - \left(\frac{t}{n} + (MO) - t\right)$

$= t - \frac{t}{n} = t\left(1 - \frac{1}{n}\right)$

$$\therefore \boxed{\text{Shift} = t\left(1 - \frac{1}{n}\right)}$$

towards dirn of incident ray

Note:



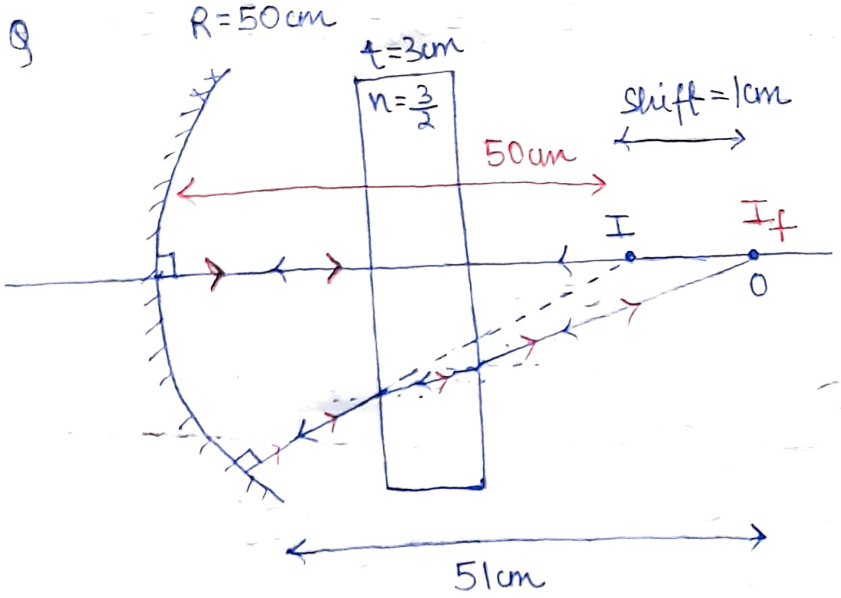
$$\text{Shift} = t\left(1 - \frac{1}{n_{rel}}\right)$$

$$= t\left(1 - \frac{1}{(n_g/n_s)}\right)$$

$$= t\left(1 - \frac{n_s}{n_g}\right)$$

Shift is along incident ray.

Q



Find position of final image.

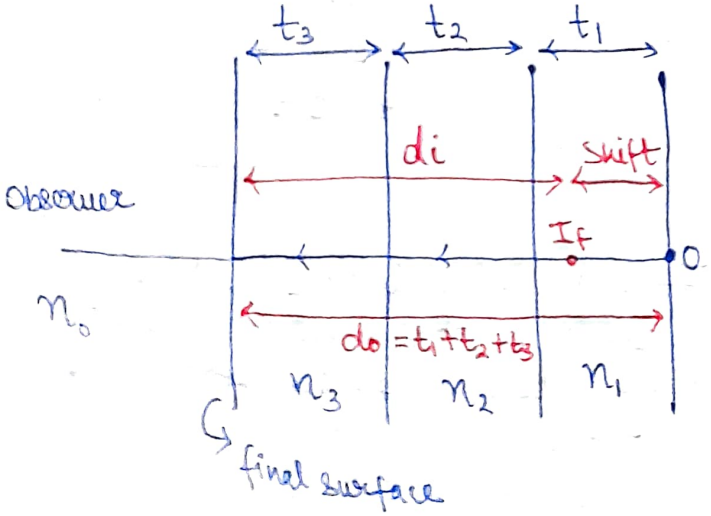
$$\text{Shift} = 3\text{cm}\left(1 - \frac{1}{3/2}\right)$$

Shift = 1cm

For mirror, I will act as object.
Also since I is at C of mirror, rays from I will strike the mirror normally.

After reflection, rays will again refract through slab.
After shift of 1cm again, the final image I_f will form at O itself, 51cm from the mirror.

Refraction by parallel slabs:



Distance of image from final surface $= d_i$

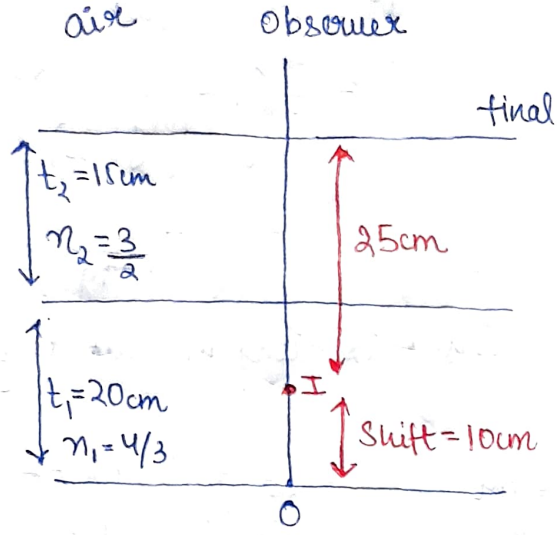
$$d_i = \frac{t_1}{n_1/n_0} + \frac{t_2}{n_2/n_0} + \frac{t_3}{n_3/n_0}$$

$$\text{Shift} = d_o - d_i$$

$$\text{shift} = d_o - d_i$$
$$= (t_1 + t_2 + t_3) - (d_i)$$
$$= t_1\left\{1 - \frac{1}{n_1/n_0}\right\} + t_2\left\{1 - \frac{1}{n_2/n_0}\right\} + t_3\left\{1 - \frac{1}{n_3/n_0}\right\}$$

Q

Find distance of image from final surface.



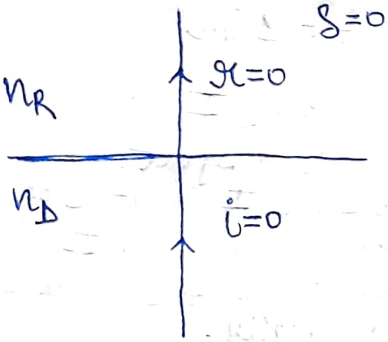
$$d_i = \frac{20}{4/3} \times 1 + \frac{15}{3/2} \times 1$$

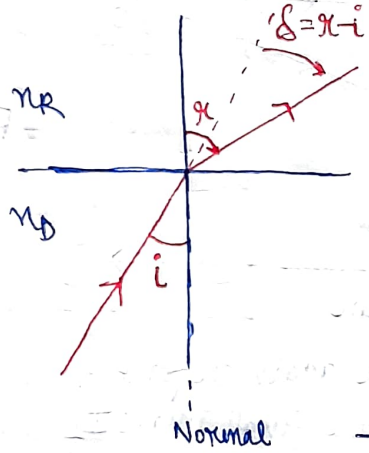
$$d_i = 15cm + 10cm$$

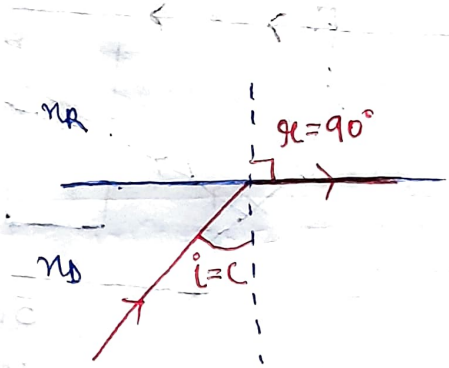
$$\boxed{d_i = 25cm}$$

$$\text{Shift} = d_o - d_i = 35 - 25 = 10cm$$

Total internal reflection:







$$n_R \sin r = n_D \sin i$$

$$\sin r = \frac{n_D}{n_R} \sin i$$

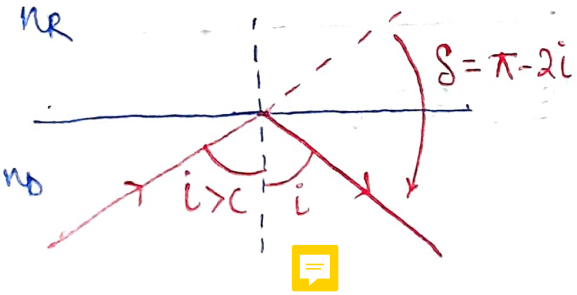
$$\therefore \boxed{r > i}$$

→ Angle of incidence in denser medium for which angle of refraction becomes 90°, is known as critical angle.

$$n_D \sin C = n_R \sin 90°$$

$$\star \boxed{C = \sin^{-1}(n_R/n_D)}$$

Now,

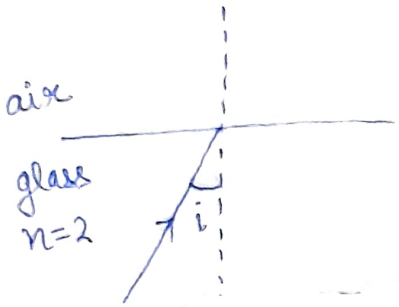


If angle of incidence in denser medium is more than critical angle, then there will be no refraction, but the ray will reflect back to the denser medium.
This phenomenon is known as total internal reflection.

Q

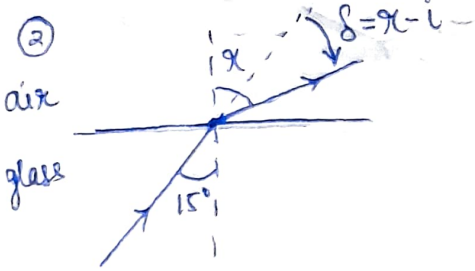
Find ① C ② δ if $i = 15°$

③ δ if $i = 30°$ ④ δ if $i = 45°$



① $C = \sin^{-1}\left(n_R/n_D\right) = \sin^{-1}\left(\frac{1}{2}\right) = 30°$

②



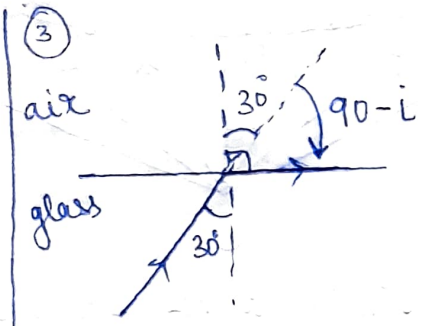
$2\sin 15° = 1 \times \sin r$

$\sin r = 2 \times \frac{\sqrt{3}-1}{2\sqrt{2}}$

$\boxed{r = \sin^{-1}\left(\frac{\sqrt{3}-1}{\sqrt{2}}\right)}$

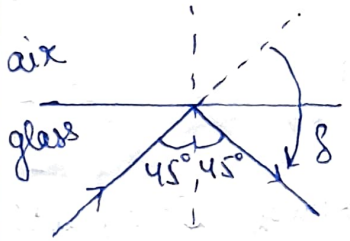
$\delta = r - i = \sin^{-1}\left(\frac{\sqrt{3}-1}{\sqrt{2}}\right) - 15°$

③



$\delta = 90° - 30°$

$\delta = 60°$

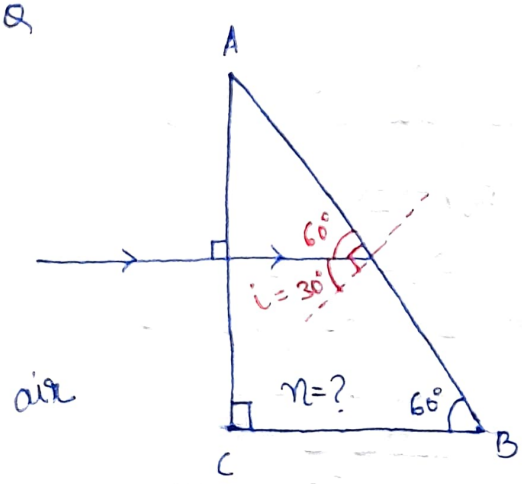


$\delta = \pi - 2i$

$= \pi - 2 \times 45°$

$\delta = 90°$

Q



Find n such that ray doesnt cross the face AB

Ray must suffer total internal reflection so that it doesnt cross the face AB.

For TIR:

$i > C$

$\Rightarrow 30° > C$

$\Rightarrow \sin 30° > \sin C$

$\Rightarrow \frac{1}{2} > \frac{1}{n}$

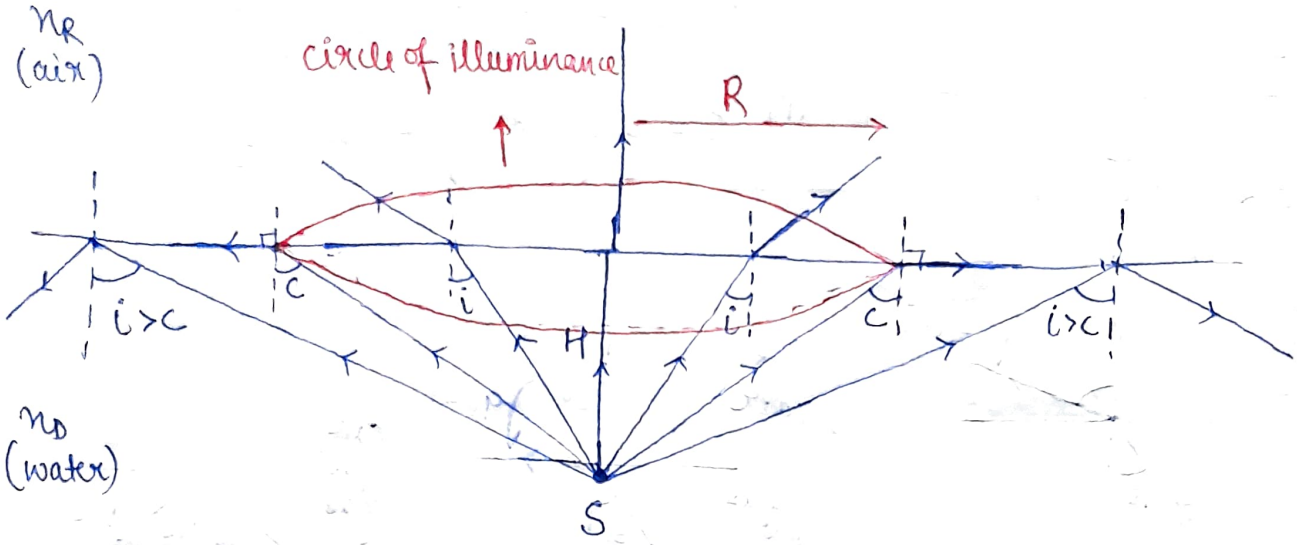
$\therefore \boxed{n > 2}$

Also at $n = 2$, ray will just graze the surface AB and not cross.

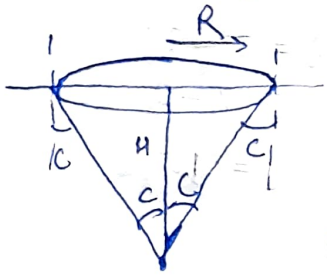
$\therefore$ Final Ans $\Rightarrow \boxed{n \geq 2}$

Circle of illuminance:

S = Source of light



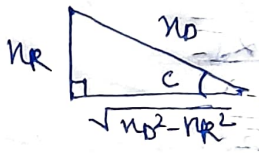
→ Only the rays which incident inside the circle, will go out.



$$\tan C = \frac{R}{H}$$

$$R = H \tan C$$

$$\star \quad \boxed{R = \frac{H n_R}{\sqrt{n_D^2 - n_R^2}}}$$



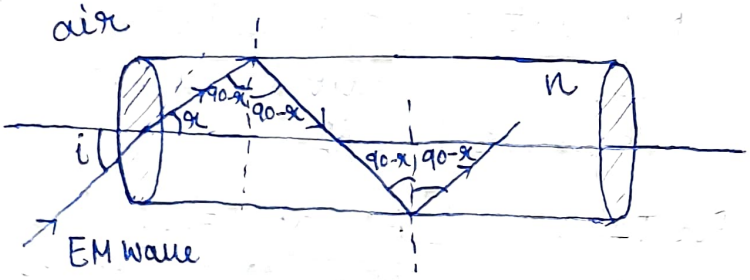
$$\sin C = \frac{n_R}{n_D}$$

$$\boxed{\text{Area of circle} = \pi R^2 = \frac{\pi H^2 n_R^2}{n_D^2 - n_R^2}}$$

Optical Fiber:

→ It is extremely thin wires made of polymers or glasses.

→ It is used to transmit EM wave from one place to another with minimum attenuation.



By snells law

$$1 \times \sin i = n \sin r$$

$$\sin r = \frac{\sin i}{n}$$

For TIR:

$$90 - r > C$$

$$\Rightarrow \sin(90 - r) > \sin C$$

$$\Rightarrow \cos r > \frac{1}{n}$$

$$= \sqrt{1 - \sin^2 r} > \frac{1}{n}$$

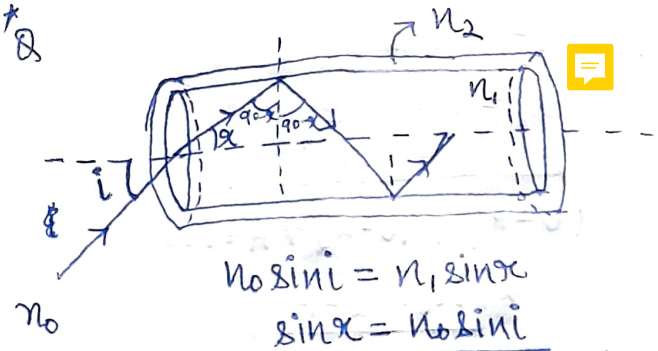
$$= \sqrt{1 - \frac{\sin^2 i}{n^2}} > \frac{1}{n}$$

$$= 1 - \frac{\sin^2 i}{n^2} > \frac{1}{n^2}$$

$$= \boxed{n^2 > 1 + \sin^2 i} \quad (\text{Max value of } 1 + \sin^2 i = 2)$$

$$\Rightarrow n^2 > 2 \Rightarrow \boxed{n > \sqrt{2}} \star$$

Q



$n_0 \sin i = n_1 \sin r$

$\sin r = \frac{n_0 \sin i}{n_1}$

Find values of i, for which ray will not come out of curved surface of optical fiber.

For TIR: $90 - r > C$

$\Rightarrow \sin(90-r) > \sin C$

$\Rightarrow \cos r > n_2/n_1$

$\Rightarrow \sqrt{1 - \frac{n_0^2 \sin^2 i}{n_1^2}} > \frac{n_2}{n_1}$

$= n_1^2 - n_0^2 \sin^2 i > n_2^2$

$= n_1^2 - n_2^2 > n_0^2 \sin^2 i$

$= \sin^2 i < \frac{n_1^2 - n_2^2}{n_0^2}$

$= \sin i < \sqrt{\frac{n_1^2 - n_2^2}{n_0^2}}$

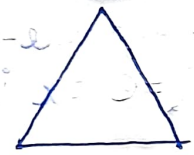
Note: Numerical Apperture of optical fiber = $\sin i_{max}$

$= \frac{\sqrt{n_1^2 - n_2^2}}{n_0}$

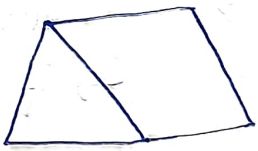
$$\boxed{i < \sin^{-1}\left(\frac{\sqrt{n_1^2 - n_2^2}}{n_0}\right)} \Leftarrow$$

Prism:

→ It is a transparent, homogenous, refracting medium bounded by two plane surfaces, inclined at some angle.

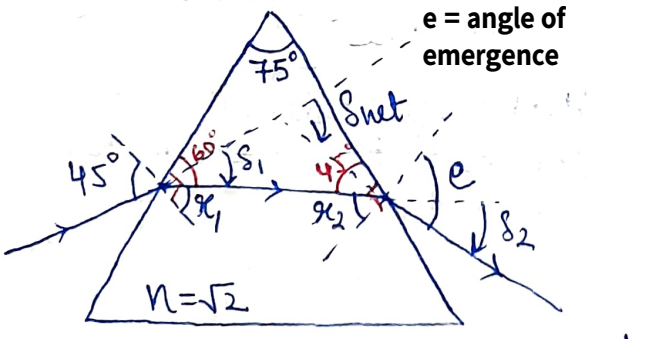


2-D



3-D

Q



Find ① r_1 ② δ_1 ③ r_2 ④ e ⑤ δ_2 ⑥ δ_{net}

① $1 \times \sin 45° = \sqrt{2} \sin r_1 \Rightarrow r_1 = 30°$

② $\delta_1 = |i - r| = |45 - 30| = 15°$

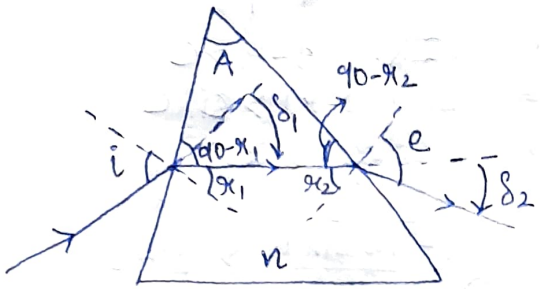
③ By geometry, $r_2 = 45°$

④ $\sqrt{2} \sin 45° = 1 \times \sin e \Rightarrow e = 90°$

⑤ $\delta_2 = |i - r| = |45 - 90| = 45°$

⑥ $\delta_{net} = \delta_1 + \delta_2 = 15° + 45° = 60°$

Theory:



$1 \times \sin i = n \sin r_1$

$\therefore \boxed{r_1 = \sin^{-1}\left(\frac{\sin i}{n}\right)}$

Now, $90 - r_1 + 90 - r_2 + A = 180$

$\star \boxed{r_1 + r_2 = A}$

$\boxed{r_2 = A - \sin^{-1}\left(\frac{\sin i}{n}\right)}$

Also, $\delta = \delta_1 + \delta_2$

$\delta = (i - r_1) + (e - r_2)$

$= i + e - (r_1 + r_2)$

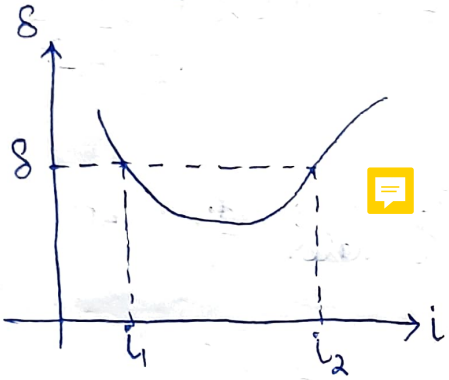
$\star \boxed{\delta = i + e - A}$

Now, $n \sin r_2 = 1 \sin e$

$e = \sin^{-1}(n \sin r_2)$

$e = \sin^{-1}\left(n \sin\left(A - \sin^{-1}\left(\frac{\sin i}{n}\right)\right)\right)$

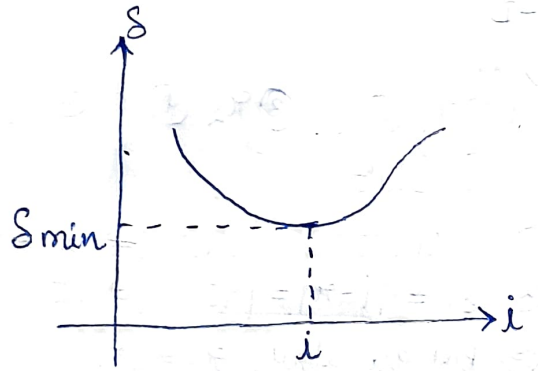
$\delta = i + \sin^{-1}\left(n \sin\left(A - \sin^{-1}\left(\frac{\sin i}{n}\right)\right)\right) - A$



→ For same deviation, we may have two possible angle of incidences. These angle of incidences are related such that, if one of them is angle of incidence other will be angle of emergence and vice-versa.

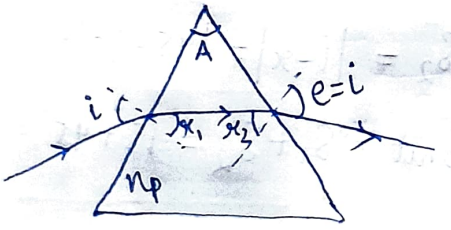
So, if $i_1 = i \Rightarrow i_2 = e$ or $i_1 = e \Rightarrow i_2 = i$

Minimum Deviation:



→ Minimum deviation will occur only for one angle of incidence.

For minⁿ deviation:
Angle of incidence = Angle of emergence



By snells law

$n_s \sin i = n_p \sin r_1$ — (1)

$n_p \sin r_2 = n_s \sin i$ — (2)

From (1) and (2)

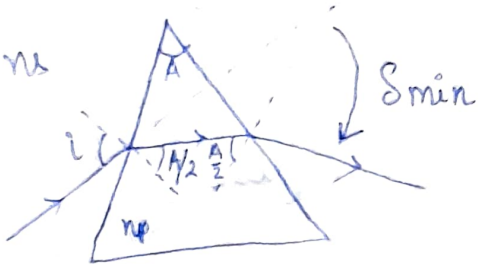
$n_p \sin r_1 = n_p \sin r_2$

$\therefore \boxed{r_1 = r_2}$

Since, $r_1 + r_2 = A$

$\Rightarrow \boxed{r_1 = r_2 = \frac{A}{2}}$

i = Angle of incidence for which δ is min



$$n_s \sin i = n_p \sin\frac{A}{2} \quad \text{— (1)}$$

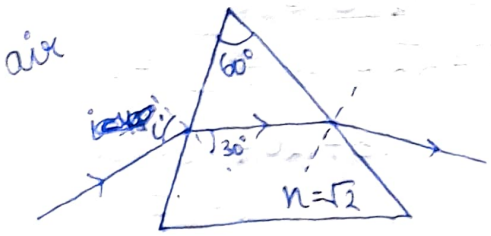
Now, $\delta_{min} = i + i - A$

$$\star \quad i = \frac{\delta_{min} + A}{2}$$

From (1)

$$n_s \sin\left(\frac{\delta_{min}+A}{2}\right) = n_p \sin\frac{A}{2} \Rightarrow \frac{n_p}{n_s} = n_{rel} = \frac{\sin\left(\frac{\delta_{min}+A}{2}\right)}{\sin(A/2)}$$

Q



Find i for δ_{min} and also find δ_{min}

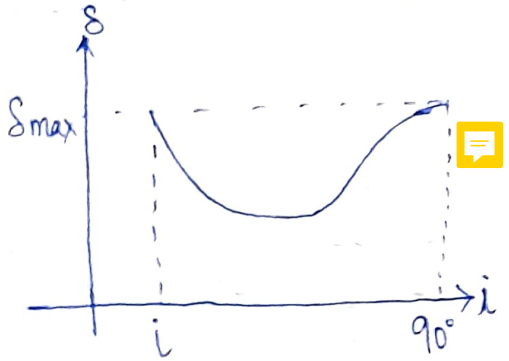
$$r = \frac{A}{2} = \frac{60}{2} = 30°$$

Now, $1 \times \sin i = \sqrt{2} \sin 30°$

$$i = 45°$$

$$\delta_{min} = 45 + 45 - 60 = 30°$$

Maximum Deviation (through refraction):

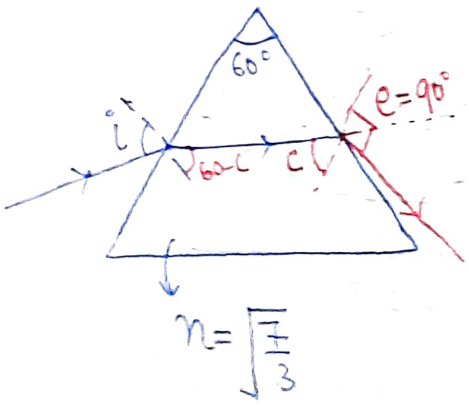


→ Maximum deviation will occur for 2 angle of incidence:

① $i = 90°$

② i for which $e = 90°$

Q



If ray is refracted by the two surfaces of prism, what will be the values of i for which δ will be maximum?

Ans ① $i = 90$

2. when e=90

$$\sin C = \frac{1}{\sqrt{\frac{7}{3}}} = \sqrt{\frac{3}{7}}$$

Now, $1 \times \sin i = \sqrt{\frac{7}{3}} \sin(60-c)$

$= \sin i = \sqrt{\frac{7}{3}} (\sin 60 \cos C - \cos 60 \sin C)$

$= \sin i = \sqrt{\frac{7}{3}} \left(\frac{\sqrt{3}}{2} \times \frac{2}{\sqrt{7}} - \frac{1}{2} \sqrt{\frac{3}{7}} \right)$

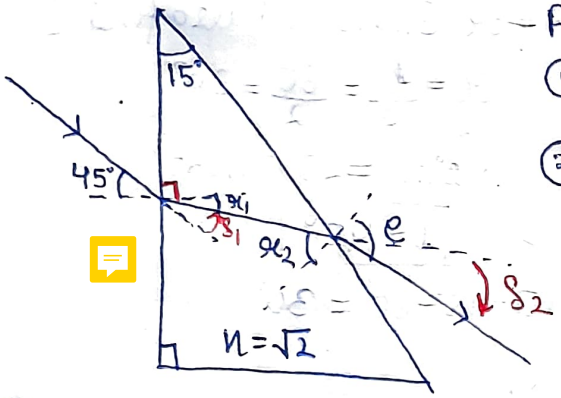
$= \sin i = 1 - \frac{1}{2} = \frac{1}{2} \Rightarrow \sin i = \frac{1}{2} \Rightarrow \boxed{i = 30^\circ}$

$\left\{ \begin{array}{l} \sin C = \sqrt{\frac{3}{7}} \\ \cos C = \sqrt{1 - \frac{3}{7}} \\ \cos C = \frac{2}{\sqrt{7}} \end{array} \right\}$

$\delta_{max} = i + e - A = 30 + 90 - 60 = 60^\circ$

Oblique incidence:

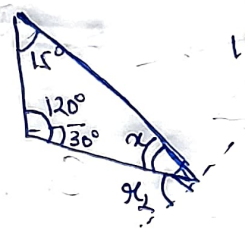
Q



Find ① r_1 ② r_2 ③ e ④ δ_{net}

① $1 \times \sin 45^\circ = \sqrt{2} \sin r_1 \Rightarrow \boxed{r_1 = 30^\circ}$

②



$15 + 120 + x = 180$

$x = 45^\circ$

$r_2 = 90 - x$

$\boxed{r_2 = 45^\circ}$

③ $\sqrt{2} \sin 45^\circ = 1 \sin e \Rightarrow \boxed{e = 90^\circ}$

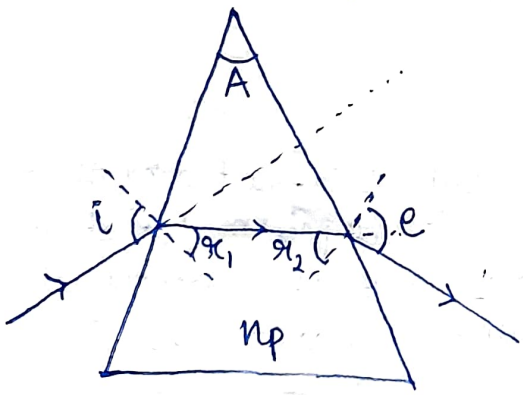
④ $\delta_{net} = \delta_1 + \delta_2 = 15^\circ \curvearrowright + 45^\circ \curvearrowright$

$= 30^\circ \curvearrowright$

Small angled prism:

n_s

A ≈ small



Since A is small, r_1, r_2 and i can also be taken as small.

$n_s i = n_p r_1$ — ①

$n_s e = n_p r_2$ — ②

$(\sin\theta \approx \theta)$

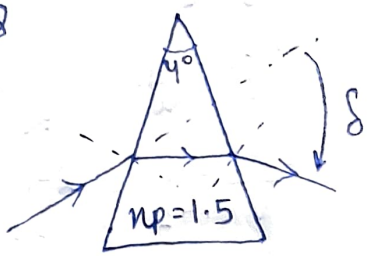
Add ① and ②

$n_s (i + e) = n_p (r_1 + r_2)$

$i + e = \frac{n_p}{n_s} A$

$\star \boxed{\delta = i + e - A = \left(\frac{n_p}{n_s} - 1 \right) A = (n_{rel} - 1) A}$

Q

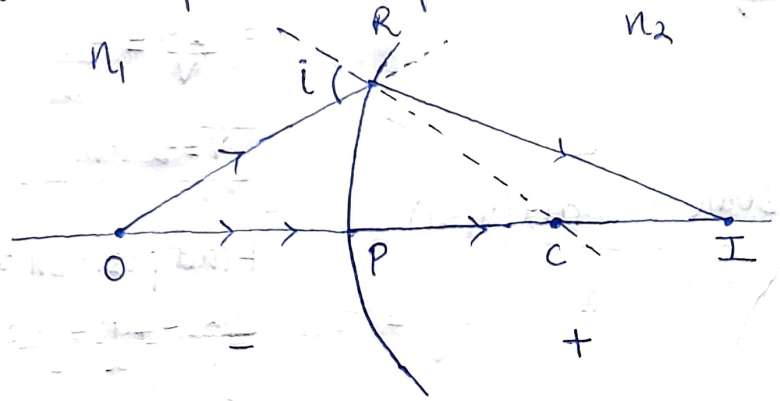


Find δ.

$$\delta = (n_{rel} - 1)A = (1.5 - 1)\,4^\circ$$

$$\delta = 2^\circ$$

Refraction at Spherical Surface:



Let $n_1 < n_2$

$$\star \quad \boxed{\frac{n_2}{v} - \frac{n_1}{u} = \frac{n_2 - n_1}{R}}$$

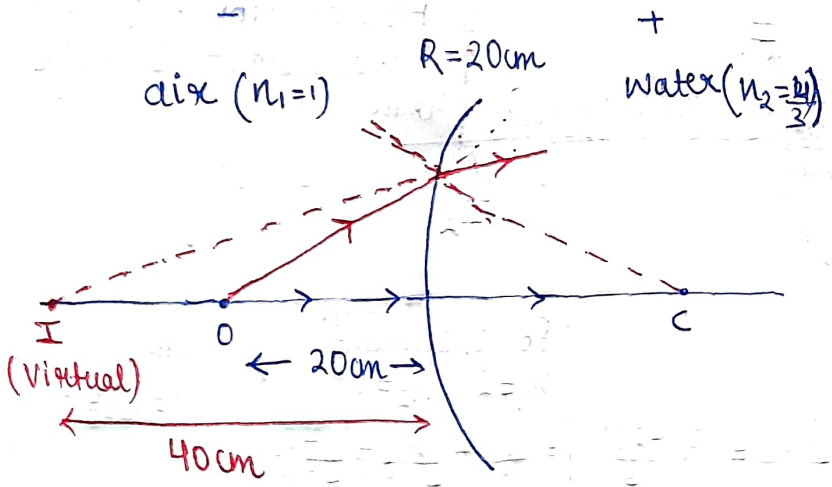
v = x-coordinate of image
u = x-coordinate of object
R = x-coordinate of center of curvature

n_1 = Refractive index of medium from which rays are coming
n_2 = Medium to which rays are going.

Lateral / Transverse Magnification:

$$\boxed{m = \frac{h_i}{h_o} = \frac{v - R}{u - R} = \frac{n_1 v}{n_2 u}}$$

Q



Find position of image

$$\frac{n_2}{v} - \frac{n_1}{u} = \frac{n_2 - n_1}{R}$$

$$= \frac{4/3}{v} - \frac{1}{(-20)} = \frac{\frac{4}{3} - 1}{20}$$

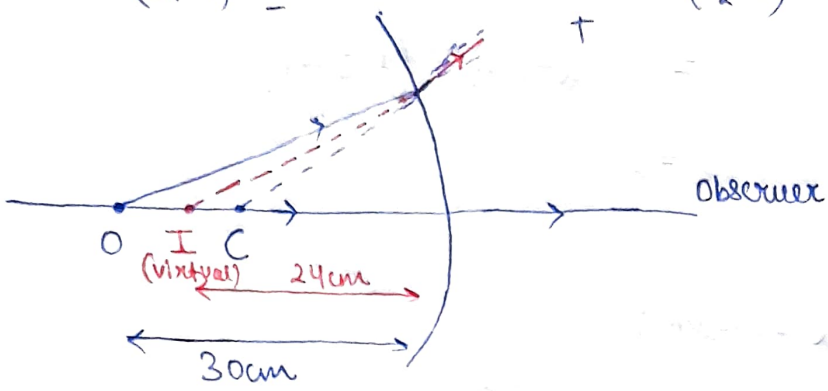
$$= \frac{4}{3v} + \frac{1}{20} = \frac{1}{3(20)}$$

$$= \frac{4}{3v} = \frac{1}{60} - \frac{1}{20}$$

$$= \frac{4}{3v} = \frac{-2}{60} = \frac{-1}{30}$$

$$\boxed{v = -40\,cm} \Leftarrow \frac{1}{v} = -\frac{1}{40} \Leftarrow$$

Q air ($n_1=1$) − R=20cm Medium ($n_2=2$) +



Find position of image

$$\frac{n_2}{v} - \frac{n_1}{u} = \frac{n_2 - n_1}{R}$$

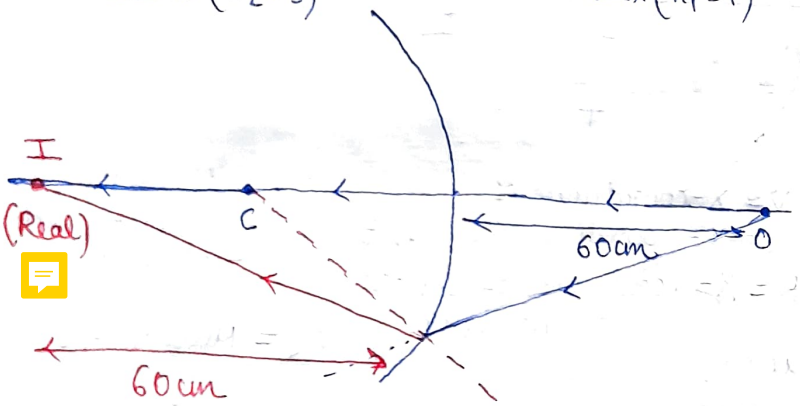
$$= \frac{2}{v} - \frac{1}{-30} = \frac{2-1}{-20}$$

$$= \frac{2}{v} = -\frac{1}{20} - \frac{1}{30}$$

$$= \frac{2}{v} = \frac{-1}{12}$$

$$\boxed{v = -24\text{cm}}$$

Q + Medium ($n_2=3$) R=30cm air ($n_1=1$) −



Find position of image

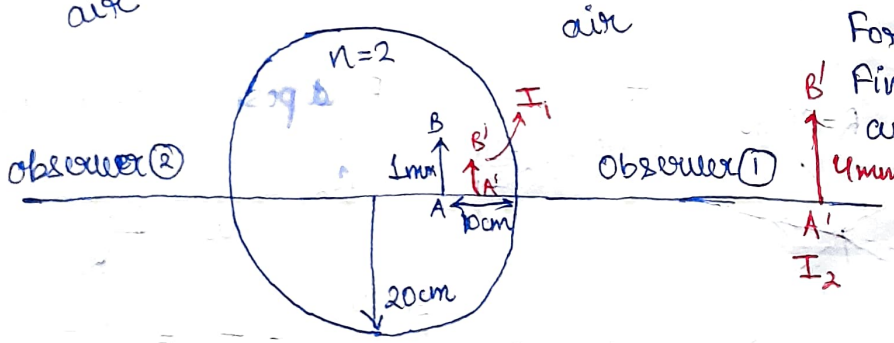
$$\frac{n_2}{v} - \frac{n_1}{u} = \frac{n_2 - n_1}{R}$$

$$= \frac{3}{v} - \frac{1}{-60} = \frac{3-1}{+30}$$

$$= \frac{3}{v} = \frac{2}{30} - \frac{1}{60}$$

$$= \frac{3}{v} = \frac{3}{60} \Rightarrow \boxed{v = 60\text{cm}}$$

Q



For both the observers Find position of image and height of image.

For observer ①

$n_1=2$, $n_2=1$

$$\frac{n_2}{v} - \frac{n_1}{u} = \frac{n_2 - n_1}{R}$$

$$= \frac{1}{v} - \frac{2}{-10} = \frac{1-2}{-20}$$

$$= \frac{1}{v} + \frac{2}{10} = \frac{+1}{+20}$$

$$= \frac{1}{v} = \frac{1}{20} - \frac{2}{10} = \frac{-3}{20}$$

$$\boxed{v = -20/3\text{ cm}}$$

Now,

$$m = \frac{hi}{1mm} = \frac{2\left(\frac{-20}{3}\right)}{(1)(-10)}$$

$$\boxed{hi = \frac{4}{3}\text{ mm}}$$

For observer ②

$n_1=2$, $n_2=1$

$$= \frac{1}{v} - \frac{2}{-30} = \frac{1-2}{-20}$$

$$= \frac{1}{v} + \frac{1}{15} = \frac{1}{20}$$

$$= \frac{1}{v} = \frac{1}{20} - \frac{1}{15}$$

$$= \frac{1}{v} = \frac{-1}{60}$$

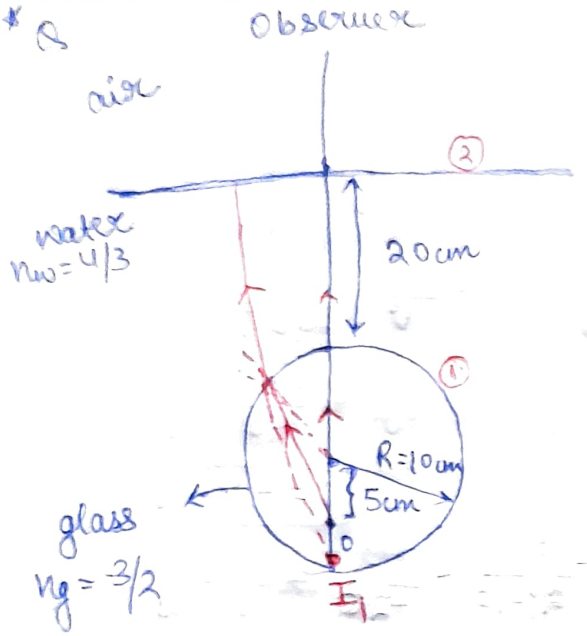
$$\boxed{v = -60\text{cm}}$$

Now,

$$m = \frac{hi}{1mm} = \frac{2(-60)}{1(-30)}$$

$$\boxed{hi = 4mm}$$

Q



Find final position of image for observer.

Refraction by surface ①:

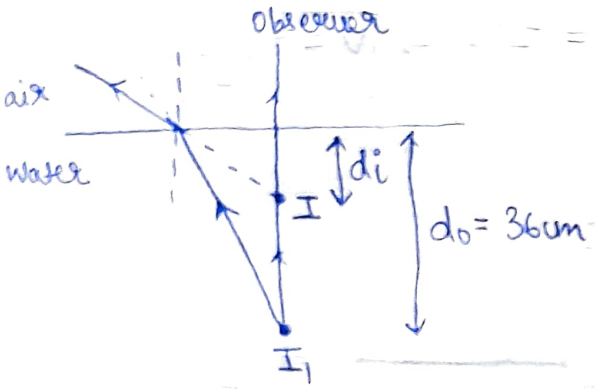
$n_1 = 3/2$, $n_2 = 4/3$

$$= \frac{4}{3v} - \frac{3}{2(-15)} = \frac{\frac{4}{3} - \frac{3}{2}}{-10}$$

$$= \frac{4}{3v} + \frac{1}{10} = \frac{1}{60} \Rightarrow \frac{4}{3v} = \frac{1}{60} - \frac{1}{10}$$

$$= \frac{4}{3v} = \frac{-5}{60} \Rightarrow \boxed{v = -16\text{cm}}$$

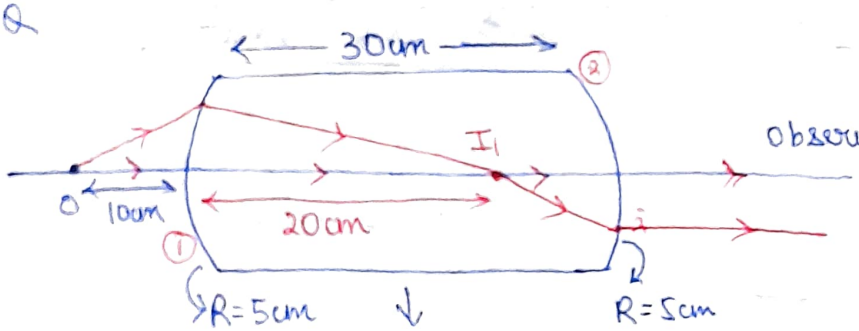
Now, I_1 acts as object for surface ②



$$d_i = \frac{d_o}{n_{rel}} = \frac{36}{4/3/1} = \frac{36}{4/3}$$

$$d_i = \frac{36}{4} \times 3 = 27\text{cm}$$

Q



Transparent glass ($n = 2$)

Find position of final image for observer.

(Situation in the question is also known as thick lens)

Refraction by surface ①:

$n_1 = 1$, $n_2 = 2$

$$\frac{2}{v} - \frac{1}{-10} = \frac{2-1}{5}$$

$$= \frac{2}{v} = \frac{1}{5} - \frac{1}{10}$$

$$= \frac{2}{v} = \frac{1}{10}$$

$$\boxed{v = 20\text{cm}}$$

Refraction by surface ②

I_1 is object for surface ②

$n_1 = 2$, $n_2 = 1$

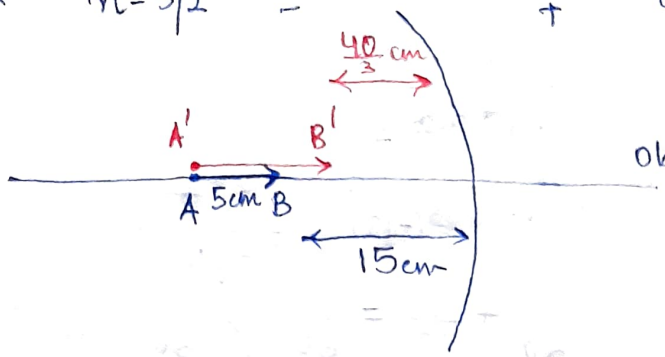
$$= \frac{1}{v} - \frac{2}{-10} = \frac{1-2}{-5}$$

$$= \frac{1}{v} = \frac{1}{5} - \frac{1}{5} = 0$$

$$\boxed{v \to \infty}$$

Longitudinal Magnification :

Q $n = 3/2$ — $R = 20\,cm$ + air



Find length of image.

A: $\frac{1}{v} + \frac{3}{2(-20)} = \frac{+1/2}{+20}$

$= \frac{1}{v} + \frac{3}{40} = \frac{1}{40}$

$= \frac{1}{v} = \frac{-2}{40} = \frac{-1}{20}$

$\boxed{v = -20\,cm}$

$l = A'B' = 20 - \frac{40}{3}$

$\therefore \boxed{l = \frac{20}{3}\,cm}$

B: $\frac{1}{v} - \frac{3}{2(-15)} = \frac{+1/2}{+20} \Rightarrow \frac{1}{v} = \frac{1}{40} - \frac{3}{30}$

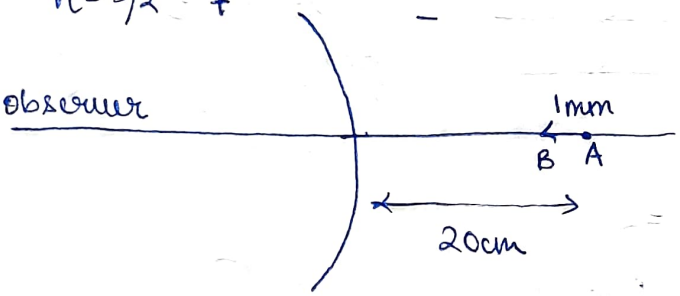
$\frac{1}{v} = \frac{-3}{40} \Rightarrow \boxed{v = -\frac{40}{3}}$

If object is very small

$$\frac{n_2}{v^2}dv - \frac{n_1}{u^2}du = 0$$

$$dv = \frac{n_1}{u^2}\cdot\frac{v^2}{n_2}du \Rightarrow^{*} \boxed{m_l = \frac{dv}{du} = \frac{n_1 v^2}{n_2 u^2}}$$

Q $n = 3/2$ + $R = 20\,cm$ air



Find position of image and length of image.

$$\frac{n_2}{v} - \frac{n_1}{u} = \frac{n_2 - n_1}{R}$$

$= \frac{3}{2v} - \frac{1}{-20} = \frac{\frac{3}{2} - 1}{20} = \frac{1/2}{20}$

$= \frac{3}{2v} = \frac{1}{40} - \frac{1}{20}$

$= \frac{3}{2v} = \frac{1-2}{40} = \frac{-1}{40}$

$= \frac{1}{v} = \frac{-1}{60}$

$\boxed{v = -60\,cm}$

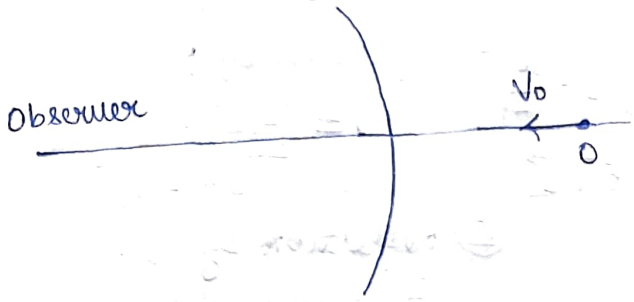
Now, $m_l = \frac{l_i}{l_o} = \frac{n_1 v^2}{n_2 u^2}$

$= \frac{l_i}{1mm} = \frac{(1)(-60)^2}{\frac{3}{2}(-20)^2}$

$\boxed{l_i = 6mm}$

Velocity of image:

① Object is moving along principal axis:



$$\frac{n_2}{v} - \frac{n_1}{u} = \frac{n_2 - n_1}{R}$$

Diff both sides wrt time

$$= \frac{-n_2}{v^2}\frac{dv}{dt} + \frac{n_1}{u^2}\frac{du}{dt} = 0$$

$$\Rightarrow \frac{dv}{dt} = \frac{n_1 v^2}{n_2 u^2}\left(\frac{du}{dt}\right)$$

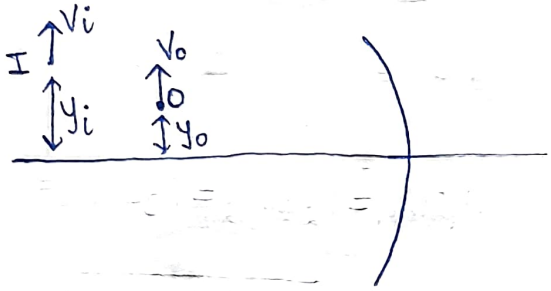
If surface is at rest

$$\boxed{V_i = \frac{n_1 v^2}{n_2 u^2}(V_0)}$$

$$\therefore \star \boxed{V_{is} = \frac{n_1 v^2}{n_2 u^2}(V_{0s})}$$

s=surface

② If object is moving perpendicular to principal axis:

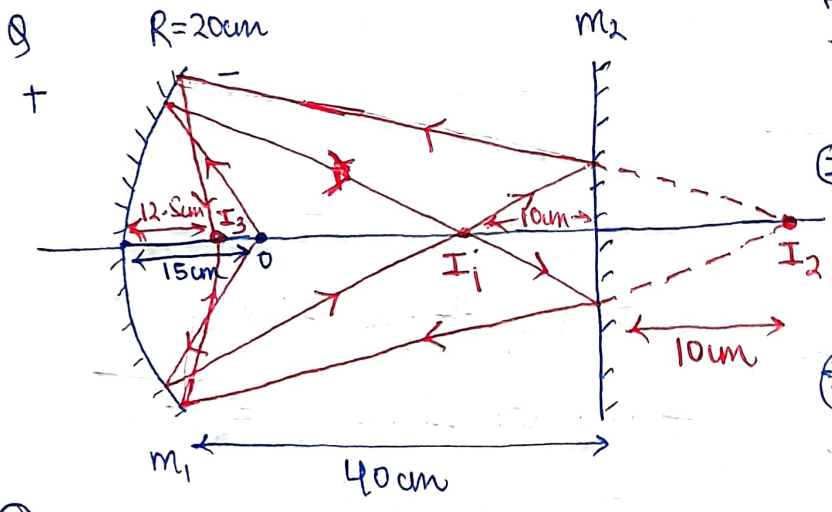


$$\frac{y_i}{y_0} = m_t = \frac{n_1 v}{n_2 u}$$

$$\Rightarrow y_i = \frac{n_1 v}{n_2 u} y_0$$

$$= \frac{dy_i}{dt} = \frac{n_1 v}{n_2 u}\left(\frac{dy_0}{dt}\right)$$

$$\therefore \boxed{V_i = \frac{n_1 v}{n_2 u}(V_0)}$$



Find position of image after three reflections, considering first reflection on m_1.

(I) Reflection by m_1:

$$v = \frac{uf}{u-f} = \frac{(-15)(-10)}{-15-(-10)} = -30\text{cm}$$

(II) Reflection by m_2:

I_1 acts as object for m_2

for plane mirror,

$d_i = d_o$

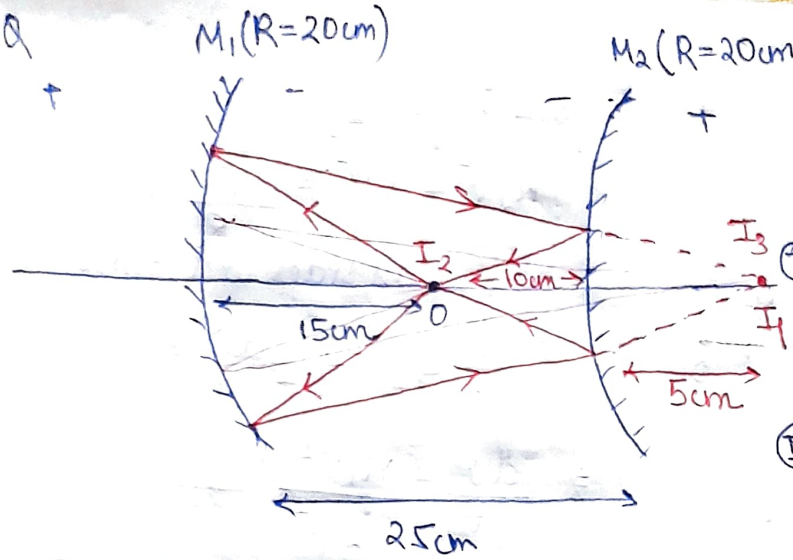
$\therefore I_2$ is at 10cm

(III) Reflection by m_1: I_2 acts as object

$$v = \frac{uf}{u-f} = \frac{(-50)(-10)}{(-50)-(-10)} = -12.5\text{ cm}$$

(I_3)

Q



Find position of image after 3 reflections, considering first reflection on M_1.

(I) Reflection by M_1:

$$v = \frac{(-15)(-10)}{-15-(-10)} = -30\text{ cm}$$

(II) Reflection by M_2:

I_4 acts as object

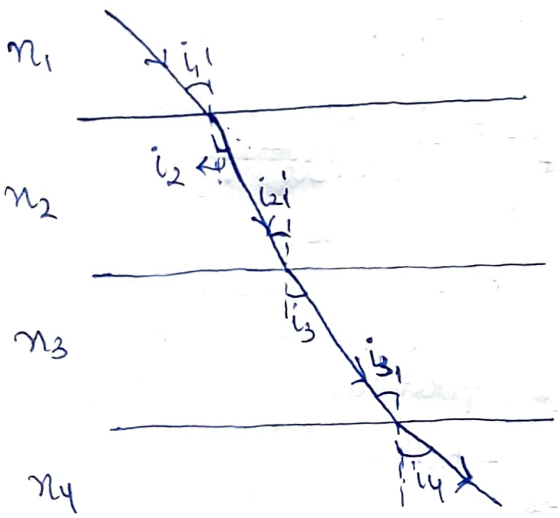
$$v = \frac{(5)(10)}{-5-10} = -10\text{ cm}$$

(III) Reflection by M_1: I_2 is object

$\boxed{v = -30\text{ cm}}$ (I_3)

#Note: For multiple reflection, magnification, $\boxed{m = m_1 \times m_2 \times m_3 \cdots}$

Composite slabs:

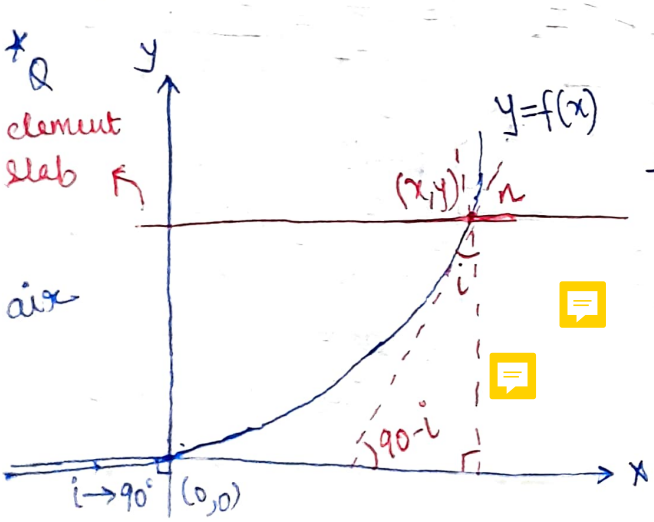


$$n_1 \sin i_1 = n_2 \sin i_2 = n_3 \sin i_3 = n_4 \sin i_4$$

$$\therefore \boxed{n_1 \sin i_1 = n_4 \sin i_4}$$

Parallel

Q



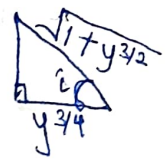
Find eqn of path of light.

transparent medium: $n(y) = \sqrt{1+y^{3/2}}$

Using snells law:

$$1 \times \sin 90° = \sqrt{1+y^{3/2}} \sin i$$

$$\sin i = \frac{1}{\sqrt{1+y^{3/2}}}$$



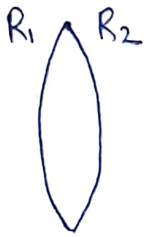
Slope of tangent $= -\tan\theta = -\tan(90-i) = \frac{dy}{dx}$

$$\cot i = \frac{dy}{dx} = \frac{y^{3/4}}{1} \Rightarrow \int_0^y y^{-3/4}\,dy = \int_0^x dx$$

$$\Rightarrow x = 4y^{1/4} \Rightarrow \boxed{y = \frac{x^4}{256}}$$

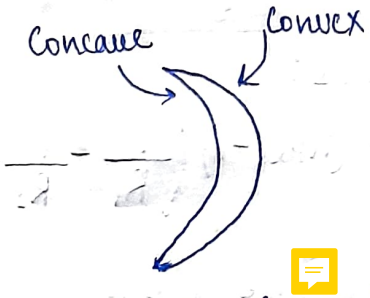
Thin lens: It is a homogeneous, transparent, refracting medium bounded by atleast one spherical surface, other surface can be spherical or planar.

Convex lens: Thick at middle



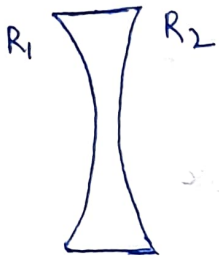
Biconvex
Doubly convex
Equi convex ($R_1 = R_2$)

Plano - Convex lens



Concavo - Convex lens
(nature is convex)

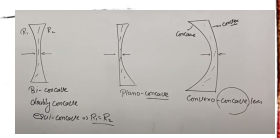
Concave lens: Thin at middle



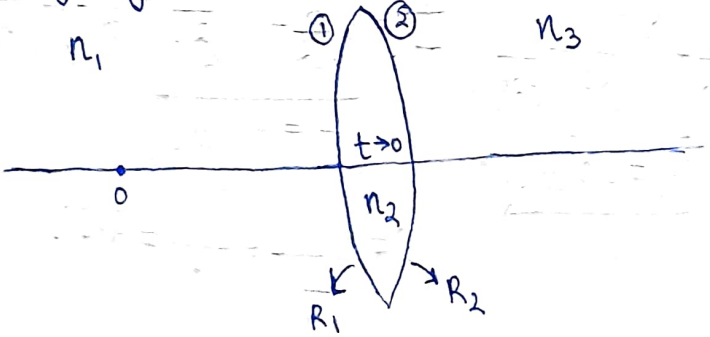
Bi - concave
Doubly Concave
Equi - Concave ($R_1 = R_2$)

Plano - Concave

Convexo - Concave lens
(nature is concave)



Position of image by thin lens:



From surface (1)

$$\frac{n_2}{v'} - \frac{n_1}{u} = \frac{n_2 - n_1}{R_1} \quad -(1)$$

From surface (2)

$$\frac{n_3}{v} - \frac{n_2}{v'} = \frac{n_3 - n_2}{R_2} \quad -(2)$$

Add ① and ②

$$\star \quad \boxed{\frac{n_3}{v} - \frac{n_1}{u} = \frac{n_2 - n_1}{R_1} + \frac{n_3 - n_2}{R_2}}$$

Thin lens in homogeneous medium:

$$n_1 = n_3 = n_s$$

$$\Rightarrow \frac{n_s}{v} - \frac{n_s}{u} = \frac{n_1 - n_s}{R_1} + \frac{n_s - n_1}{R_2}$$

$$= \frac{1}{v} - \frac{1}{u} = \frac{(n_1 - n_s)}{n_s}\left(\frac{1}{R_1} - \frac{1}{R_2}\right)$$

$$= \frac{1}{v} - \frac{1}{u} = \left(\frac{n_1}{n_s} - 1\right)\left(\frac{1}{R_1} - \frac{1}{R_2}\right)$$

$$= \boxed{\frac{1}{v} - \frac{1}{u} = (n_{rel} - 1)\left(\frac{1}{R_1} - \frac{1}{R_2}\right)}$$

Power of lens:

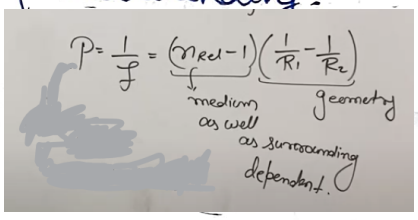
$$P = \frac{1}{f} = (n_{rel} - 1)\left(\frac{1}{R_1} - \frac{1}{R_2}\right)$$

Unit = Diopter or m^{-1} of Power

Note: For mirror, f only depends on geometry of mirror but for lens f depends on geometry as well as medium of surrounding.

Lens Maker Formula:

$$\boxed{\frac{1}{v} - \frac{1}{u} = \frac{1}{f}} \qquad \boxed{v = \frac{uf}{u+f}}$$



Q Find focal length.

① air (−) | $n = \frac{3}{2}$ | air (+)

direction of incident ray

R_1, R_2, $R = 20$ cm

$$\frac{1}{f} = \left(\frac{n_1}{n_s} - 1\right)\left(\frac{1}{R_1} - \frac{1}{R_2}\right) = \left(\frac{3}{2} - 1\right)\left(\frac{1}{20} - \frac{1}{(-20)}\right)$$

$$\frac{1}{f} = \frac{1}{2} \times \frac{2}{20} = \frac{1}{20} \Rightarrow \boxed{f = 20 \text{ cm}}$$

$$P = \frac{1}{f} = \frac{1}{0.2} = 5D$$

② air (−) | $\frac{3}{2}$ | air (+)

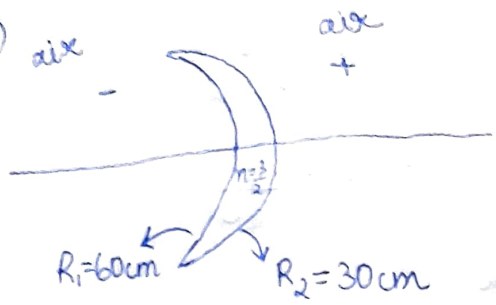
R_1, R_2, $R = 20$ cm

$$\frac{1}{f} = \left(\frac{3}{2} - 1\right)\left(\frac{1}{20} - \frac{1}{\infty}\right) = \frac{1}{2} \times \frac{1}{20}$$

$$\boxed{f = 40 \text{ cm}}$$

$$P = \frac{1}{f} = \frac{1}{0.4} = 2.5D$$

③

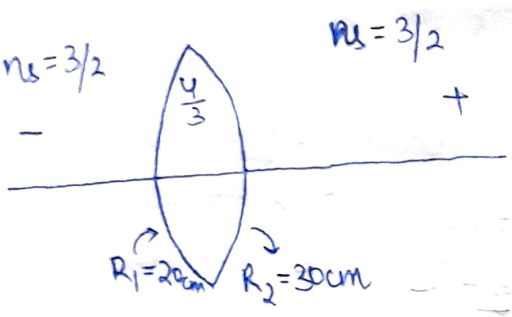


$$\frac{1}{f} = \left(\frac{3}{2}-1\right)\left(\frac{1}{-60} - \frac{1}{-30}\right) = \frac{1}{2} \times \frac{1}{60}$$

$$\boxed{f = 120\,cm}$$

$$P = \frac{1}{1.2} = \frac{5}{6}\,D$$

④

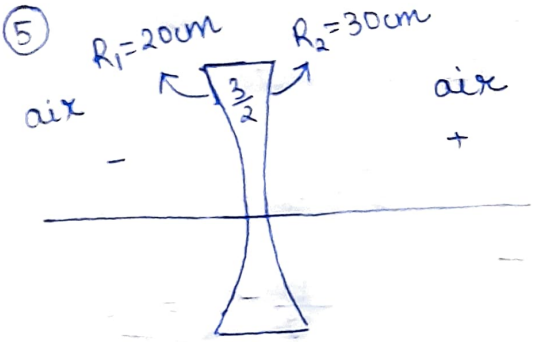


$$\frac{1}{f} = \left(\frac{4/3}{3/2}-1\right)\left(\frac{1}{20} - \frac{1}{-30}\right)$$

$$\frac{1}{f} = -\frac{1}{9} \times \frac{5}{60} = -\frac{1}{9} \times \frac{1}{12} = -\frac{1}{108}$$

$$\boxed{f = -108\,cm}$$

⑤

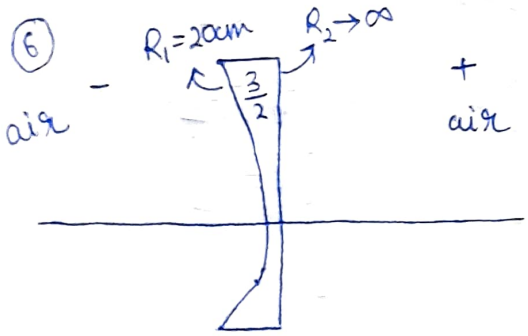


$$\frac{1}{f} = \left(\frac{3}{2}-1\right)\left(\frac{1}{-20} - \frac{1}{30}\right)$$

$$\frac{1}{f} = \frac{1}{2} \times \frac{-1}{12} = \frac{-1}{24}$$

$$\boxed{f = -24\,cm}$$

⑥

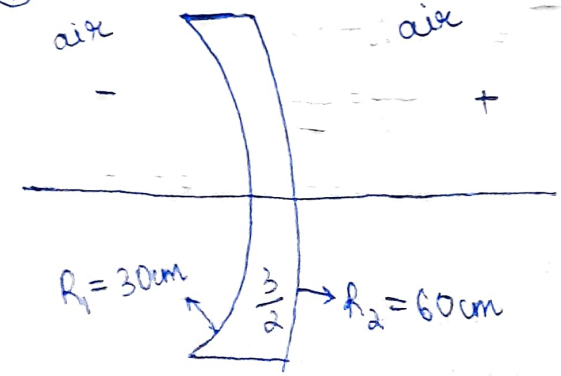


$$\frac{1}{f} = \left(\frac{3}{2}-1\right)\left(\frac{1}{-20} - \frac{1}{\infty}\right)$$

$$\frac{1}{f} = \frac{1}{2} \times \frac{-1}{20} = -\frac{1}{40}$$

$$\boxed{f = -40\,cm}$$

⑦

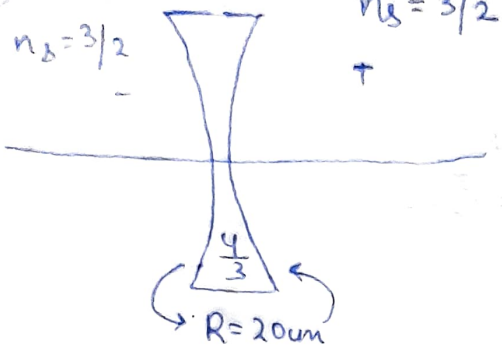


$$\frac{1}{f} = \left(\frac{3}{2}-1\right)\left(\frac{1}{-30} - \frac{1}{-60}\right)$$

$$\frac{1}{f} = \frac{1}{2}\left(\frac{-1}{60}\right) = \frac{-1}{120}$$

$$\boxed{f = -120\,cm}$$

(8)



$n_s = 3/2$

$$\frac{1}{f} = \left(\frac{4/3}{3/2} - 1\right)\left(\frac{1}{-20} - \frac{1}{20}\right)$$

$$\frac{1}{f} = -\frac{1}{9}\left(-\frac{1}{10}\right) = \frac{1}{90}$$

$$\boxed{f = 90\text{cm}}$$

#Note:

$$\frac{1}{f} = \left(\frac{n_l}{n_s} - 1\right)\left(\frac{1}{R_1} - \frac{1}{R_2}\right)$$

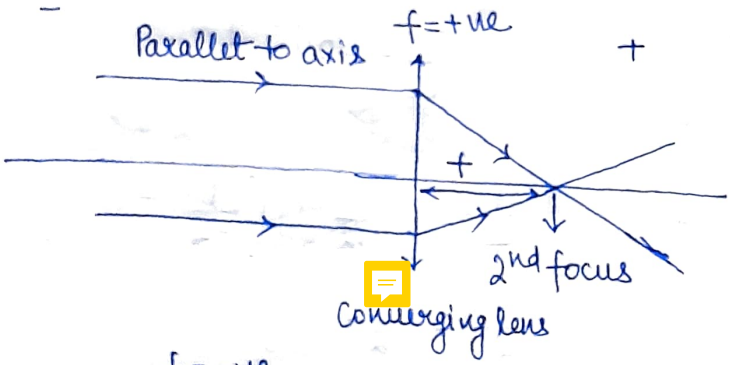
- $n_l > n_s \Rightarrow$ +ve
  - Convex $\Rightarrow$ +ve
  - Concave $\Rightarrow$ -ve
- $n_l < n_s \Rightarrow$ -ve

Focii of lens:

① If $f = +ve \Rightarrow$ Converging lens

② If $f = -ve \Rightarrow$ Diverging lens

(I)



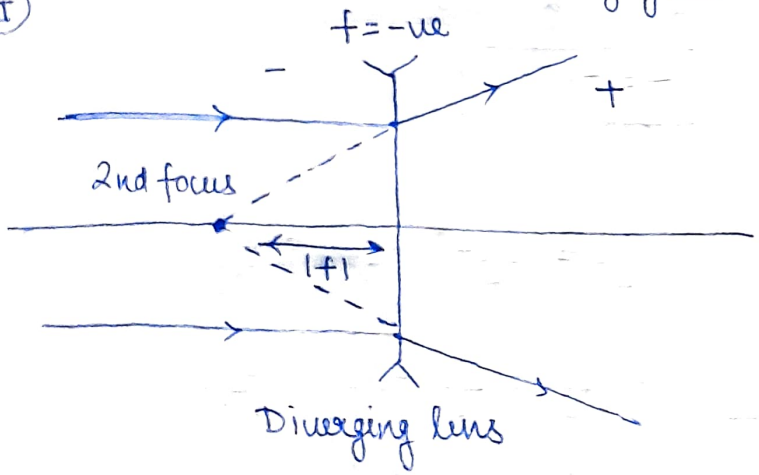
$$\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$$

$u \to -\infty$

$$\frac{1}{v} = \frac{1}{f}$$

$$\boxed{v = f}$$ as f is +ve, hence v is also +ve

(II)



$$\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$$

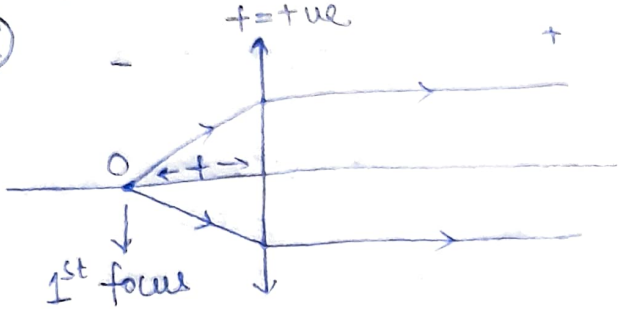
$u \to \infty$

$$\frac{1}{v} = \frac{1}{f}$$

Since $f = -ve$, $v = -ve$

→ If incident rays are parallel to principal axis, then after refraction they will strike principal axis at 2nd focus

(III)

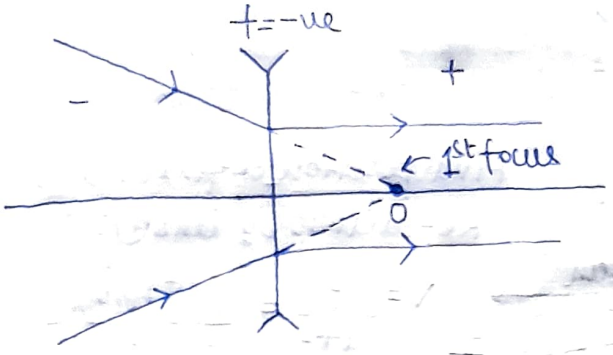


$\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$

$v \to \infty$

$\therefore \boxed{u = -f}$

(IV)



$\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$

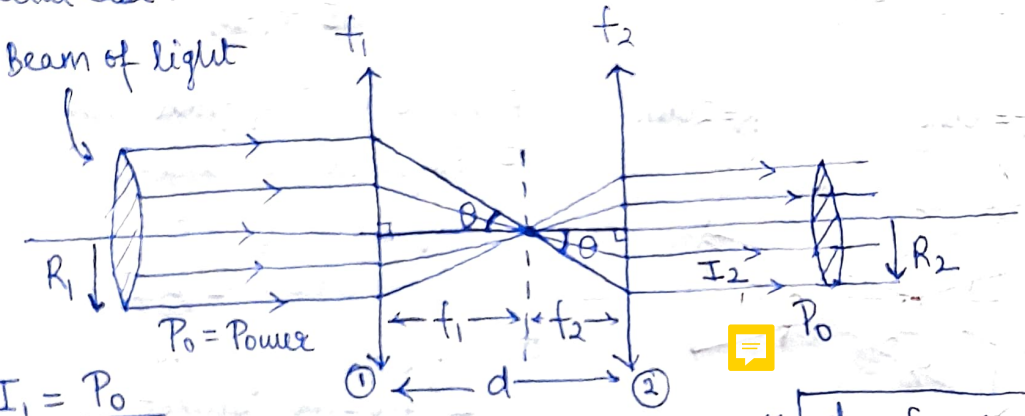
$v \to \infty$

$u = -f$

But $f = -ve \Rightarrow$ $\boxed{u \text{ is +ve}}$

→ 1st focus is the point at which object is kept, then rays emerging from it become parallel to principal axis after refraction.

Special Case:



$I_1 = \frac{P_0}{\pi R_1^2}$

$\star \boxed{d = f_1 + f_2}$

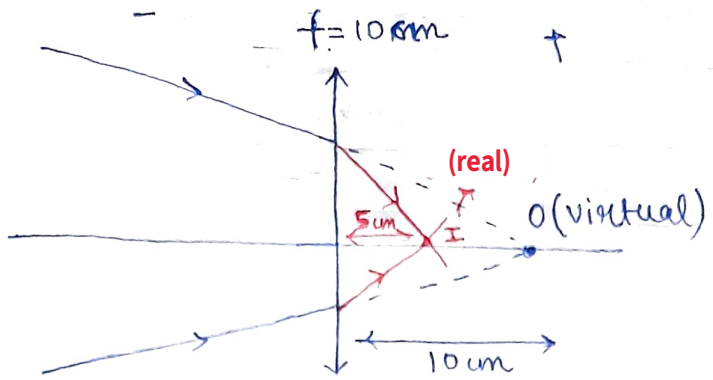
→ For final emergent ray to be parallel to principal axis, 2nd focus of lens ① should coincide with 1st focus of lens ②

$\frac{R_2}{f_2} = \frac{R_1}{f_1} \Rightarrow \boxed{R_2 = \frac{R_1 f_2}{f_1}}$ (By using similar triangles)

$I_2 = \frac{P_0}{\pi R_2^2} = \frac{P_0 f_1^2}{\pi R_1^2 f_2^2} = \frac{I_1 f_1^2}{f_2^2} \Rightarrow$

$\boxed{\frac{I_2}{I_1} = \left(\frac{f_1}{f_2}\right)^2}$

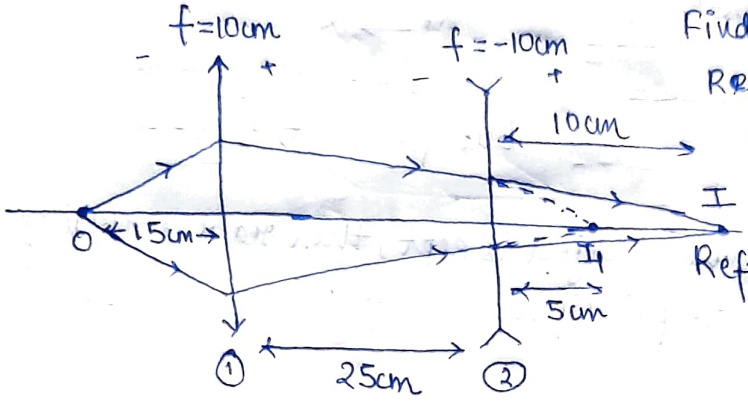
Q



Find position of image

$$v = \frac{(10)(10)}{10+10} = 5\,cm$$

Q



Find position of final image

Refraction by lens (1):

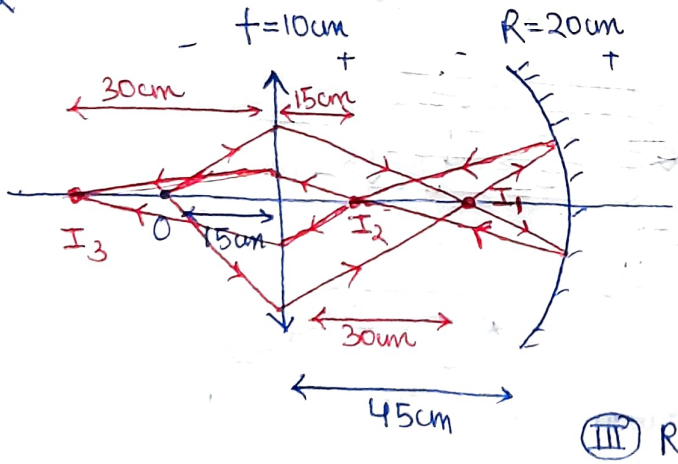
$$v = \frac{uf}{u+f} = \frac{(-15)(10)}{-15+10} = 30\,cm$$

Refraction by lens (2):

I_1 is object

$$v = \frac{(5)(-10)}{5+(-10)} = 10\,cm$$

Q



Find position of final image.

(I) Refraction by lens:

$$v = \frac{uf}{u+f} = \frac{(-15)(10)}{-15+10} = 30\,cm$$

(II) Reflection by mirror:

I_1 acts as object

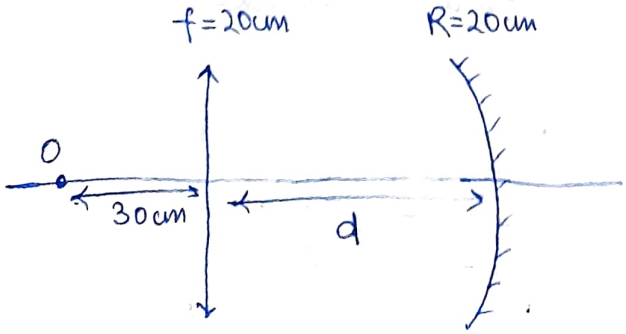
$$v = \frac{uf}{u-f} = \frac{(-15)(10)}{-15-(-10)} = -30\,cm$$

(III) Refraction by mirror:

I_2 acts as object

$$v = \frac{uf}{u+f} = \frac{(-15)(10)}{-15+10} = 30\,cm$$

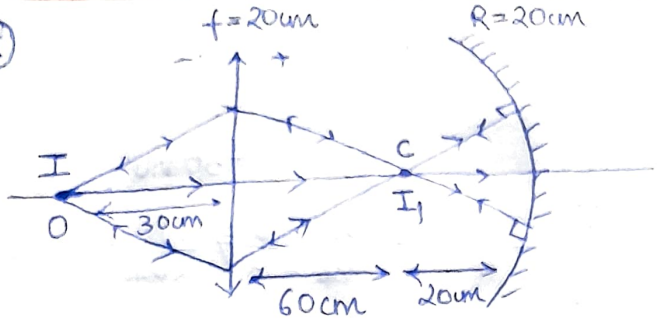
Q



Find d such that final image is on object itself.

concept: For final image to be on object itself, image by lens should co-incide with center of curvature of mirror or pole of the mirror.

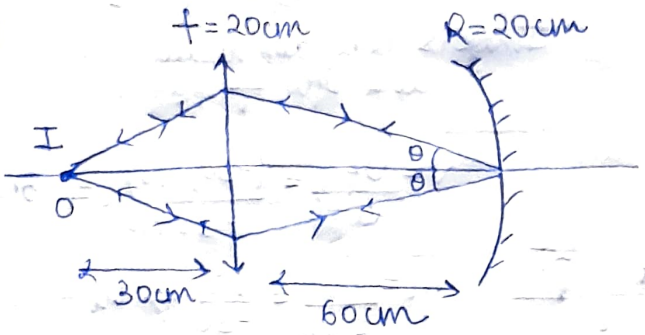
(I)



$$v = \frac{uf}{u+f} = \frac{(-30)(20)}{-30+20} = 60\text{cm}$$

$$\therefore d = 60+20 = 80\text{cm}$$

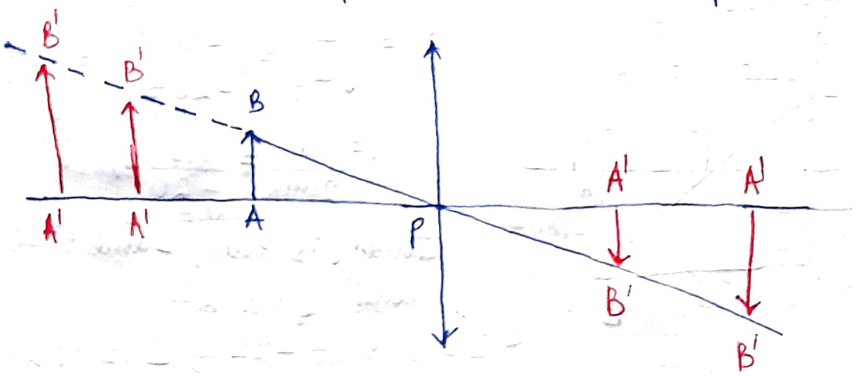
(II)



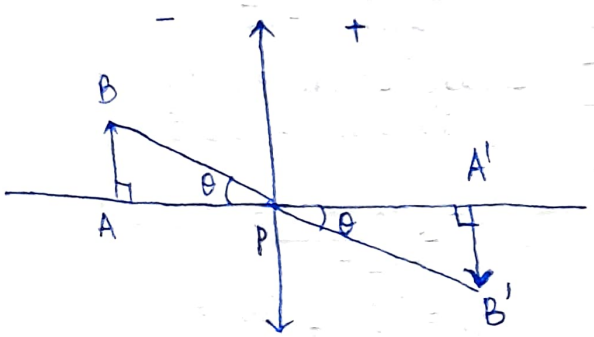
$$\therefore d = 60\text{cm}$$

Lateral/Transverse Magnification:

→ Object is placed perpendicular to principal axis.



→ Object and Image are on same side ⇒ $m = +u$ ⇒ erect
→ Object and Image are on opposite side ⇒ $m = -u$ ⇒ inverted
→ Farther the image, larger it is.



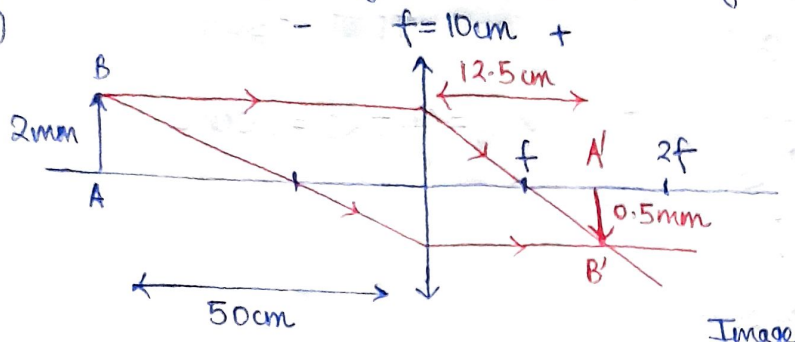
$$\Delta ABP \sim \Delta A'B'P$$

$$\Rightarrow \frac{A'B'}{A'P} = \frac{AB}{AP}$$

$$m = -\frac{A'B'}{AB} = \frac{A'P}{AP} = -\left(\frac{v}{-u}\right) = \frac{v}{u}$$

$$\therefore \boxed{m = \frac{v}{u} = \frac{f}{f+u}}$$

Q Find position of image and height of image. (diagrams not to scale)

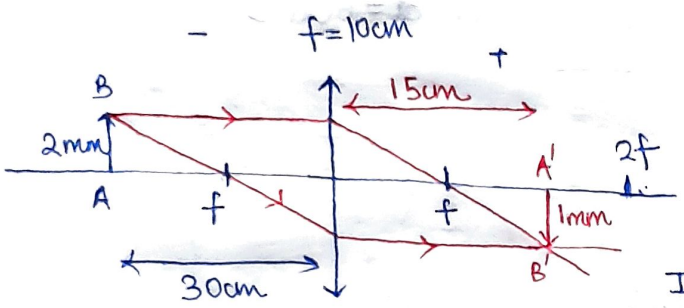
① 

$$v = \frac{uf}{u+f} = \frac{(-50)(10)}{-50+10} = 12.5\text{cm}$$

$$m = \frac{h_i}{h_o} = \frac{v}{u} \Rightarrow \frac{h_i}{2\text{mm}} = \frac{12.5}{-50}$$

$$h_i = -\frac{1}{2}\text{mm}$$

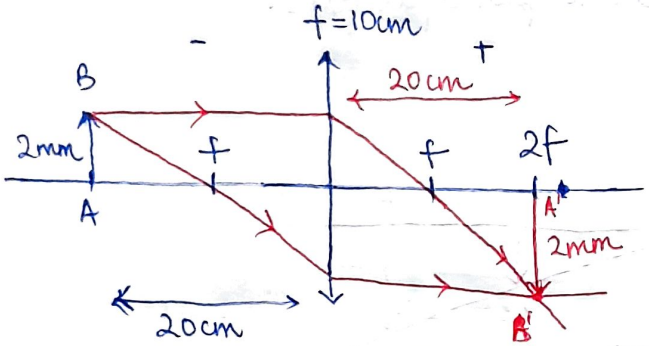
Image is real, diminished, inverted and b/w f & 2f.

② 

$$v = \frac{uf}{u+f} = \frac{(-30)(10)}{-30+10} = 15\text{cm}$$

$$m = \frac{h_i}{2\text{mm}} = \frac{15}{-30} \Rightarrow h_i = -1\text{mm}$$

Image is real, diminished, inverted, and b/w f and 2f.

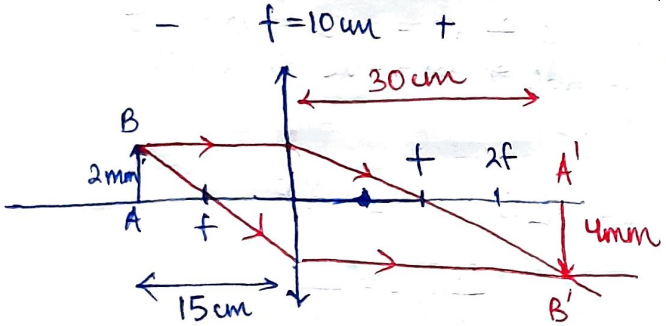
③ 

$$v = \frac{uf}{u+f} = \frac{(-20)(10)}{-20+10} = \frac{-200}{-10} = 20\text{cm}$$

$$m = \frac{h_i}{2\text{mm}} = \frac{20}{-20} \Rightarrow h_i = -2\text{mm}$$

Image is real, same size, inverted and ~~b~~ at 2f.

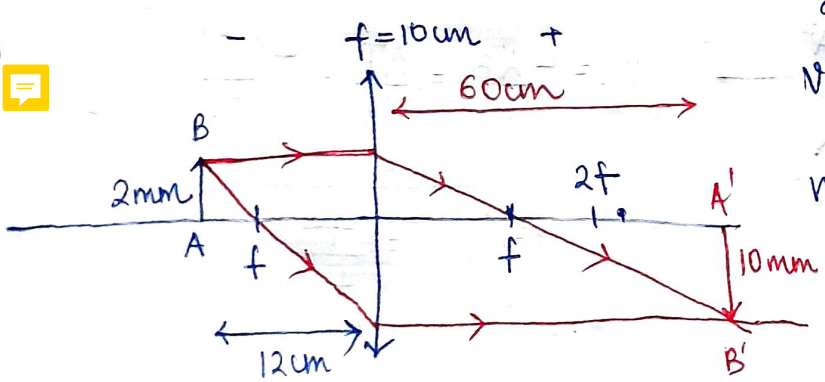
* If O is at $-2f \Rightarrow$ I is at 2f.

④ 

$$v = \frac{(-15)(10)}{-15+10} = 30\text{cm}$$

$$m = \frac{h_i}{2\text{mm}} = \frac{30}{-15} \Rightarrow h_i = -4\text{mm}$$

Image is real, enlarged, inverted and b/w 2f and ∞

⑤ 

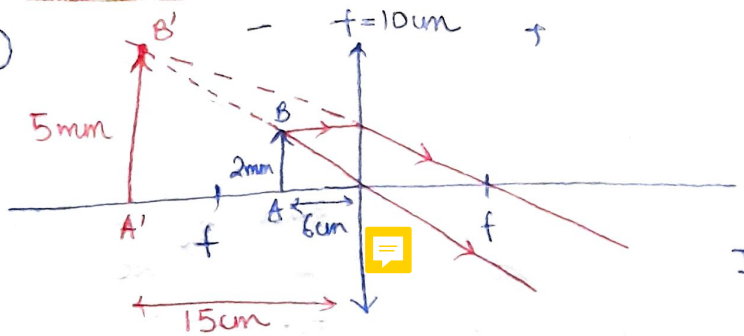
$$v = \frac{(-12)(10)}{-12+10} = 60\text{cm}$$

$$m = \frac{h_i}{2\text{mm}} = \frac{60}{-12} \Rightarrow h_i = -10\text{mm}$$

Image is real, enlarged, inverted and b/w 2f and ∞

5.b. when the object is at 1st focus, image will be at infinity

⑥



$v = \frac{(-6)(10)}{-6+10} = -15\text{cm}$

$m = \frac{h_i}{2\text{mm}} = \frac{-15}{-6} \Rightarrow h_i = 5\text{mm}$

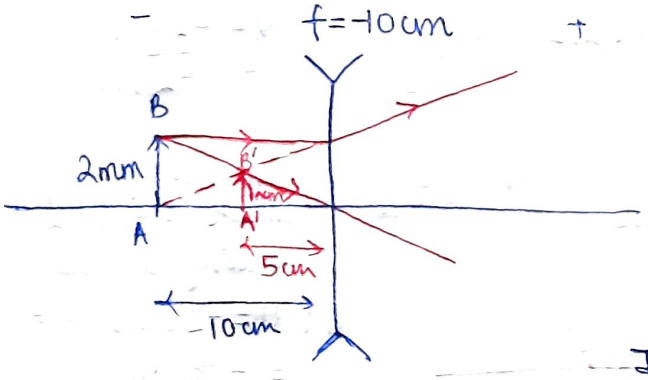
Image is virtual, enlarged, erect and behind the lens.

→ object or image : Farther ⇒ larger ⇒ Faster

at each instant, velocity of object and image are in the same direction

→ Minimum separation b/w real object and real image is 4f which occurs when $u = -2f$ and $v = 2f$

Q



Find position and height of image.

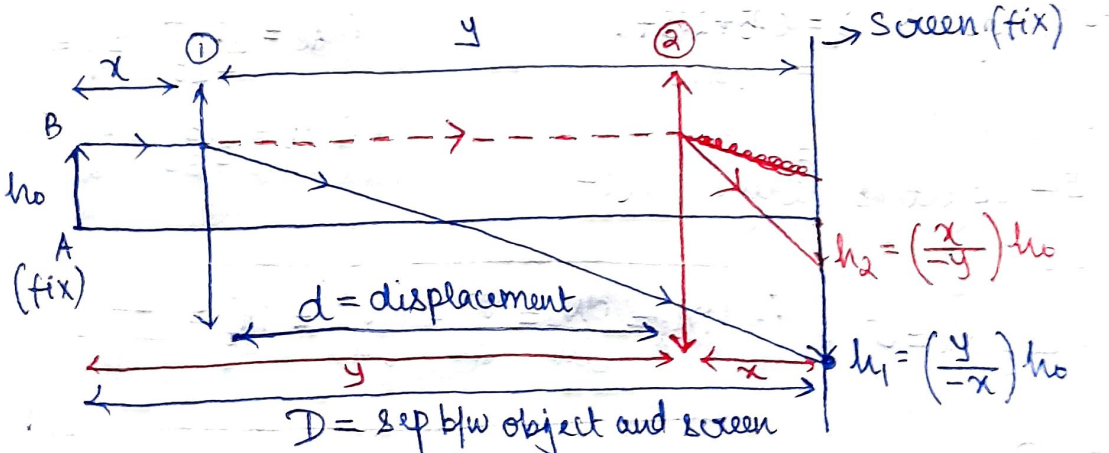
$v = \frac{uf}{u+f} = \frac{(-10)(-10)}{-10-10} = -5\text{cm}$

$m = \frac{h_i}{2\text{mm}} = \frac{-5}{-10} \Rightarrow h_i = 1\text{mm}$

Image is virtual, diminished, erect and b/w pole and focus (1st).

Displacement Method :

→ It is an experiment used to find focal length of converging lens.



① and ② are different positions of ~~lens~~ same lens where sharp image is obtained

$x + y = D$

$y - x = d$

$\Rightarrow y = \frac{D+d}{2}$

$x = \frac{D-d}{2}$

Now using lens formula (on ①)

$\frac{1}{y} - \frac{1}{-x} = \frac{1}{f}$

$\Rightarrow \frac{1}{y} + \frac{1}{x} = \frac{1}{f}$

$f = \frac{xy}{x+y} = \frac{D^2-d^2}{D \times 4}$

$\therefore \boxed{f = \frac{D^2-d^2}{4D}}$

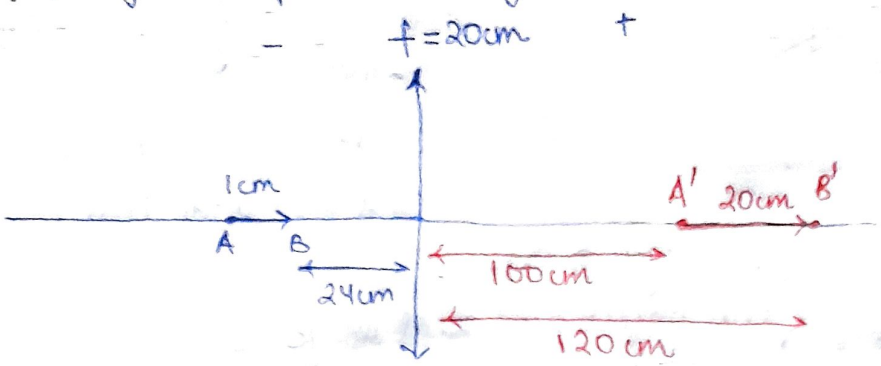
Also, $\boxed{h_1 h_2 = h_o^2}$

$h_1 h_2 = \left(\frac{x}{-y} h_o\right)\left(\frac{y}{-x} h_o\right)$

Longitudinal Magnification:

→ If object is placed along principal axis.

Q



$$m = \frac{v_2 - v_1}{u_2 - u_1} = \frac{120 - 100}{-24 - (-25)} = \frac{20}{1} = 20\text{cm}$$

Find length of image AB.

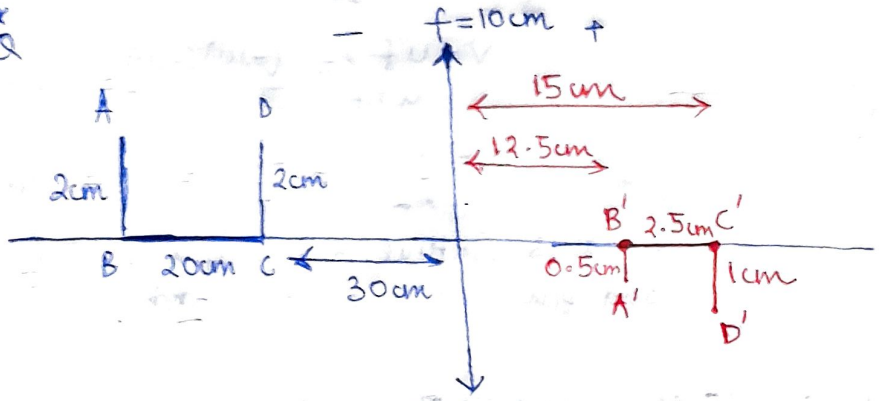
A: $u = -25$, $f = 20\text{cm}$

$$v = \frac{-25 \times 20}{-25 + 20} = 100\text{cm}$$

B: $u = -24\text{cm}$, $f = 20\text{cm}$

$$v = \frac{-24 \times 20}{-24 + 20} = 120\text{cm}$$

Q



Length of image $A'B'C'D' = 0.5 + 2.5 + 1$
$= 4\text{cm}$

Find length of image of this 24cm long wire.

B: $u = -50\text{cm}$, $f = 10\text{cm}$

$$v = \frac{-50 \times 10}{-50 + 10} = 12.5\text{cm}$$

C: $u = -30\text{cm}$, $f = 10\text{cm}$

$$v = \frac{-30 \times 10}{-30 + 10} = 15\text{cm}$$

$$m_{AB} = \frac{12.5}{-50} = \frac{-1}{4} \Rightarrow h_i = -\frac{1}{2}\text{cm}$$

$$m_{CD} = \frac{15}{-30} = \frac{-1}{2} \Rightarrow h_i = -1\text{cm}$$

Note: If object is very small:-

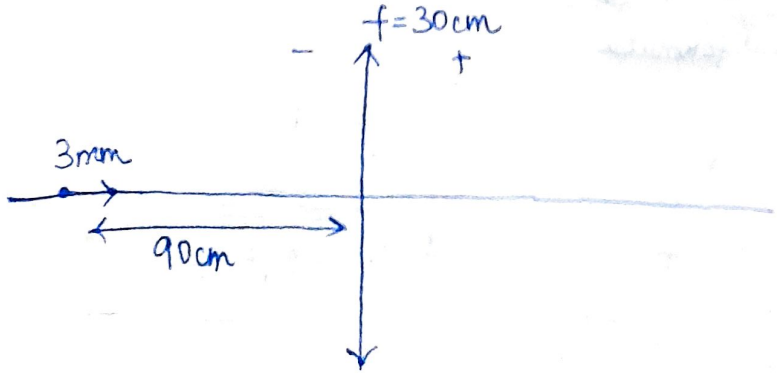
$$v^{-1} - u^{-1} = f^{-1}$$

$$\frac{-1}{v^2}dv + \frac{1}{u^2}du = 0$$

$$\frac{dv}{du} = \frac{v^2}{u^2}$$

$$\therefore \boxed{m_l = \frac{dv}{du} = \frac{v^2}{u^2} = \left(\frac{v}{u}\right)^2 = m_t^2 = \left(\frac{f}{f+u}\right)^2}$$

Q



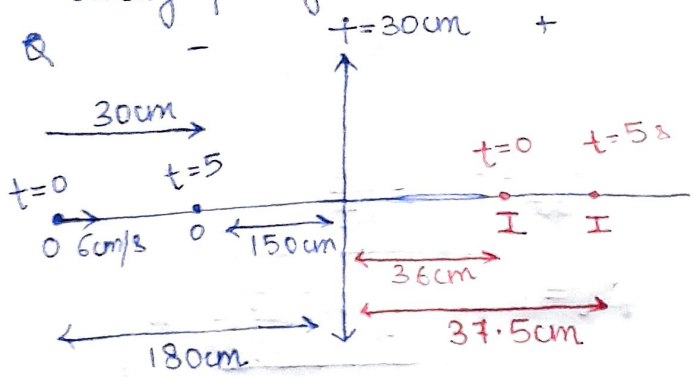
Find length of image.

$$m_l = \frac{l_i}{3\text{mm}} = \left(\frac{f}{f+u}\right)^2$$

$$= \frac{l_i}{3\text{mm}} = \left(\frac{30}{-60}\right)^2 = \frac{1}{4}$$

$$\boxed{l_i = 0.75\text{mm}}$$

Velocity of image:



$$V_{avg} = \frac{37.5-36}{5-0} = \frac{1.5\,cm}{5\,s} = 0.3\ cm/s$$

In time interval $t=0$ to $t=5s$
Find avg speed of image

At $t=0$: $u=-180\,cm$, $f=30\,cm$

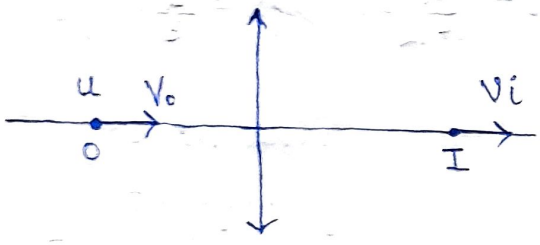
$$v = \frac{(-180)(30)}{-180+30} = 36\,cm$$

In 5s object travels, $6\frac{cm}{s} \times 5s = 30\,cm$

At $t=5$: $u=-150\,cm$, $f=30\,cm$

$$v = \frac{(-150)(30)}{-150+30} = 37.5\,cm$$

① Object moving along principal axis:



If $V_i=0$, $\Rightarrow$

$$V_i = \frac{v^2}{u^2}(V_o) = m^2 V_o = \left(\frac{f}{f+u}\right)^2 V_o$$

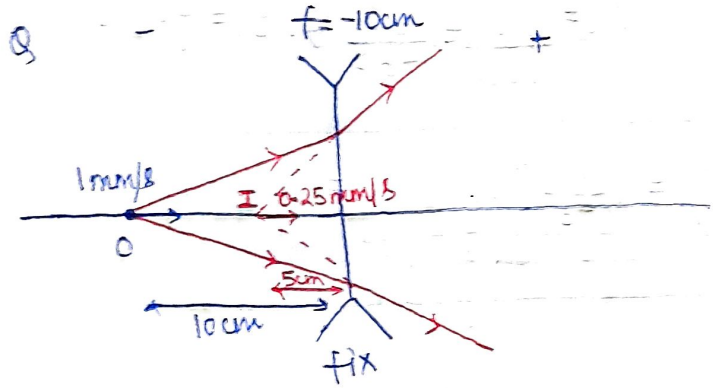
$$v^{-1} - u^{-1} = f^{-1}$$

$$\Rightarrow \frac{-1}{v^2}\frac{dv}{dt} + \frac{1}{u^2}\frac{du}{dt} = 0$$

$$= \frac{dv}{dt} = \frac{v^2}{u^2}\left(\frac{du}{dt}\right)$$

$$\therefore \boxed{V_{il} = \frac{v^2}{u^2}(V_{ol})}$$

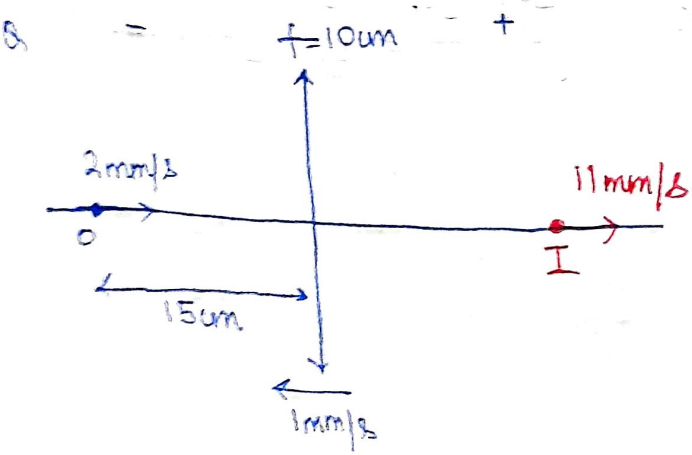
→ wrt lens, O and I will have velocity in same direction



Find velocity of image at this instant.

$$V = \frac{(-10)(-10)}{-10-10} = -5\,cm$$

$$V_i = \frac{(-5)^2}{(-10)^2} \times 1mm/s = \frac{1}{4}\,mm/s$$



Find velocity of image at this instant.

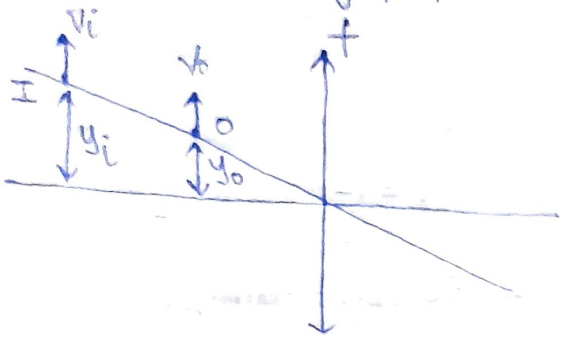
$$V_{il} = \frac{v^2}{u^2}(V_{ol})$$

$$V_{il} = \left(\frac{f}{f+u}\right)^2 (V_{ol})$$

$$= V_i - (-1mm/s) = \left(\frac{10}{10-15}\right)^2 (2-(-1))\ mm/s$$

$$V_i = 11\ mm/s$$

(2) Object is moving perpendicular to principal axis:

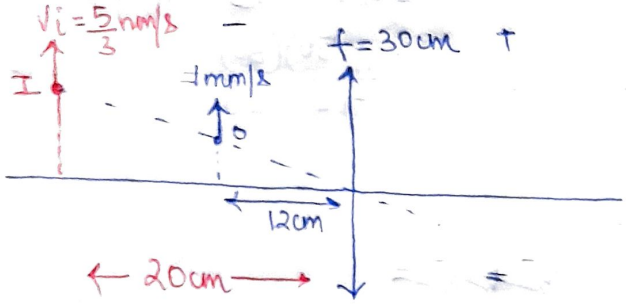


$$m_t = \frac{y_i}{y_o} \Rightarrow y_i = m y_o$$

$$\Rightarrow \frac{dy_i}{dt} = m\frac{dy_o}{dt}$$

$$= \boxed{V_i = mV_o = \frac{v}{u}V_o = \left(\frac{f}{f+u}\right)V_o}$$

Q

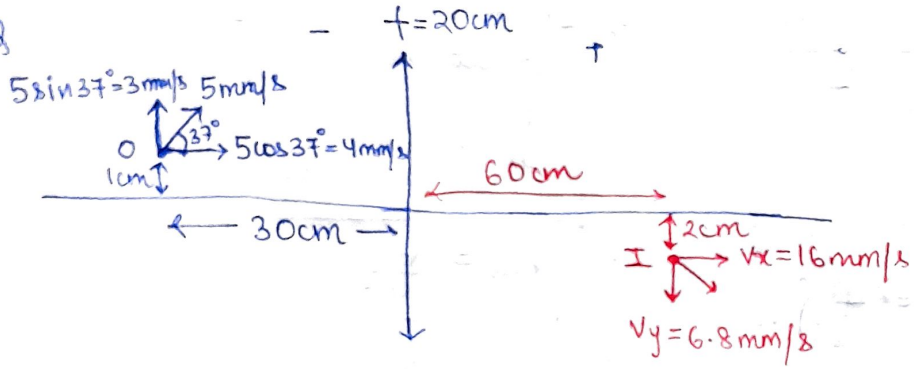


Find velocity of image

$$v = \frac{(-12)(30)}{-12+30} = -20\text{cm}$$

$$V_i = \frac{-20}{-12} \times 1\text{mm/s} = \frac{5}{3}\text{mm/s}$$

*Q



Find velocity of image at this instant

$$v = \frac{uf}{u+f} = \frac{(-30)(20)}{-30+20} = 60\text{cm}$$

$$m = \frac{y_i}{1\text{cm}} = \frac{60}{-30} \Rightarrow y_i = -2\text{cm}$$

Now, $$V_x = \frac{dv}{dt} = \frac{v^2}{u^2}\left(\frac{du}{dt}\right) = \frac{v^2}{u^2}(V_o)_x = \left(\frac{60}{-30}\right)^2 \times 4\text{mm/s} = 16\text{mm/s}$$

Also, $$y_i = my_o = \frac{f}{f+u}y_o$$

$$= \frac{dy_i}{dt} = f\left\{\frac{(f+u)\frac{dy_o}{dt} - y_o\left(\frac{dy}{dt}\right)}{(f+u)^2}\right\} = 20\text{cm} \times \left\{\frac{-100 \times 3\text{mm/s} - 1\text{cm} \times 4\text{mm/s}}{100\text{cm}^2}\right\}$$

$$V_y = -\frac{34}{5}\text{mm/s} = -6.8\text{mm/s}$$

$$\therefore V_I = \sqrt{V_x^2 + V_y^2} = \sqrt{16^2 + 6.8^2}\ \text{mm/s}$$

Graph b/w $\frac{1}{u}$ and $\frac{1}{v}$:

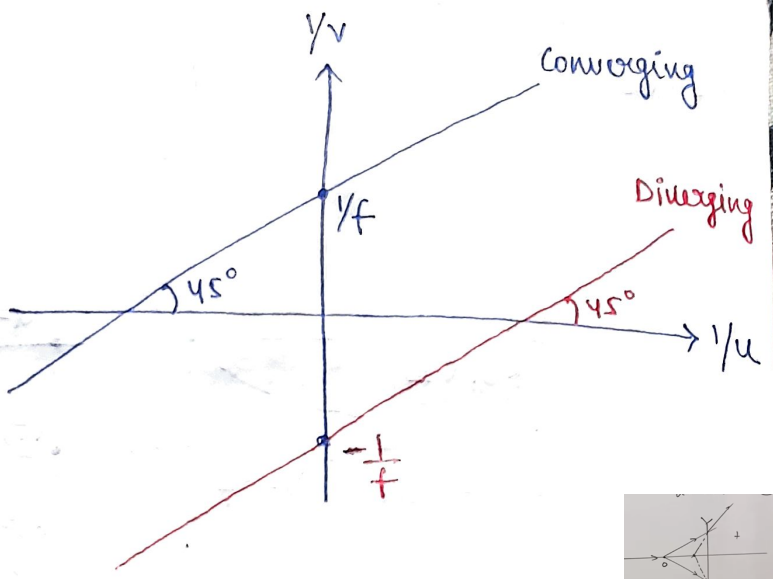
$$\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$$

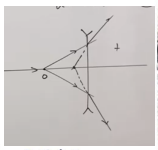
$$\frac{1}{v} = \frac{1}{u} + \frac{1}{f}$$

$$\Rightarrow \boxed{y = x + \frac{1}{f}}$$

For converging $\Rightarrow f = +ve$

For diverging $\Rightarrow f = -ve$





→ From graph we observe that, for diverging lens, for $u = -ve$ we can never get $v = +ve$. So, diverging lens will never form real image of real object.

→ From graph we observe that, for converging lens, for $u = +ve$ we can never get $v = -ve$. So, converging lens will never form virtual image of virtual object.

Graph b/w u and v :

① Converging lens :

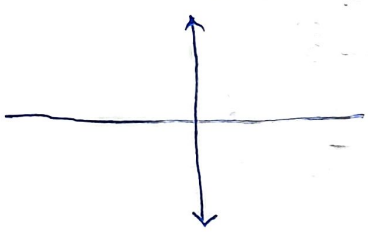
$f = +ve$

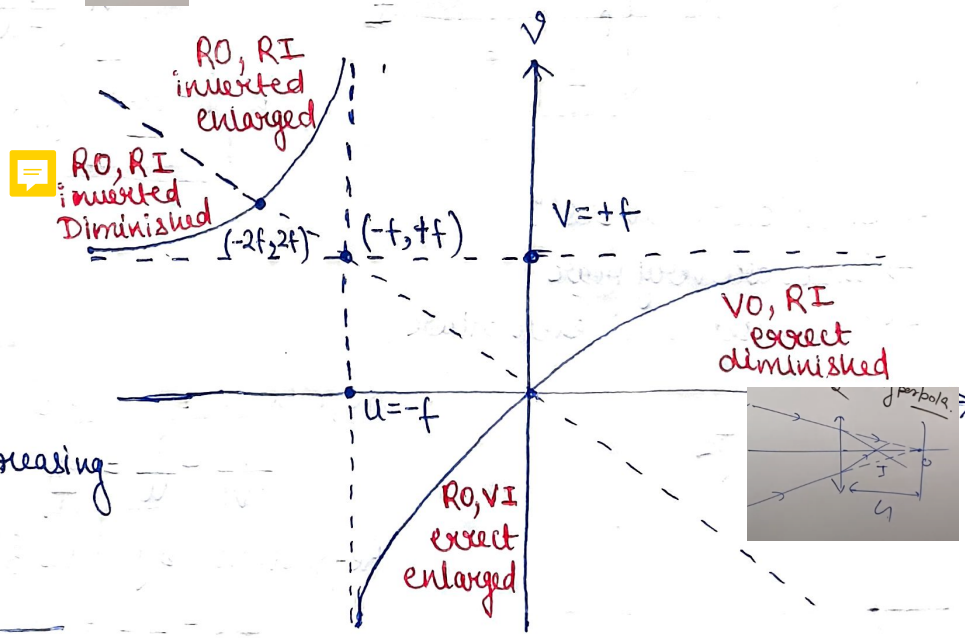
$$v = \frac{uf}{u+f} \quad (u \neq -f)$$

$$\frac{dv}{du} = f\left(\frac{(u+f) - u}{(u+f)^2}\right)$$

$$= \frac{f^2}{(u+f)^2} = +ve$$

∴ Monotonically increasing





We keep objects at different u and find v.

$$\boxed{v = \frac{f}{1 + \frac{f}{u}}}$$

If $u \to -\infty \Rightarrow v = f$

If $u \to -f \Rightarrow v \to \infty$

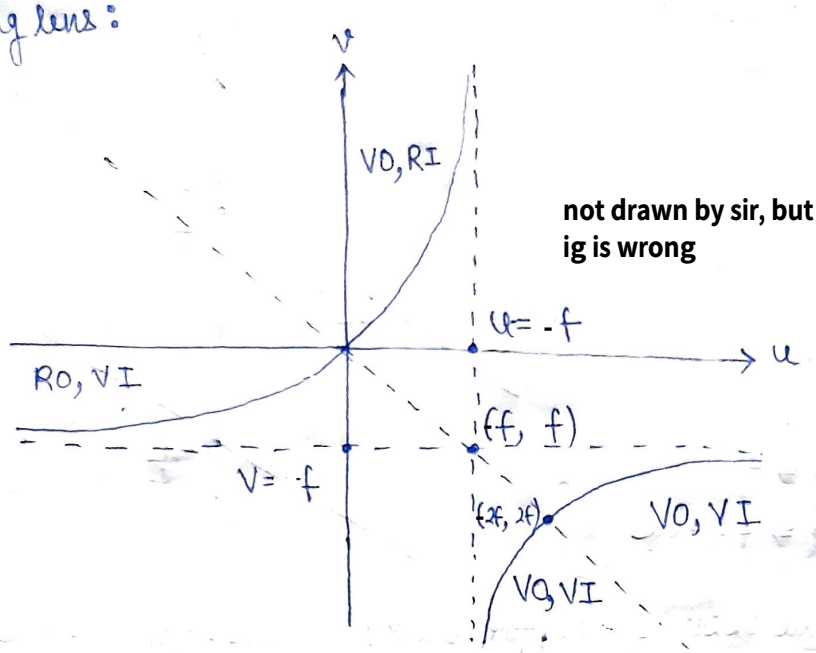
If $u \to \infty \Rightarrow v = f$

If $u \to -f + h \Rightarrow v \to -\infty$

∴ Graph is rectangular hyperbola.

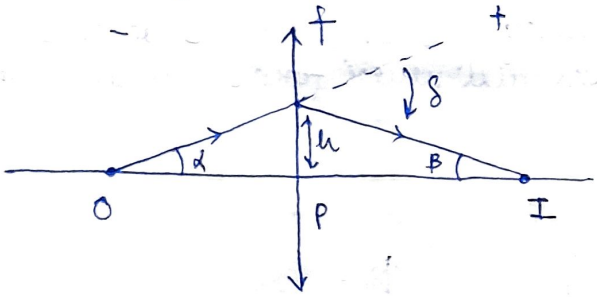
② Diverging lens:

$f = -ve$



not drawn by sir, but by Abhinav
ig is wrong

Deviation by lens:



$\delta = \alpha + \beta$ ($\alpha, \beta \approx$ small)

$= \frac{h}{OP} + \frac{h}{IP} = h\left(\frac{1}{OP} + \frac{1}{IP}\right)$

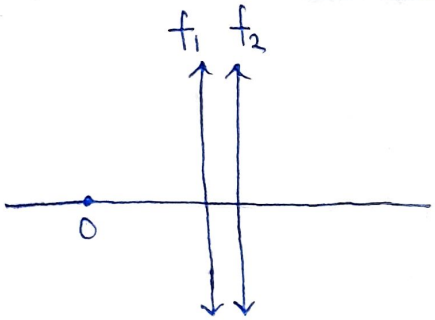
$= h\left(\frac{1}{-u} + \frac{1}{v}\right) = h\left(\frac{1}{v} - \frac{1}{u}\right)$

* $\boxed{\delta = \frac{h}{f}}$ (in radian)

Combination of lenses:

→ lenses are very near

→ or lenses touch each other



Refraction by lens ①

$$\frac{1}{v'} - \frac{1}{u} = \frac{1}{f_1} \quad -①$$

Refraction by lens ②

$$\frac{1}{v} - \frac{1}{v'} = \frac{1}{f_2} \quad -②$$

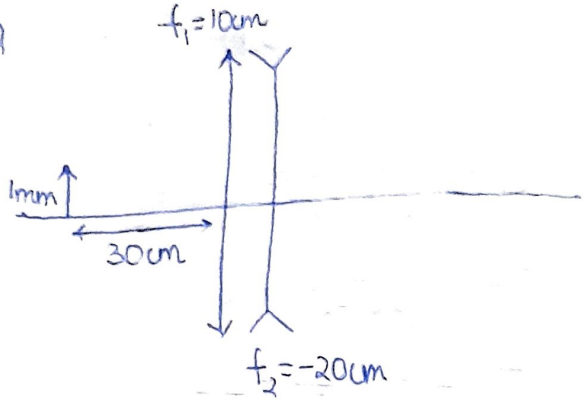
Add ① and ②

* $\boxed{\frac{1}{v} - \frac{1}{u} = \frac{1}{f_1} + \frac{1}{f_2} = \frac{1}{f_{eq}}}$

* $\boxed{\frac{1}{f_{eq}} = \frac{1}{f_1} + \frac{1}{f_2}}$

* $\boxed{P_{eq} = P_1 + P_2}$

Q



Find position and height of image.

$$\frac{1}{f_{eq}} = \frac{1}{10} + \frac{1}{-20} = \frac{1}{20}$$

$$\boxed{f_{eq} = 20\text{cm}}$$

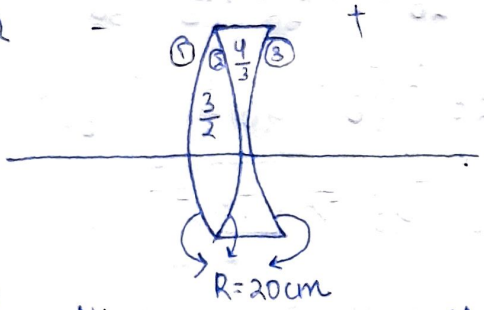
∴ We can replace these lens by a lens of f_{eq} = 20cm.

$$v = \frac{(-30)(20)}{-30+20} = 60\text{cm}$$

$$m = \frac{h_i}{1\text{mm}} = \frac{60}{-30}$$

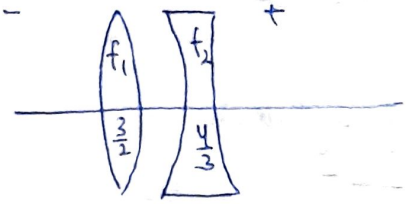
$$\boxed{h_i = -2\text{mm}}$$

Q



Find f_{eq} of this system.

*Method 1: We can assume both lens to be slightly away and apply formula $\frac{1}{f_{eq}} = \frac{1}{f_1} + \frac{1}{f_2}$



$$\frac{1}{f_{eq}} = \frac{1}{f_1} + \frac{1}{f_2}$$

$$\frac{1}{f_{eq}} = \left(\frac{3}{2}-1\right)\left(\frac{1}{20} - \frac{1}{-20}\right) + \left(\frac{4}{3}-1\right)\left(\frac{1}{-20} - \frac{1}{20}\right)$$

$$= \frac{1}{f_{eq}} = \frac{1}{2}\times\frac{2}{20} + \frac{1}{3}\times\frac{-2}{20} = \frac{1}{20} - \frac{1}{30} = \frac{3-2}{60} = \frac{1}{60}$$

$$\therefore \boxed{f_{eq} = 60}$$

Method 2:

Refraction by curved surface ①

$$\frac{n_2 - n_1}{v} - \frac{n_2 - n_1}{u} = \frac{n_2 - n_1}{R}$$

$$= \frac{3}{2v_1} - \frac{1}{u} = \frac{1}{2(20)} \quad -①$$

Refraction by curved surface ②

$$= \frac{4}{3v_2} - \frac{3}{2v_1} = \frac{4/3 - 3/2}{-20}$$

$$= \frac{4}{3v_2} - \frac{3}{2v_1} = \frac{1}{120} \quad -②$$

Refraction by curved surface ③

$$\frac{1}{v} - \frac{4}{3v_2} = \frac{-1/3}{20}$$

$$= \frac{1}{v} - \frac{4}{3v_2} = \frac{-1}{60} \quad -③$$

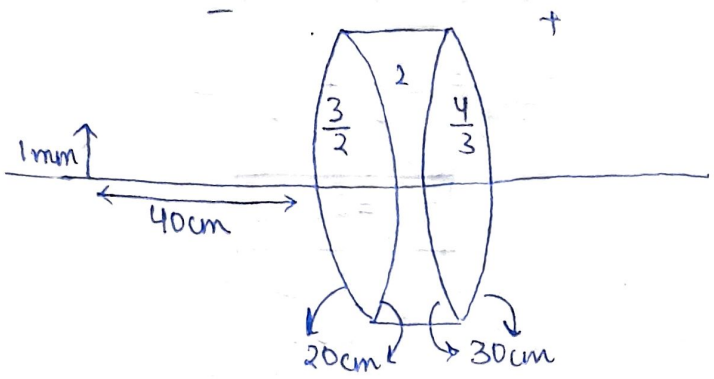
Now add ①, ② and ③

$$\frac{1}{v} - \frac{1}{u} = \frac{1}{40} + \frac{1}{120} - \frac{1}{60} = \frac{3+1-2}{120} = \frac{2}{120} = \frac{1}{60}$$

$$\frac{1}{v} - \frac{1}{u} = \frac{1}{60} \Rightarrow \frac{1}{f} = \frac{1}{60}$$

$$\therefore \boxed{f = 60\text{cm}}$$

Q



Find ① f_{eq} ② position of image ③ height of image.

① $\frac{1}{f_{eq}} = \frac{1}{f_1} + \frac{1}{f_2} + \frac{1}{f_3}$

$= \left(\frac{3}{2}-1\right)\left(\frac{1}{20}-\frac{1}{-20}\right) + (2-1)\left(\frac{1}{-20}-\frac{1}{30}\right) + \left(\frac{4}{3}-1\right)\left(\frac{1}{30}-\frac{1}{-30}\right)$

$= \frac{1}{2}\times\frac{2}{20} + 1\times\frac{-5}{60} + \frac{1}{3}\times\frac{2}{30}$

$= \frac{1}{20} - \frac{5}{60} + \frac{2}{90} = \frac{9-15+4}{180}$

$\frac{1}{f_{eq}} = \frac{-2}{180} = \frac{-1}{90} \Rightarrow \boxed{f_{eq} = -90\,cm}$

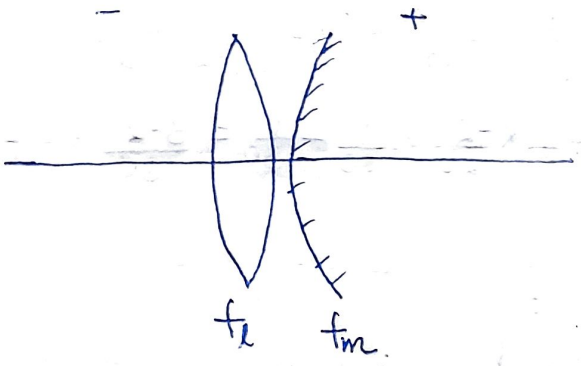
② $v = \frac{uf}{u+f} = \frac{-40\times-90}{-40+90} = \frac{-360}{13}\,cm$

③ $m = \frac{h_i}{1mm} = \frac{+360/13}{+40} = \frac{9}{3}$

$\therefore h_i = \frac{9}{3}\,mm$

Combination of lens and mirror:

If lens and mirror are very near



Refraction by lens:

$\frac{1}{v_1} - \frac{1}{u} = \frac{1}{f_l}$ — ①

Reflection by mirror:

$\frac{1}{v_2} + \frac{1}{v_1} = \frac{1}{f_m}$ — ②

Refraction by lens:

$\frac{1}{v} - \frac{1}{v_2} = -\frac{1}{f_l}$ { -ve as sign convention changes while ray comes back }

Now, ② − ① + ③

$\frac{1}{v} + \frac{1}{u} = \frac{1}{f_m} - \frac{1}{f_l} - \frac{1}{f_l}$

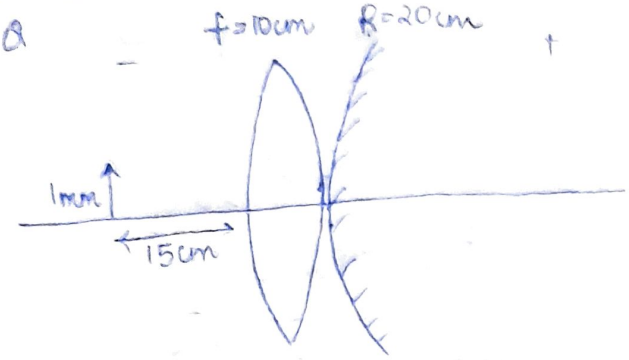
$\Rightarrow \boxed{\frac{1}{v} + \frac{1}{u} = \frac{1}{f_m} - \frac{2}{f_l} = \frac{1}{f_{eq}}}$

↓ Mirror formula

→ Hence, this arrangement will behave like mirror whose f_{eq} will be given by

$\boxed{\frac{1}{f_{eq}} = \frac{1}{f_m} - \frac{2}{f_l}}$

Q



Find position and height of image.

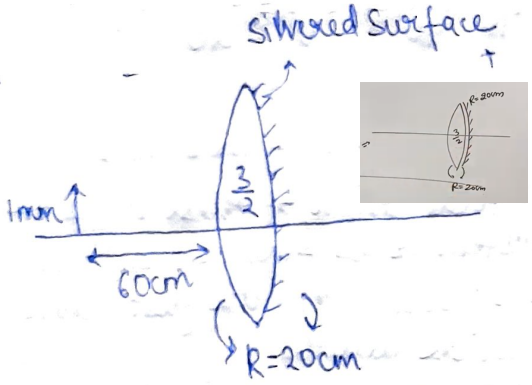
$$\frac{1}{f_{eq}} = \frac{1}{f_m} - \frac{2}{f_l} = \frac{1}{10} - \frac{2}{10}$$

$$\therefore \boxed{f_{eq} = -10\,cm}$$

$$v = \frac{uf}{u-f} = \frac{(-15)(-10)}{-15+10} = -30\,cm$$

$$m = \frac{h_i}{1mm} = -\frac{(-30)}{-15} = -2 \Rightarrow h_i = -2\,mm$$

Q



(Silvered surface can be replaced with mirror)

Find position and height of image.

$$\frac{1}{f_{eq}} = \frac{1}{f_m} - \frac{2}{f_l}$$

$$= \frac{+1}{-10} - 2\left\{\left(\frac{3}{2}-1\right)\left(\frac{1}{20} - \frac{1}{-20}\right)\right.$$

$$= \frac{-1}{10} - 2\times\frac{1}{2}\times\frac{2}{20} = \frac{-1}{10} - \frac{1}{10}$$

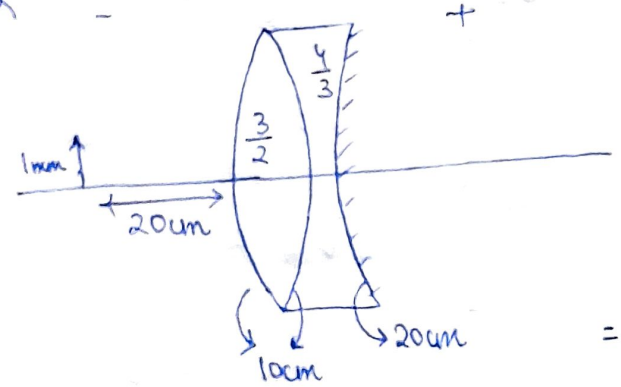
$$\boxed{f_{eq} = -5\,cm}$$

$$v = \frac{uf}{u-f} = \frac{(-60)(-5)}{-60+5} = \frac{-60}{11}\,cm$$

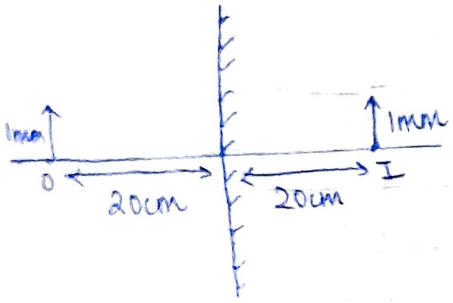
$$m = \frac{h_i}{1mm} = \frac{-\left(\frac{-60}{11}\right)}{-60}$$

$$h_i = -\frac{1}{11}\,mm$$

Q



III



Find position and height of image.

$$\frac{1}{f_{eq}} = \frac{1}{f_m} - \frac{2}{f_l}$$

$$= \frac{1}{f_{eq}} = \frac{1}{f_m} - 2\left\{\frac{1}{f_1} + \frac{1}{f_2}\right\}$$

$$= \frac{1}{10} - 2\left\{\left(\frac{3}{2}-1\right)\left(\frac{1}{10} - \frac{1}{-10}\right) + \left(\frac{4}{3}-1\right)\left(\frac{1}{-10} - \frac{1}{20}\right)\right.$$

$$= \frac{1}{10} - 2\left\{\frac{1}{2}\times\frac{2}{10} + \frac{1}{3}\left(\frac{-3}{20}\right)\right\}$$

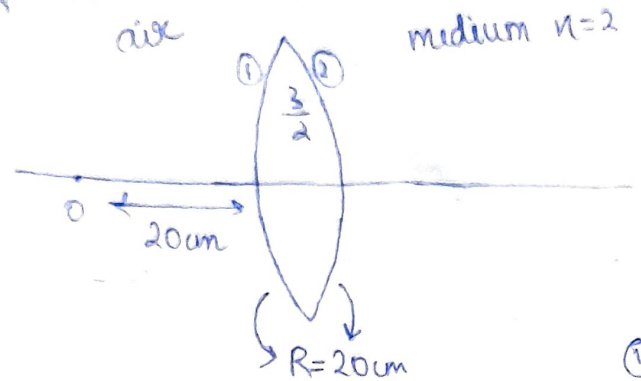
$$= \frac{1}{10} - 2\left\{\frac{1}{10} - \frac{1}{20}\right\} = \frac{1}{10} - 2\left\{\frac{1}{20}\right\}$$

$$\frac{1}{f_{eq}} = \frac{1}{10} - \frac{1}{10} = 0$$

$$\therefore \boxed{f_{eq} \to \infty}$$

∴ Entire arrangement acts as plane mirror.

Q



Find position of final image

Refraction by surface ①:

$$\frac{3}{2v_1} - \frac{1}{-20} = \frac{1/2}{20} \quad -①$$

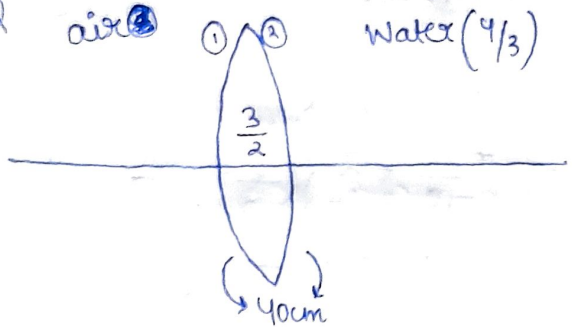
Refraction by surface ②:

$$\frac{2}{v} - \frac{3}{2v_1} = \frac{1/2}{-20} \quad -②$$

① + ②

$$\Rightarrow \frac{2}{v} + \frac{1}{20} = \frac{1}{40} - \frac{1}{40} \Rightarrow \frac{2}{v} = -\frac{1}{20} \Rightarrow \boxed{v = -40\text{cm}}$$

* Q



Find focal length of this lens,

Note: Focal length of these types of system is as such not defined. Here by focal length we just mean that where will the rays coming from ∞ parallel to principal axis meet.

Case ①: Rays from ∞, coming parallel to principal axis strike surface ① first.

S①: $\frac{3}{2v_1} - \frac{1}{\infty} = \frac{1/2}{40} \quad -①$

S②: $\frac{4}{3v} - \frac{3}{2v_1} = \frac{4/3 - 3/2}{40} = \frac{1}{240} \quad -②$

Now, ① + ②: $\frac{4}{3v} - \frac{1}{\infty} = \frac{1}{80} + \frac{1}{240}$

$\Rightarrow \frac{4}{3v} = \frac{4}{240} \Rightarrow \boxed{v = 80\text{cm}}$

↳ Focal length

Case ②: Rays from ∞, coming parallel to principal axis strike surface ② first.

S②: $\frac{3}{2v_1} - \frac{4}{3\times\infty} = \frac{\frac{3}{2} - \frac{4}{3}}{40} = \frac{1}{240} \quad -①$

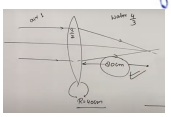
S①: $\frac{1}{v} - \frac{3}{2v_1} = \frac{-1/2}{-40} = \frac{1}{80} \quad -②$

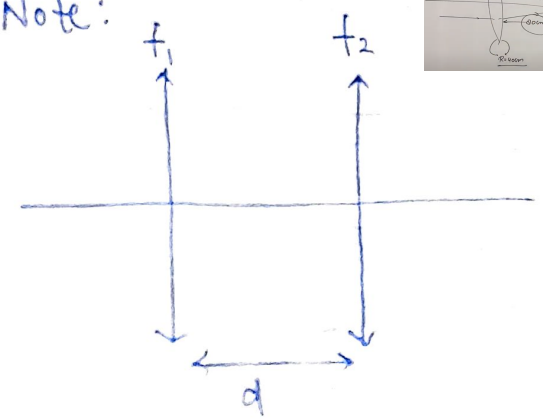
Now ① + ②: $\frac{1}{v} - \frac{4}{\infty} = \frac{1}{240} + \frac{1}{80}$

$\Rightarrow \frac{1}{v} = \frac{4}{240} \Rightarrow \boxed{v = 60\text{cm}}$

↳ focal length

* Note:





$$\boxed{\frac{1}{f_{eq}} = \frac{1}{f_1} + \frac{1}{f_2} - \frac{d}{f_1 f_2}}$$

Do not use this formula untill required, use traditional method!

Dispersion:

→ Angular splitting of white light into its constituents (VIBGYOR) is known as dispersion.

→ Reason: Speed of light in a medium (other than vacuum) depends on wavelength of light

→ $n = \frac{c}{v}$, refractive index of a medium depends on wavelength of light too.

Cauchy's formula: $\boxed{n(\lambda) = a + \frac{b}{\lambda^2}}$

In vacuum, $a = 1$ and $b = 0 \Rightarrow n = 1$ (for any wavelength)

In any other medium, $b = +ve \Rightarrow \boxed{n \propto \frac{1}{\lambda}}$ $\lambda \uparrow\ n \downarrow$ $\lambda \downarrow\ n \uparrow$

VIBGYOR

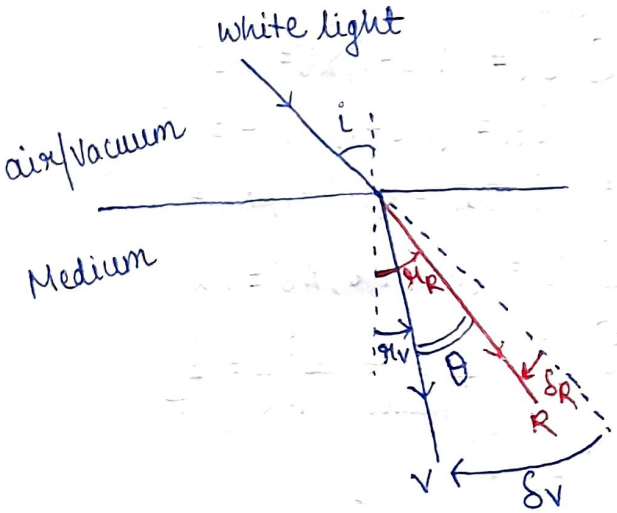
$\lambda \uparrow$

$n \downarrow$

$\therefore \lambda_R > \lambda_V$

But $n_V > n_R$ (for any medium)

$\boxed{n_{yellow} = n_{avg} = \frac{n_V + n_R}{2}}$

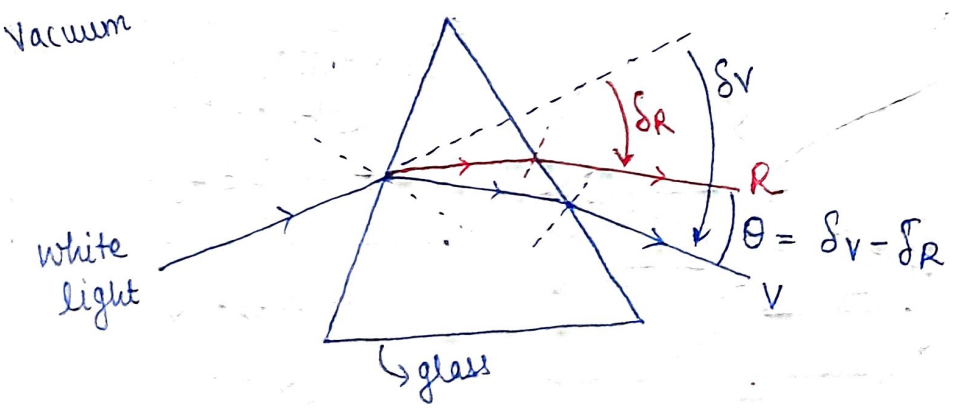


$1 \times \sin i = n \sin r$

$\boxed{\sin r = \frac{\sin i}{n}}$

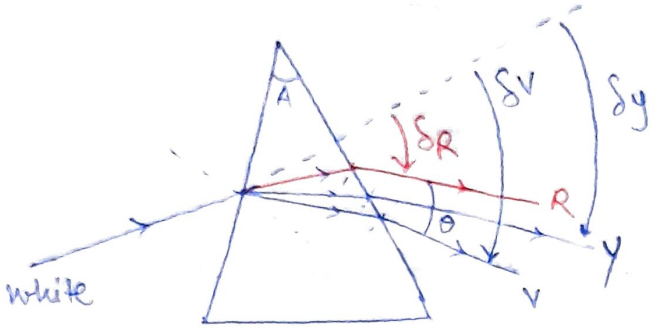
θ = Angle of dispersion.

$\boxed{\theta = \delta_V - \delta_R}$



$\theta = \delta_V - \delta_R$ = angle of dispersion

Dispersion in small angled prism :



$$\delta_R = (n_R - 1)A$$
$$\delta_V = (n_V - 1)A$$

$$\boxed{\theta = \delta_V - \delta_R = (n_V - n_R)A}$$

$$\boxed{\delta_y = \delta_{mean} = (n_y - 1)A}$$

If n_y not given then $n_y = \frac{n_R + n_V}{2}$

Refractive index of prism, for violet $= n_V$
for red $= n_R$
for yellow $= n_y$

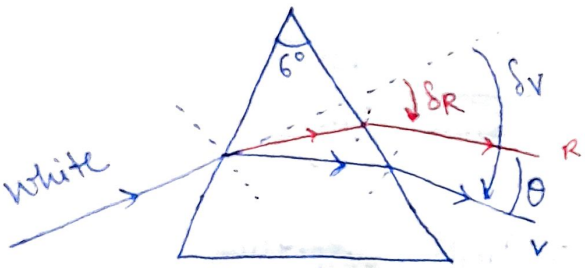
Dispersive Power (ω) :

$$\boxed{\omega = \frac{\theta}{\delta_{mean}} = \frac{(n_V - n_R)A}{(n_y - 1)A} = \frac{n_V - n_R}{n_y - 1}}$$

- → Ratio
- → Unitless, dimensionless
- → property of medium only

Also, $\boxed{\omega = \frac{dn}{n-1}}$

Q



$n_V = 1.8$
$n_R = 1.6$
$n_y = \frac{1.8 + 1.6}{2} = 1.7$

Find

① $\delta_V = (1.8 - 1) \times 6^\circ = 4.8^\circ$

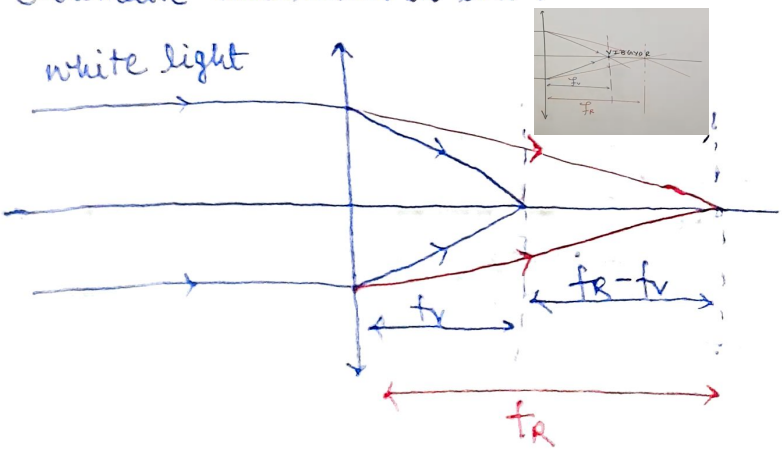
② $\delta_R = (1.6 - 1) \times 6^\circ = 3.6^\circ$

③ $\delta_{mean} = (1.7 - 1) \times 6^\circ = 4.2^\circ$

④ $\theta = (1.8 - 1.6) \times 6^\circ = 1.2^\circ$

⑤ $\omega = \frac{\theta}{\delta_{mean}} = \frac{1.2}{4.2} = \frac{2}{7}$

Chromatic aberration in lens :



Longitudinal chromatic aberration $= -df = f_R - f_V$

$n_V \uparrow \Rightarrow f_V \downarrow$
$n_R \downarrow \Rightarrow f_R \uparrow$

Now, $\frac{1}{f} = (n-1)\left\{\frac{1}{R_1} - \frac{1}{R_2}\right\}$ — ①

Diff both sides

$-\frac{df}{f^2} = dn\left\{\frac{1}{R_1} - \frac{1}{R_2}\right\}$ — ②

Now, Divide ① and ②

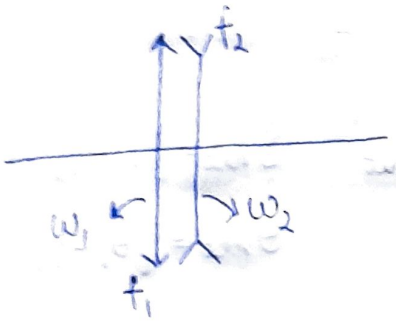
$\frac{-df}{f} = \frac{dn}{n-1} = \omega$

$\therefore * \boxed{df = -\omega f}$

(f is f_{avg} or f_y)

How to remove chromatic aberration?

→ Use combination of lenses of opposite nature



$\frac{1}{f_{eq}} = \frac{1}{f_1} + \frac{1}{f_2}$

Diff both sides

$-\frac{df_{eq}}{f_{eq}^2} = -\frac{df_1}{f_1^2} - \frac{df_2}{f_2^2} \Rightarrow \frac{df_{eq}}{f_{eq}^2} = \frac{df_1}{f_1^2} + \frac{df_2}{f_2^2}$

$= \frac{df_{eq}}{f_{eq}^2} = \frac{-\omega_1 f_1}{f_1^2} + \frac{-\omega_2 f_2}{f_2^2} = \frac{-\omega_1}{f_1} - \frac{\omega_2}{f_2}$

Now, for no chromatic aberration, $df_{eq} = 0$

$\Rightarrow \frac{\omega_1}{f_1} = \frac{-\omega_2}{f_2} \Rightarrow * \boxed{\frac{f_2}{f_1} = \frac{-\omega_2}{\omega_1}}$

Q Given $f_R = 100cm$, $f_v = 98cm$, $f_y = 99cm$

Find dispersive power of lens.

We know, $\frac{1}{f} = (n-1)\underbrace{\left(\frac{1}{R_1} - \frac{1}{R_2}\right)}_{\text{let } k \text{ (say)}}$

Red: $\frac{1}{100} = (n_R - 1)k$ — ①

Violet: $\frac{1}{98} = (n_v - 1)k$ — ②

Yellow: $\frac{1}{99} = (n_y - 1)k$ — ③

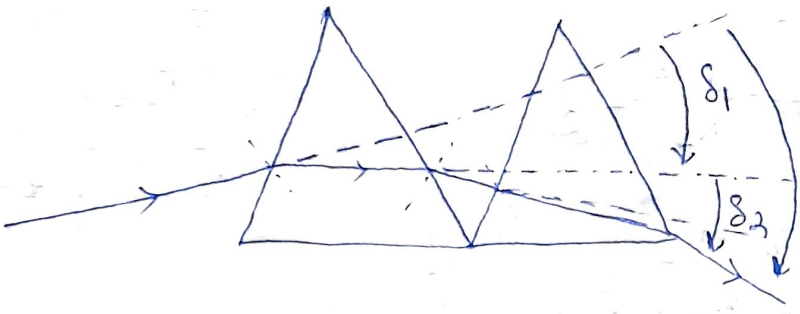
$\frac{② - ①}{③} = \frac{k(n_v - n_R)}{k(n_y - 1)} = \frac{n_v - n_R}{n_y - 1} = \omega$ (RHS)

$\omega = \frac{\frac{1}{98} - \frac{1}{100}}{\frac{1}{99}} = \frac{\frac{2}{98 \times 100}}{1/99}$

$\omega = \frac{2 \times 99}{98 \times 100} = \frac{99}{4900}$

Combination of prism:

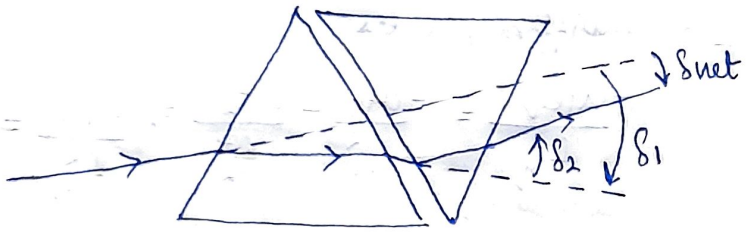
①



$\delta_{net} = \delta_1 + \delta_2$

$\theta_{net} = \theta_1 + \theta_2$

②

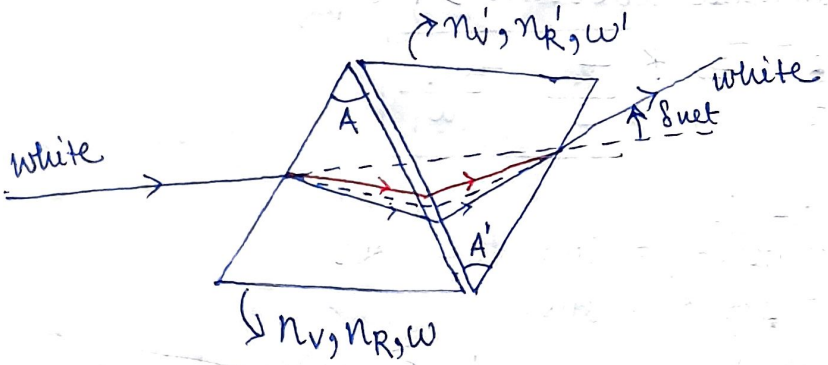


$\delta_{net} = |\delta_1 - \delta_2|$

$\theta_{net} = |\theta_1 - \theta_2|$

Achromatic Combination:

→ Deviation without dispersion ($\delta_{net} \neq 0$, $\theta_{net} = 0$)



$\theta_{net} = 0$

$\theta_1 - \theta_2 = 0$

$(n_v - n_R)A - (n_v' - n_R')A' = 0$

$$\frac{A'}{A} = \frac{(n_v - n_R)}{(n_v' - n_R')}$$

↳ Condition for Achromatic combination.

Now, $\omega = \frac{\theta_1}{\delta_1}$ $\omega' = \frac{\theta_2}{\delta_2}$

But $\theta_1 = \theta_2 \Rightarrow \omega = \frac{\theta}{\delta_1}$ $\omega' = \frac{\theta}{\delta_2}$

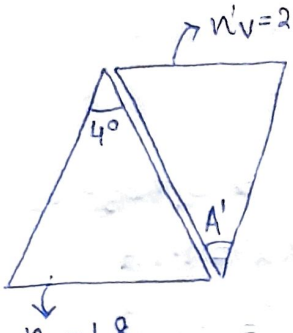
Now, $\delta_{net} = |\delta_1 - \delta_2|$

$$= \left|\frac{\theta}{\omega} - \frac{\theta}{\omega'}\right| = \theta\left|\frac{1}{\omega} - \frac{1}{\omega'}\right|$$

→ If two prisms are of different material then $\omega \neq \omega' \Rightarrow \delta_{net} \neq 0$

$\theta_{net} = 0$

Q



① Find A' for achromatic combination
② Also find δ_{net}

① $\theta_{net} = \theta_1 - \theta_2 = 0$

$= 0.1 \times 4° - (0.2)A' = 0$

$\boxed{A' = 2°}$

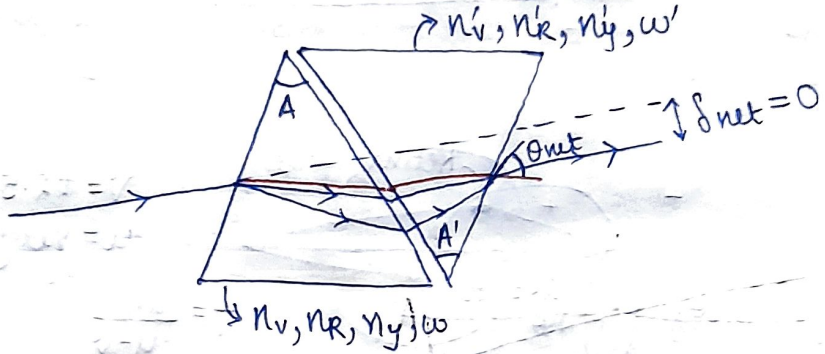
② $n_y = \frac{1.8+1.7}{2} = 1.75$ $\quad n'_y = \frac{2+1.8}{2} = 1.9$

$\delta_{net} = |\delta_1 - \delta_2| = |(1.75-1)4° - (1.9-1)2°|$

$\delta_{net} = |3° - 1.8°| = 1.2°$

Direct Vision Combination:

→ Dispersion without deviation ($\theta_{net} \neq 0$, $\delta_{net} = 0$)



$\delta_{net} = 0$

$\Rightarrow |\delta_1 - \delta_2| = 0$

$= (n_y - 1)A - (n'_y - 1)A' = 0$

$= \boxed{\frac{A'}{A} = \frac{n_y - 1}{n'_y - 1}}$

↳ Condition for direct vision combination

$\omega = \frac{\theta_1}{\delta_1}$ $\quad \omega' = \frac{\theta_2}{\delta_2}$

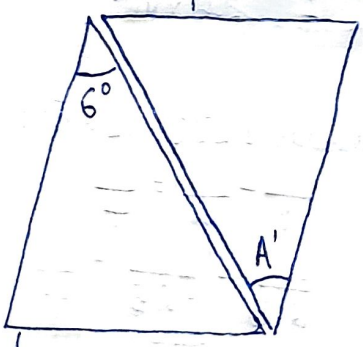
But $\delta_1 = \delta_2 \Rightarrow \omega = \frac{\theta_1}{\delta}$ $\quad \omega' = \frac{\theta_2}{\delta}$

Now, $\theta_{net} = |\theta_1 - \theta_2| = |\omega\delta - \omega'\delta|$

$\theta_{net} = \delta|\omega - \omega'|$

→ If prisms are of different medium then $\omega \neq \omega' \Rightarrow \theta_{net} \neq 0$, $\delta_{net} = 0$

Q



① Find A' for DVC
② Find θ_{net}

① $\delta_1 = \delta_2$

$(1.4-1)6° = (1.75-1)A'$

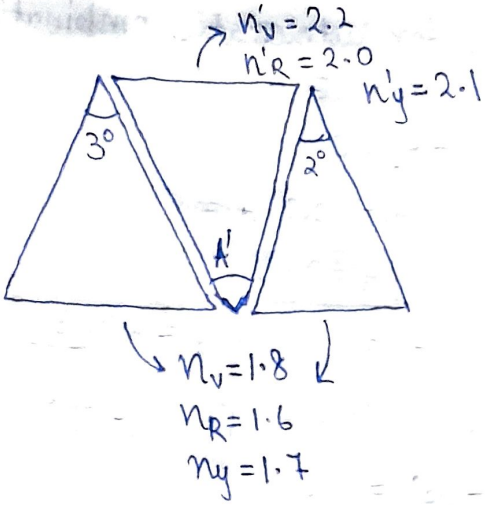
$= 2.4 = 0.75A'$

$= A' = \left(\frac{2.4}{0.75}\right)° = 3.2°$

② $\theta_{net} = |\theta_1 - \theta_2| = |0.2 \times 6° - 0.1 \times 3.2|$

$= |1.2 \times 0.32| = 0.88°$

Q



① Find A' for DVC
② Find θ_{net}.

① $\delta_{net} = 0$

$\delta_1 - \delta_2 + \delta_3 = 0 \Rightarrow \delta_1 + \delta_3 = \delta_2$

$= 0.7 \times 3° + 0.7 \times 2° = 1.1\,A'$

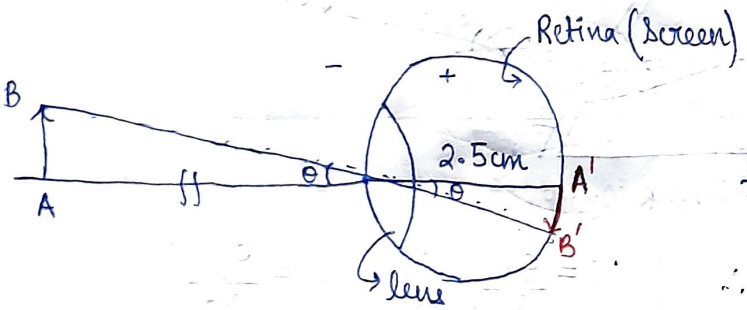
$A' = \left(\frac{35}{11}\right)° = 3.18°$

② $\theta_{net} = \theta_1 - \theta_2 + \theta_3$

$= 0.2 \times 3° - \left(0.2 \times \frac{35}{11}\right) + 0.2 \times 2° = 0.36°$

Optical Instruments

Human Eye:



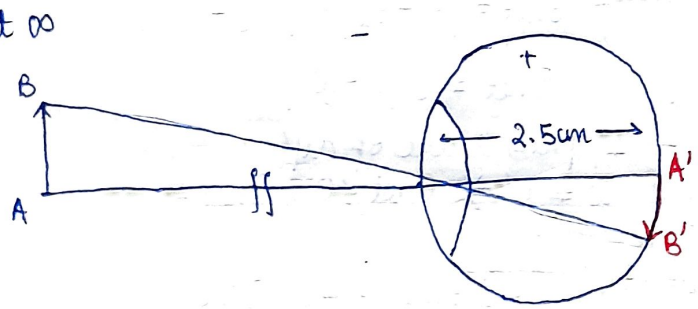
$v = +2.5\,cm$ (fixed)
u = varying.

$f = \frac{uv}{u-v}$

$\therefore \boxed{f = \frac{2.5u}{u-2.5}\,cm}$

① Case when eyes are least strained (relaxed):

at ∞



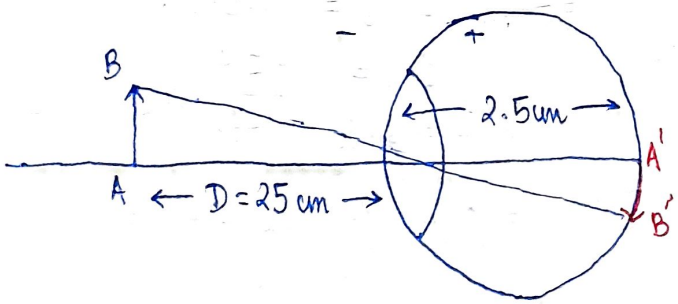
$\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$

$\Rightarrow \frac{1}{2.5} - \frac{1}{\infty} = \frac{1}{f}$

$\boxed{f = 2.5\,cm}$

$P = \frac{1}{f} = \frac{1}{2.5 \times 10^{-2}} = 40\,D$

② Object at least distance for clear vision:
↳ eyes are most stressed at LDCV



Generally LDCV is 25cm.

$\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$

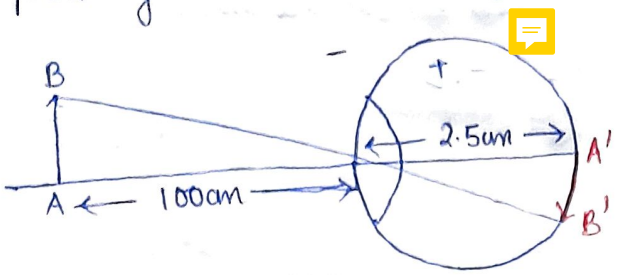
$= \frac{1}{2.5} - \frac{1}{-25} = \frac{1}{f}$

$\boxed{f = \frac{25}{11}\,cm}$

$P = \frac{1}{f} = \frac{1}{\frac{25}{11} \times 10^{-2}} = 44\,D$

→ Those who dont use spectacles have eye lens power varying from 40D to 44D. This phenomenon is known as accomodation.

Q If human eye observes an object 1m away from itself, then find focal length and power of eye lens.



$$\frac{1}{v} - \frac{1}{u} = \frac{1}{f} \Rightarrow \frac{1}{2.5} - \frac{1}{-100} = \frac{1}{f}$$

$$\frac{1}{f} = \frac{40+1}{100} = \frac{41}{100} \Rightarrow f = \frac{100}{41}\,cm$$

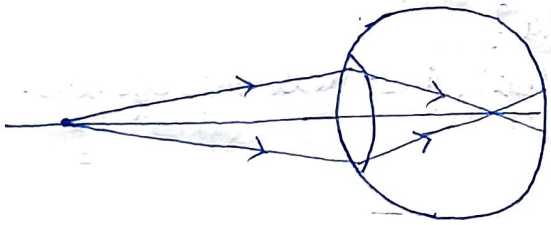
$$\therefore \boxed{f = \frac{1}{41}\,m}$$

$$P = \frac{1}{f} = \frac{1}{1/41} = 41D$$

Defects in human eyes:

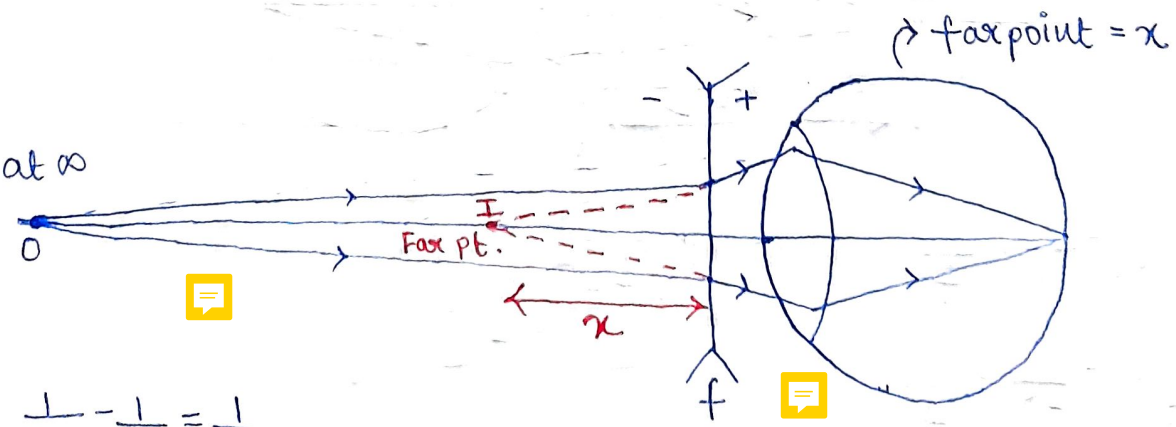
① Myopia:

→ Short-sightedness / Near sightedness
→ Unable to see objects clearly if they are very far.
→ For a person suffering from myopia, far point 'x' is less than infinity.
→ For far away points, an eye having myopia defect will converge the rays before retina.



How to remove myopia?

→ Use a diverging lens, just near our eye lens (concave lens)



$$\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$$

$$= \frac{1}{-x} - \frac{1}{\infty} = \frac{1}{f}$$

$$\therefore \boxed{f = -x}$$

→ A person suffering from myopia having far point at distance x from eye, should use a lens of focal length, $f = -x$ (concave lens)

Q A boy suffering from myopia cant see object clear if it is more than 100cm away from eyes. Find power of lens he should use.

$f = -x$

$\therefore f = -100cm = -1m$

$P = \frac{1}{f} = \frac{1}{-1} = -1D$

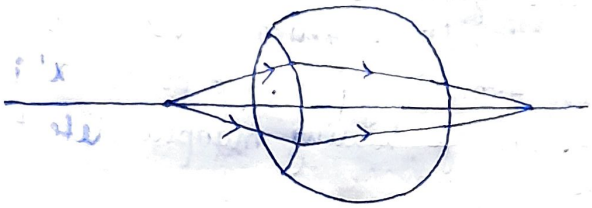
② Hypermetropia:

→ Far sightedness / long-sightedness

→ Their near point (LDCV) (x) is more than usual ie 25cm.

→ They cant see clear object if object is nearer than x (near point).

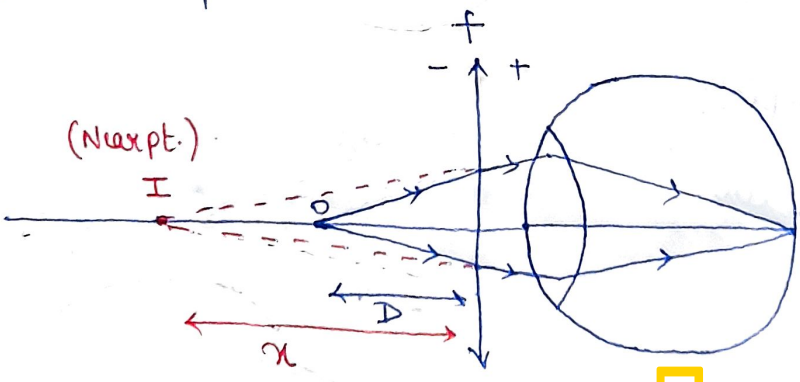
→ For any near object, eye lens will converge the rays beyond retina.



How to remove hypermetropia?

→ Use a converging lens (convex lens), just near our eye lens.

D: LDCV for normal person.



$\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$

$= \frac{1}{-x} - \frac{1}{-D} = \frac{1}{f}$

$= \frac{1}{f} = \frac{1}{D} - \frac{1}{x}$

$\therefore \boxed{f = \frac{Dx}{x - D}}$

Q A man is suffering from hypermetropia and he cant see object nearer than 50cm. Find focal length and power of lens he should use.

$D = 25cm$
$x = 50cm$

$$f = \frac{xD}{x-D} = \frac{50 \times 25}{50-25} = 50cm = 0.5m$$

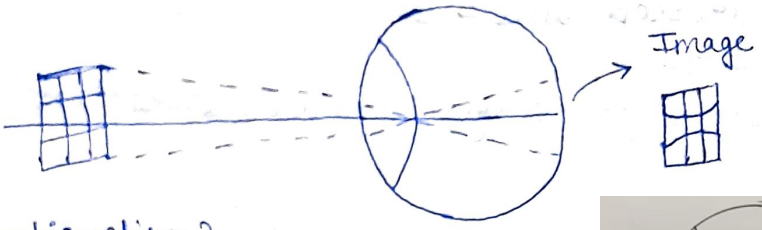
$$P = \frac{1}{f} = \frac{1}{0.5} = +2D$$

③ Presbyopia:

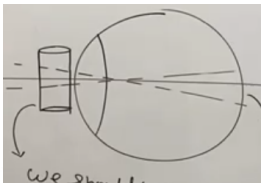
→ An old age problem
→ eye muscles are not strong enough
→ Focal length of eye lens cant adjust much.
→ He is suffering from myopia as well as hypermetropia.
→ His near point is beyond 25cm and far point is less than ∞.

④ Astigmatism:

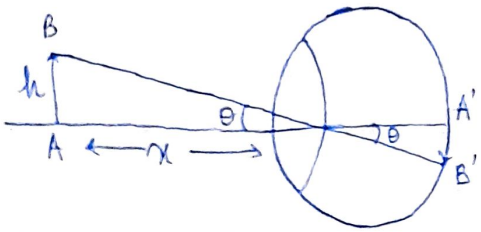
→ Eyes are not perfectly spherical
→ Eye sphere is little distorted.
→ Radius of curvature of eye lens will not be same in orthogonal directions
→ Focal length of eye lens will not be same in orthogonal directions.



How to remove astigmatism?
→ We should use cylindrical lenses.



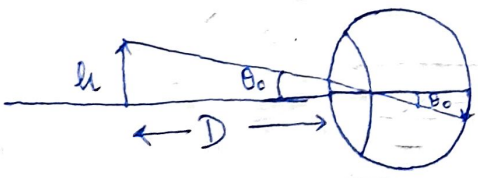
Visual angle:



Visual Angle = θ

$$\theta = \frac{h}{x} \text{ (radian)}$$

→ Max. visual angle is when object is at LDCV.



$$\theta_0 = \frac{h}{D}$$

↳ Max^m visual angle with naked eyes.

→ We use optical instrument so that visual angle $\theta > \theta_0$.

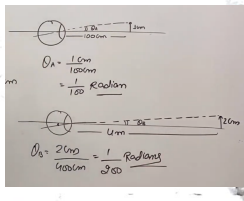
Angular magnification (m) $= \frac{\theta}{\theta_0}$

$m = \frac{\text{visual angle using optical instrument}}{\text{max}^m \text{ visual angle with naked eyes}}$

$m > 1$

Q An object A of height 1cm is 1m away from eye and another object B of height 2cm is 4m away from eye. Which object looks larger to our eyes?

$\theta_A = \frac{1\text{cm}}{100\text{cm}} = \frac{1}{100}$ Radian $\qquad \theta_B = \frac{2\text{cm}}{400\text{cm}} = \frac{1}{200}$ Radian



As $\theta_A > \theta_B \Rightarrow$ Object A looks larger.

Simple Microscope (Magnifier):

→ A convex lens having small focal length.

→ Using microscope we can increase visual angle made by object.

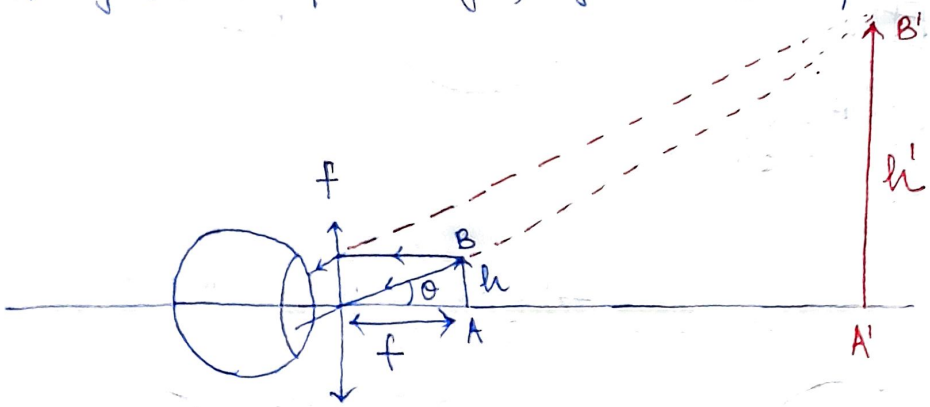
(I) Normal Adjustment:

→ In this case eyes are least strained / most relaxed

→ Object should be placed at focal length from simple microscope.

→ Simple microscope will form image very far which will act like object for eye.

→ As object is at ∞ for our eyes, eyes are relaxed / least strained.



$\theta = \frac{h}{f}$

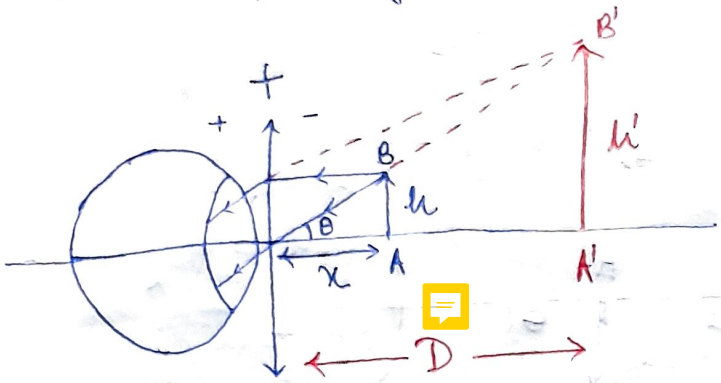
$\therefore$ **$m = \frac{D}{f}$**

↳ As $f \downarrow$, $m \uparrow$

$m = \frac{\theta}{\theta_0} = \frac{h/f}{h/D} = \frac{D}{f}$

(II) Maximum Magnification:

→ In this case, our eyes are most strained / least relaxed



$$\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$$

$$\Rightarrow \frac{1}{-D} - \frac{1}{-x} = \frac{1}{f}$$

$$\Rightarrow \frac{1}{f} + \frac{1}{D} = \frac{1}{x}$$

$$\boxed{x = \frac{fD}{f+D}}$$

$$\boxed{\theta = \frac{h}{x}}$$

$$m = \frac{\theta}{\theta_0} = \frac{h/x}{h/D} = \frac{D}{x} = \frac{D(f+D)}{fD}$$

$$= \frac{f+D}{f} = 1 + \frac{D}{f}$$

$$\therefore \boxed{m_{(max)} = 1 + \frac{D}{f}}$$

→ Object should be placed at $x = \frac{fD}{f+D}$.

→ Image by lens which is object for eye will form at D. (LDCV).

→ Due to this eyes are more stressed.

Q We use a magnifier of power 10D.

① For normal adjustment where should object be placed and what will be the magnification.

② For maximum magnification where should object be placed and what will be the magnification.

① $P = \frac{1}{f} \Rightarrow f = \frac{1}{P} = \frac{1}{10} = 0.1\,m = 10\,cm$

∴ Object should be kept at 10cm from lens.

$$m = \frac{D}{f} = \frac{25}{10} = 2.5\,X$$

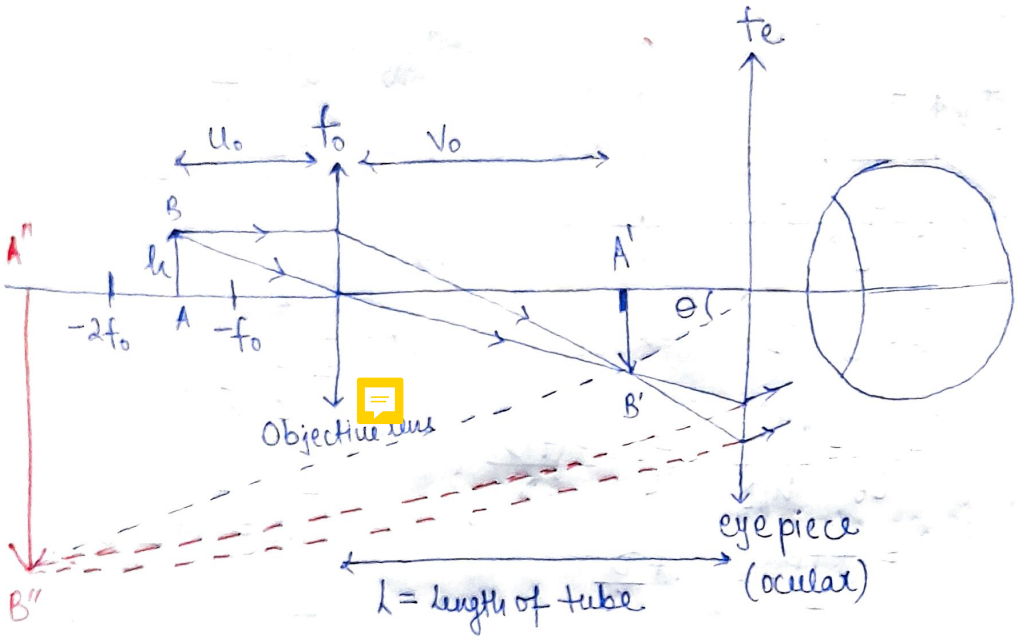
② $x = \frac{fD}{f+D} = \frac{10 \times 25}{10+25} = \frac{10 \times 25^{5}}{35_{7}} = \frac{50}{7}\,cm$

∴ Object should be kept at $\frac{50}{7}$ cm from lens.

$$m = 1 + \frac{D}{f} = 1 + \frac{25}{10} = 3.5\,X$$

Compound Microscope:

→ It is combination of two convex lenses with small focal lengths.



→ Objective lens will have smaller aperture and focal length as compared to eyepiece.

→ A'B' : Image for objective lens
Object for eye piece

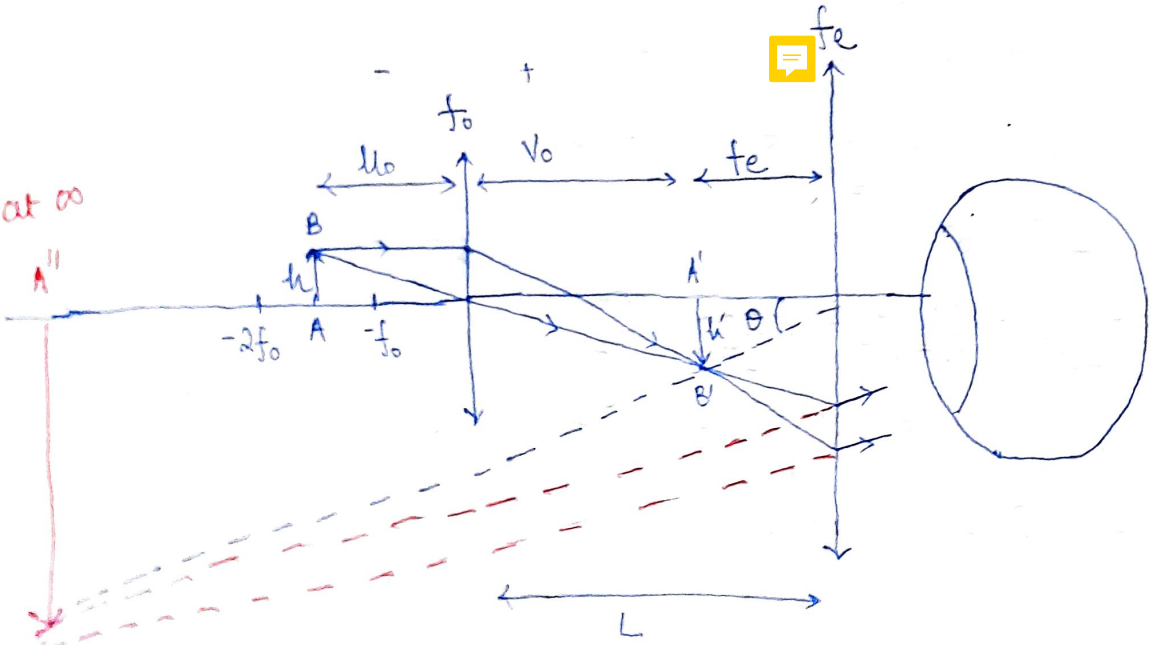
$$m = \frac{\theta}{\theta_o}$$

→ In effect eyepiece is acting like simple microscope.

→ Final image is virtual, enlarged and inverted.

(I) Normal Adjustment:

→ In this case eyes are relaxed.



We can find magnification in two ways:

M-1

$$\frac{1}{v_0} - \frac{1}{-u_0} = \frac{1}{f_0}$$

$$= \frac{1}{v_0} = \frac{1}{f_0} - \frac{1}{u_0}$$

$$\therefore \star \boxed{v_0 = \frac{f_0 u_0}{u_0 - f_0}}$$

Now,

$$m_t = \frac{v}{u} = \frac{v_0}{-u_0} = \frac{f_0}{f_0 - h_0} = \frac{h'}{h}$$

$$\therefore \boxed{h' = \frac{f_0 h}{f_0 - u_0}}$$

$$\theta = \frac{A'B'}{f_e} = \frac{h'}{f_e} = \frac{f_0 h}{(f_0 - u_0) f_e}$$

$$m = \frac{\theta}{\theta_0} = \frac{f_0 h / (f_0 - u_0) f_e}{h/D}$$

$$\therefore \star \boxed{m = \frac{f_0 D}{(f_0 - u_0) f_e}}$$

$$\star \boxed{L = v_0 + f_e = \left(\frac{f_0 u_0}{u_0 - f_0}\right) + f_e}$$

M-2

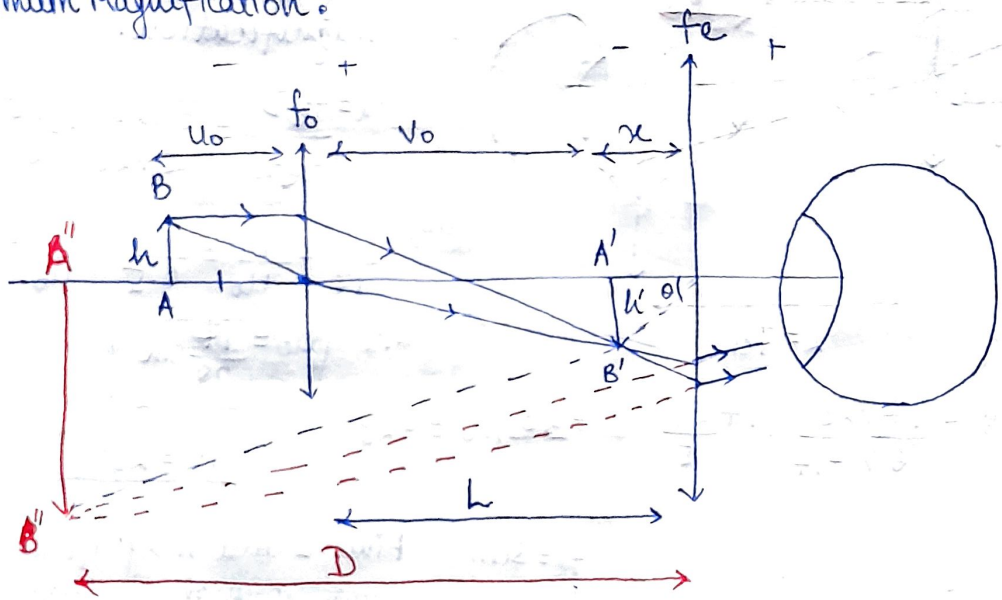
$$m = m_t \times m_\theta$$

$$= -\frac{v_0}{u_0} \times \frac{D}{f_e} = \frac{f_0}{(u_0 - f_0)} \frac{D}{f_e}$$

$$\star \boxed{m = \frac{v_0}{u_0} \frac{D}{f_e}}$$

$$\boxed{L = v_0 + f_e}$$

(II) Maximum Magnification:



$$\frac{1}{v_0} - \frac{1}{-u_0} = \frac{1}{f_0}$$

$$= \frac{1}{v_0} = \frac{1}{f_0} - \frac{1}{u_0}$$

$$\therefore \boxed{v_0 = \frac{f_0 u_0}{v_0 - f_0}}$$

$$\frac{1}{-D} - \frac{1}{-x} = \frac{1}{f_e}$$

$$= \frac{1}{x} = \frac{1}{f_e} + \frac{1}{D}$$

$$\boxed{x = \frac{f_e D}{f_e + D}}$$

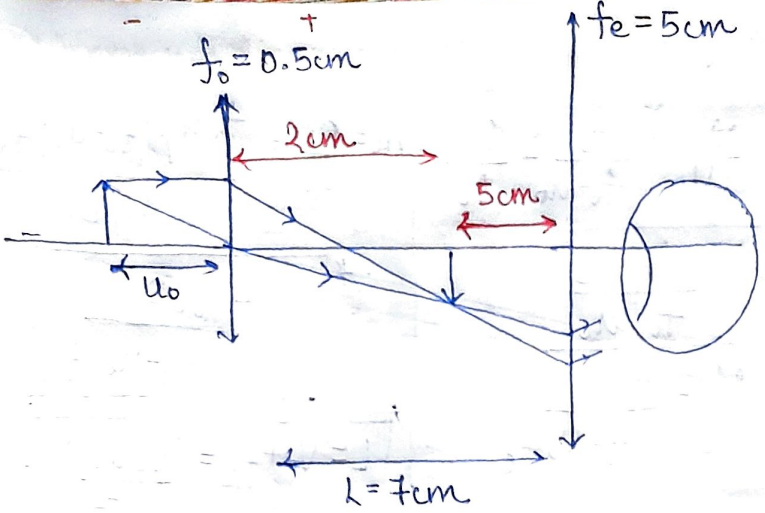
$$m = m_t \times m_\theta = \frac{v_0}{-u_0}\left(1 + \frac{D}{f_e}\right)$$

$$\boxed{m = \frac{f_0}{f_0 - u_0}\left(1 + \frac{D}{f_e}\right)}$$

$$\boxed{L = v_0 + x = v_0 + \frac{f_e D}{f_e + D}}$$

$$= \frac{f_0 u_0}{u_0 - f_0} + \frac{f_e D}{f_e + D}$$

Q



Find distance of object from objective lens for normal adjustment. Also find magnification.

$$\frac{1}{2} - \frac{1}{-u_o} = \frac{1}{0.5}$$

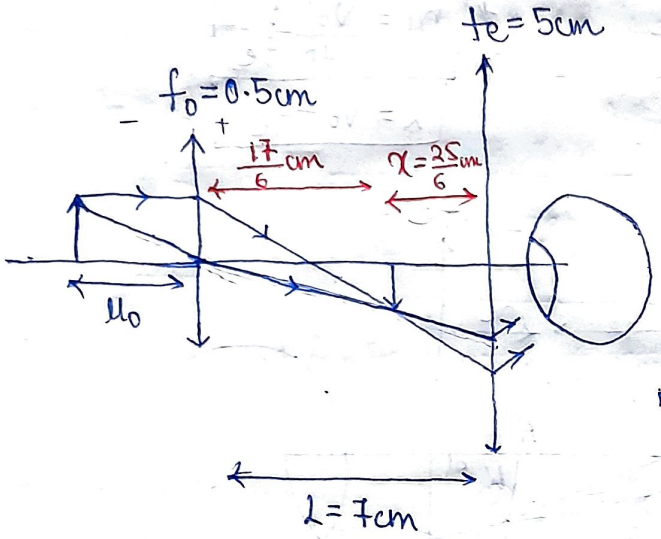
$$\Rightarrow \frac{1}{u_o} = 2 - \frac{1}{2} = \frac{3}{2}$$

$$\boxed{u_o = \frac{2}{3}cm}$$

$$m = m_t \times m_o = \frac{V}{-u} \times \frac{D}{f_e} = \frac{2}{-2/3} \times \frac{25}{5} = -15$$

$$\therefore \boxed{m = 15X}$$

Q



Find distance of object from objective lens for max^m magnification. Also find magnification.

$$x = \frac{f_e D}{f_e + D} = \frac{5 \times 25}{5+25} = \frac{25}{6}cm$$

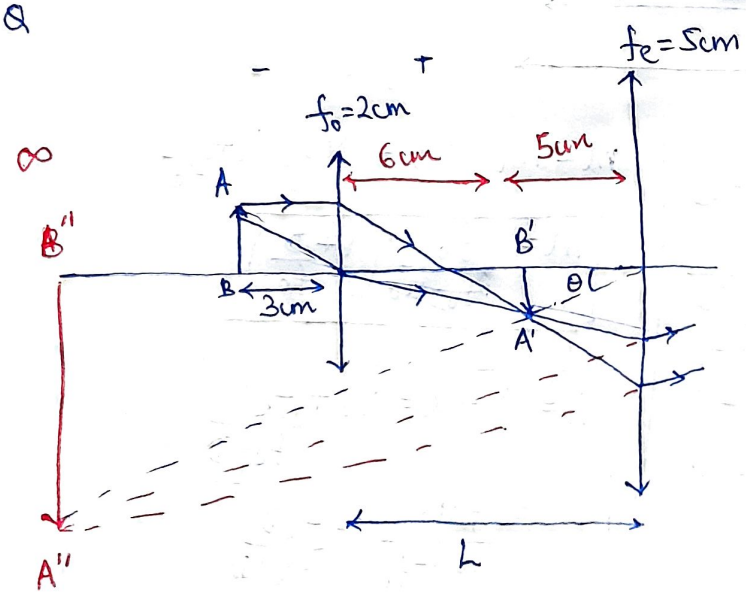
Now, $\frac{6}{17} - \frac{1}{-u_o} = \frac{1}{0.5} \Rightarrow \frac{1}{u_o} = 2 - \frac{6}{17} = \frac{28}{17}$

$$\therefore \boxed{u_o = \frac{17}{28}cm}$$

$$m = m_t \times m_o = \frac{17 \times 28}{6 \times -17}\left(1 + \frac{25}{5}\right) = \frac{-28}{6} \times 6 = -28$$

$$\therefore \boxed{m = 28X}$$

Q



Find L and magnification for normal adjustment.

For objective lens,

$$\frac{1}{v} - \frac{1}{-3} = \frac{1}{2} \Rightarrow \frac{1}{v} = \frac{1}{2} - \frac{1}{3} = \frac{1}{6}$$

$$\therefore v = 6cm$$

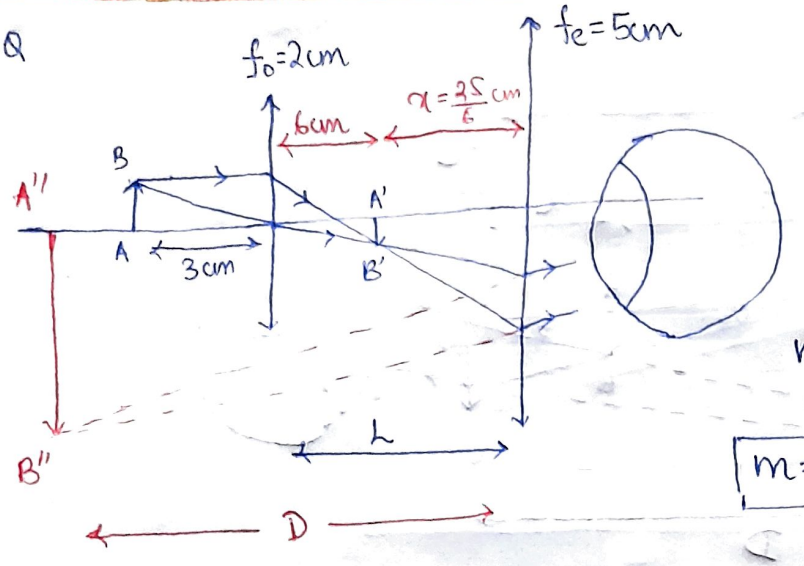
$$L = v + f_e = 6 + 5 = 11cm$$

$$m = m_t \times m_o$$

$$= \frac{6}{-3} \times \frac{25}{5} = -10$$

$$\therefore \boxed{m = 10X}$$

Q



Find L and magnification for maximum magnification.

$$x = \frac{f_e D}{f_e + D} = \frac{5 \times 25}{5 + 25} = \frac{25}{6} \text{ cm}$$

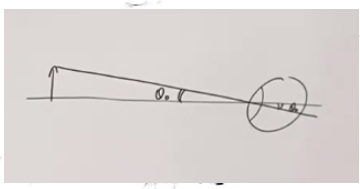
$$L = \frac{25}{6} + 6 = \frac{61}{6} \text{ cm}$$ (v_o from prev ques)

$$m = m_e \times m_o = \frac{6}{-3}\left(1 + \frac{25}{5}\right) = -2 \times 6 = -12$$

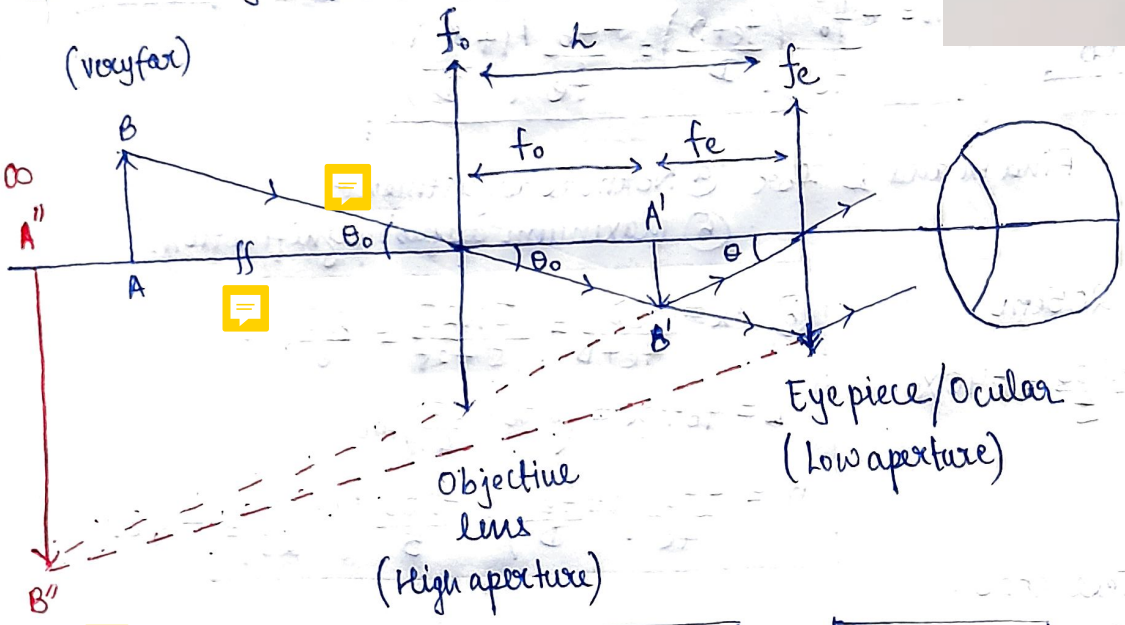
$$\boxed{m = 12X}$$

Astronomical Telescope:

→ Telescope is used to see far away objects.



(I) Normal Adjustment:



$$m = -\frac{\theta}{\theta_0} = \frac{-A'B'/f_e}{A'B'/f_o}$$

$$\therefore \boxed{m = -\frac{f_o}{f_e}}$$ ($f_o \uparrow$, $f_e \downarrow$)

$$\boxed{L = f_o + f_e}$$

→ In this case our eyes are least strained/most relaxed

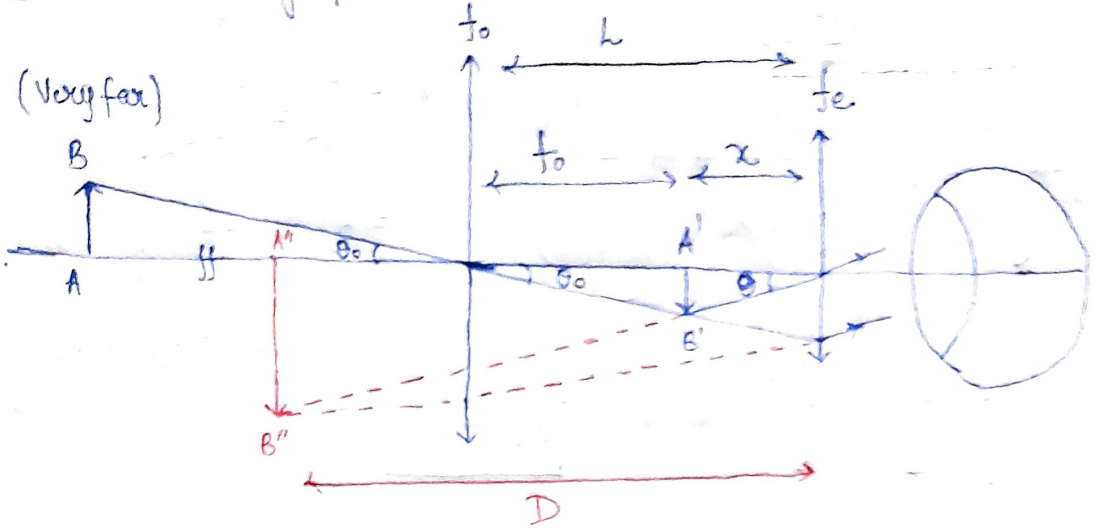
→ Object is very far.

→ Objective lens has higher aperture than eyepiece.

→ A'B': Image for objective lens
Object for eye piece

→ Final image is formed at ∞.

(II) Maximum Magnification:



$$\frac{1}{-D} - \frac{1}{-x} = \frac{1}{f_e}$$

$$\frac{1}{x} = \frac{1}{f_e} + \frac{1}{D}$$

$$\therefore \boxed{x = \frac{f_e D}{f_e + D}}$$

$$m = -\frac{\theta}{\theta_o} = \frac{-A'B'/f_e D/f_e+D}{A'B'/f_o}$$

$$\boxed{m = -\frac{f_o}{f_e}\left(\frac{f_e+D}{D}\right) = -\frac{f_o}{f_e}\left(1+\frac{f_e}{D}\right)}$$

$$\boxed{L = f_o + \frac{f_e D}{f_e + D}}$$

Q $f_o = 200\,cm$, $f_e = 5\,cm$. Find m and L for (a) Normal adjustment (b) Maximum ~~adjus~~ magnification.

(a) $L = f_o + f_e = 205\,cm$

$$m = -\frac{f_o}{f_e} = \frac{-200}{5} = -40X$$

(b) $$x = \frac{f_e D}{f_e + D} = \frac{5 \times 25}{5+25} = \frac{25}{6}\,cm$$

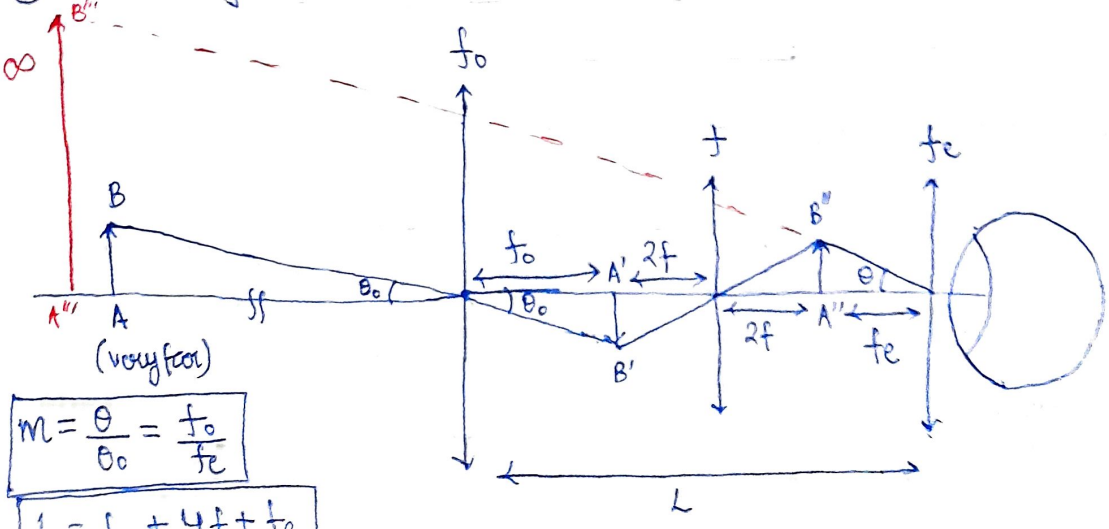
$$L = f_o + x = \left(200 + \frac{25}{6}\right)cm$$

$$m = -\frac{f_o}{f_e}\left(1+\frac{f_e}{D}\right) = \frac{-200}{5}\left(1+\frac{5}{25}\right) = -48X$$

Terrestrial Telescope:

→ It is used to give positive magnification → Final image is correct

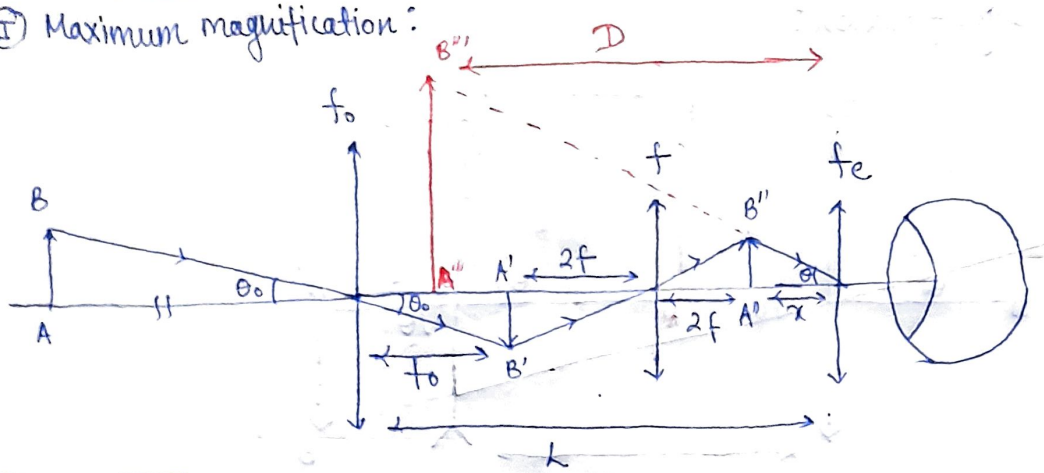
(I) Normal Adjustment:



$$\boxed{m = \frac{\theta}{\theta_o} = \frac{f_o}{f_e}}$$

$$\boxed{L = f_o + 4f + f_e}$$

(II) Maximum magnification:



$$x = \frac{f_e D}{f_e + D}$$

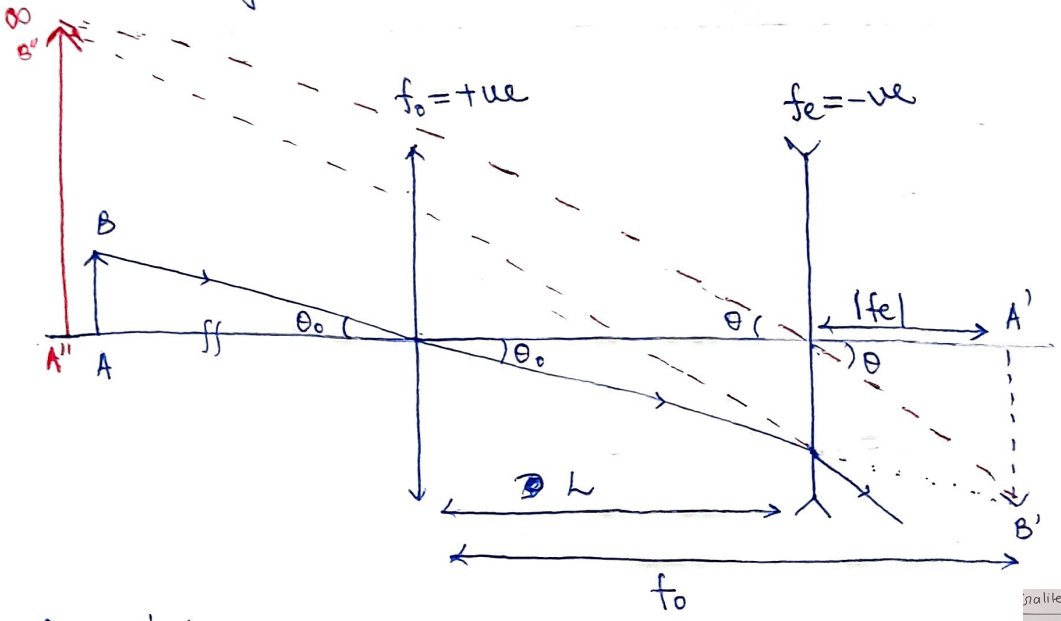
$$m = \frac{\theta}{\theta_o} = \frac{f_o}{f_e}\left(1 + \frac{f_e}{D}\right)$$

$$L = f_o + 4f + \frac{f_e D}{f_e + D}$$

Galilean Telescope:

→ It gives final image erect.
→ Tube length is small.
→ Objective lens ⇒ Converging lens ⇒ f_o = +ve
→ Eyepiece ⇒ Diverging lens ⇒ f_e = −ve

(I) Normal adjustment:



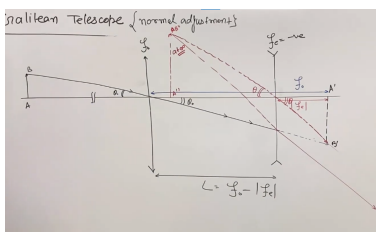
$$\theta_o = \frac{A'B'}{f_o}$$

$$\theta = \frac{A'B'}{|f_e|}$$

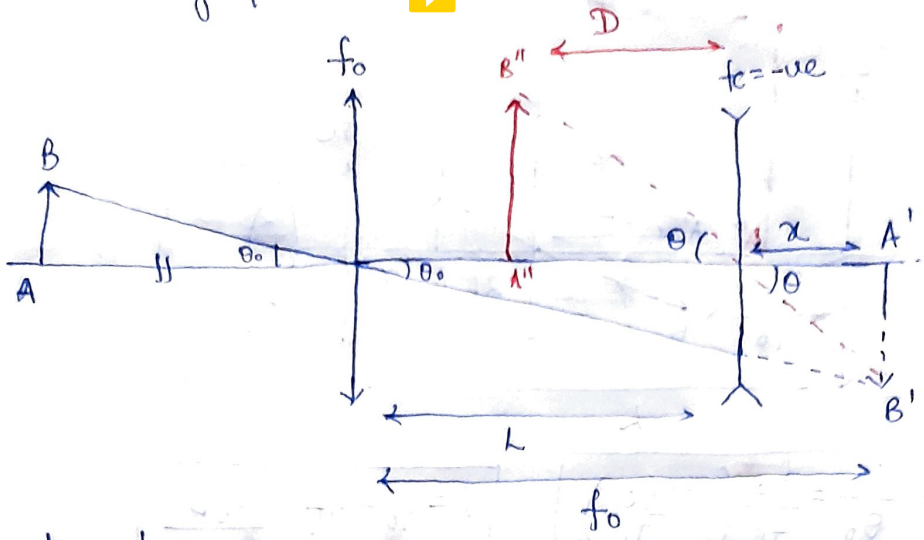
$$m = \frac{\theta}{\theta_o} = \frac{A'B'/|f_e|}{A'B'/f_o} = \frac{f_o}{|f_e|}$$

$$m = \frac{f_o}{|f_e|}$$

$$L = f_o - |f_e|$$



(II) Maximum magnification:



$$\frac{1}{-D} - \frac{1}{x} = \frac{1}{f_e}$$

$$\frac{1}{x} = -\frac{1}{f_e} - \frac{1}{D}$$

$$\boxed{x = -\frac{f_e D}{f_e + D}}$$

$$\theta_o = \frac{A'B'}{f_o} \qquad \theta = \frac{A'B'}{x} = \frac{A'B'}{\left|\frac{f_e D}{f_e + D}\right|}$$

$$m = \frac{\theta}{\theta_o} = \frac{(A'B')\left|f_e + D\right| f_o}{\left|f_e D\right| (A'B')}$$

$$\boxed{m = \left|\frac{f_o}{f_e}\right| \left(1 + \frac{f_e}{D}\right)}$$

$$L = f_o - |x|$$

$$= f_o - \left|\frac{f_e D}{f_e + D}\right|$$